# L08 — study notes

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## Class 8 — Univariate time series · Map of the notes

**One claim. A linear time-series model is a recipe for the ψ weights, and the roots decide everything.** Roots outside the unit circle give a stationary ARMA whose forecasts revert to the mean with a bounded error variance; complex roots give stochastic cycles (10.6 quarters for US GNP). **A root on the unit circle makes every shock permanent**, the forecast variance grows like $\ell\sigma^2$, and the $t$ test of $\phi = 1$ has its own Dickey–Fuller law (5% point $-2.86$, not $-1.645$). Between the two sit long memory ($\rho_k \sim k^{2d-1}$) and the seasonal unit roots that the airline model differences away. **The second half moves the same ideas to the variance**: returns are uncorrelated but their squares are not, so GARCH is an ARMA for $a_t^2$, with persistence $\alpha_1 + \beta_1$ near 1 and fat tails from Gaussian shocks; asymmetric variants add leverage; **stochastic volatility gives the variance its own shock** and needs MCMC; and **Bayesian GARCH** turns "is $\alpha_1 + \beta_1 < 1$?" into a posterior probability (0.9966 for Petrobras).

Three layers per topic:

| Layer | What it means here |
|---|---|
| **U** — intuition | permanent vs transitory shocks; volatility clusters; the news-impact curve; "two shocks per observation" in SV |
| **C** — course (examinable) | the decks', the site pages' and Tsay's results with Tsay's numbering: AR(2)/AR(p) moments, roots, cycles; PACF, AIC/BIC; forecasts and ψ-variance; MA and ARMA ACFs, invertibility; random walk, drift, trend-stationarity, ARIMA, DF/ADF; ARFIMA ACF; airline model ACF and SARIMA; ARCH/GARCH moments, likelihoods, forecasts, IGARCH; GJR/APARCH/EGARCH/GARCH-M, $t$ innovations; SV moments and the Gibbs sampler; conjugate and MCMC Bayesian GARCH |
| **G** — graduate | EACF, exact MA likelihood, the Brownian-motion limit of DF, the Gamma-function asymptotics of ARFIMA, EGARCH forecasting, ASIS and auxiliary-mixture SV samplers, Ardia–Hoogerheide's proposal. **Named or sketched, not examinable** |

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### Reading order

#### Part I — linear models and forecasting

1. L08-01-ar2-arp-cycles-identification — `ar-ma.pdf` slides 18–37; `ar-ma.html`; T §2.4.1–2.4.2 (pp. 40–49). **AR(2) moments and $\gamma_0$ step by step, the stationarity triangle, complex roots and the cycle length (GNP: 10.62 quarters; the slides'** $(1.75, -0.80)$ **is complex with** $k = 30.1$**), AR(p), the PACF cut-off, AIC vs BIC (Table 2.1: 9 vs 1).**
2. L08-02-forecasting-ma-arma — slides 38–52; `ar2-forecast.html`; T §2.4.4–2.6 (pp. 54–71). **Forecasts are conditional means; the AR(2) page's $E$ and $V$ for $h = 1..4$ derived and the**

**variance limit 7.12; MA(q) and the $\frac{1}{2}$ bound; ARMA(1,1); $V[e_h(\ell)] = \sigma^2 \sum \psi_j^2$; the identification table.**

**Part II — nonstationarity**

3. L08-03-unit-roots-dickey-fuller — `unitroot.pdf` slides 1–31; `unitroot.html`; T §2.7 (pp. 71–81); W §18.2–18.3 (pp. 574–580). **Random walk, drift as a slope, trend-stationary vs difference-stationary, DF and ADF derived (AR(3) reparametrisation), critical values, the three printed GDP statistics, and spurious regression.**
4. L08-04-long-memory-arfima — slides 32–38; T §2.11 (pp. 101–103). $(1-B)^d$, $\rho_1 = d/(1-d)$, **hyperbolic decay $k^{2d-1}$, PACF $d/(k-d)$; $d = 0.4$ keeps $\rho_{100} = 0.27$.**
5. L08-05-seasonal-airline-model — `seasonal.pdf`; `airpassengers-seasonalmodels.html`; T §2.8 (pp. 81–90); W §10.5e. **The airline model's ACF at lags 1, $s-1$, $s$, $s+1$; SARIMA as a constrained ARMA(26,13); J&J; the January effect; AirPassengers fits reproduced, BIC choice, airline vs dummies.**

**Part III — volatility**

6. L08-06-arch-garch — `garchmodeling.html`, `garch-sv-petrobras.html`; T §3.1–3.6, §3.16; W §12.6. **ARCH tests; ARCH(1) moments and kurtosis; Gaussian and $t$ likelihoods; GARCH as ARMA for $a_t^2$; kurtosis, forecasts, half-life; Tsay Ex. 3.3 and Petrobras; IGARCH and RiskMetrics.**
7. L08-07-garch-family-asymmetry-tails — glossary page; T §3.7–3.9, §3.16. **News impact; GJR, TGARCH and what `fGarch` really fits (APARCH); EGARCH; GARCH-M; $t$, skew-$t$, GED and Tiao's kurtosis formula; the log-likelihood ladder.**
8. L08-08-stochastic-volatility — SV pages; T §3.12, §12.7. **SV-AR(1) in two notations; moments, kurtosis $3e^{\sigma_h^2}$, ACF of $a_t^2$; the integral likelihood; Tsay's Gibbs sampler; GARCH vs SV on Petrobras.**
9. L08-09-bayesian-garch — `bayes-garch.pdf`; GARCH11-SVar1 §4. **Training-sample priors; the inverse-gamma update; Gibbs for M1–M2; GARCH(1,1)-$t$ with `bayesGARCH` priors;** $\Pr(\alpha_1 + \beta > 1) = 0.0034$**; Bayes vs ML.**

**Support**

10. L08-10-formulas — the site's five items and the class-plan goals mapped; every result; every number.
11. L08-11-pitfalls — 22 confusions with minimal counterexamples and an original 30-item True/False self-test.
12. L08-12-python — the reference implementation with asserts (`L08_python_checks.py`).

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## The one result to hold

$$r_t = \mu + \sum_{j\geq 0} \psi_j a_{t-j} \implies V[e_h(\ell)] = \sigma^2 \sum_{j=0}^{\ell-1} \psi_j^2 \begin{cases} \to \gamma_0 & \text{roots outside the unit circle} \\ = \ell\sigma^2 & \text{random walk, } \psi_j \equiv 1 \end{cases} \qquad \sigma_h^2(\ell) - \sigma_\infty^2 = (\alpha_1+\beta_1)^\ell$$

**The same arithmetic runs twice: once on the level, once on the variance.** In levels the $\psi$ weights decide whether forecast uncertainty is bounded (L08-02 (28) vs L08-03 (6)); in variances GARCH's ARMA form for $a_t^2$ makes $\alpha_1 + \beta_1$ play the role of the AR root (L08-06 (21), (28)), and IGARCH is its unit root.

## Numbers and cases to have memorized

| Quantity | Value | Meaning |
|---|---|---|
| AR(2) $(1.75, -0.80)$ | $\rho_1 = 0.972$, sd 7.12, complex, $k = 30.1$ | inside the triangle despite $\phi_1 > 1$ |
| GNP AR(3) cycle | **10.62** quarters (R: 10.656) | complex pair of $1 - 0.869B + 0.274B^2$ |
| MA(1) bound | $\lvert\rho_1\rvert \leq$ **0.5** | identification |
| DF 5% critical value | **−2.86** (trend: −3.41); −1.645 rejects a true unit root **46.5%** of the time | non-normal law |
| Spurious regression | **66%** rejections at $n = 50$ | independent random walks |
| ARFIMA $d = 0.4$ | $\rho_1 = 2/3$, $\rho_{100} = 0.27$ | hyperbolic decay |
| Airline (AirPassengers, $n = 108$) | $\theta = 0.386$, $\Theta = 0.589$; BIC best of 16 | $\rho_{11} = \rho_{13} = \rho_1\rho_{12} = 0.147$ |
| Intel ARCH(1) | $\alpha_1 = 0.356$, kurtosis **4.23** | fat tails from normal shocks |
| Tsay S&P 500 GARCH(1,1) | persistence **0.9727**, $\sigma_\infty = 0.0561$, half-life **25** months | Tsay misprints 0.9772 |
| Petrobras GARCH(1,1) | persistence **0.980**, half-life **34** days | daily data |
| Petrobras GJR (as fitted) | bad news **0.135** vs good news **0.058** | APARCH form, persistence 0.978 |
| SV kurtosis | $3e^{\sigma_h^2}$: **4.00** (Tsay 12.3), **5.55** (PBR) | volatility's own shock |
| Bayesian GARCH (PBR) | $\Pr(\alpha_1 + \beta > 1 \mid \text{data}) =$ **0.0034** | posterior evidence of stationarity |

## Figures

Built by `figures/make_figures.py`. Run: `python Notas/L08/figures/make_figures.py`. Every figure asserts at least one plotted number.

| Figure | Shows | Lives in |
|---|---|---|
| L08-ar2-triangle-acf | stationarity triangle, complex-root parabola, the deck's and Tsay's points with cycle lengths; exact ACFs of $(1.75, -0.80)$ and $(0.90, 0.05)$ | L08-01-ar2-arp-cycles-identification |
| L08-identification-acf-pacf | exact ACF and PACF of AR(1), AR(2), MA(1), ARMA(1,1) | L08-02-forecasting-ma-arma |
| L08-ar2-forecast-fan | ar2-forecast page: mean and 5–95% band for $h = 1..50$, converging to $0 \pm 1.645 \times 7.12$ | L08-02-forecasting-ma-arma |

| Figure | Shows | Lives in |
|---|---|---|
| L08-unit-root-bands-df | forecast bands: RW with drift vs trend-stationary; DF $t$ density (seeded MC) vs $N(0,1)$ | L08-03-unit-roots-dickey-fuller |
| L08-long-memory-acf | ARFIMA(0,d,0) ACF vs AR(1) with the same $\rho_1$, linear and log–log | L08-04-long-memory-arfima |
| L08-airline-acf | exact ACF of $w_t$ for the J&J and AirPassengers airline fits | L08-05-seasonal-airline-model |
| L08-garch-persistence-kurtosis | variance term structures (Tsay, Petrobras, IGARCH); $(\alpha_1, \beta_1)$ region with the IGARCH line and the fourth-moment boundaries | L08-06-arch-garch |
| L08-news-impact-sv-memory | news impact of GARCH, GJR, EGARCH (Petrobras fits); ACF of $a_t^2$ under GARCH vs SV | L08-07-garch-family-asymmetry-tails, L08-08-stochastic-volatility |

### Interactive companions

- The AR(2) Triangle (`companions/companion-ar2-dynamics.html`). Move $\phi_1$, $\phi_2$, $\sigma$, the last two observations $y_n$, $y_{n-1}$ and the horizon $H$, or snap to the deck's, GNP's and Tsay's parameter sets. Watch the point in the triangle, the **stationarity verdict with the root moduli and the cycle length**, the ACF and the forecast fan. Defaults reproduce the ar2-forecast page ($E$ −6.795, −7.502, −7.693, −7.460; sd → 7.121). Linked from L08-01-ar2-arp-cycles-identification and L08-02-forecasting-ma-arma.
- The Volatility Dial (`companions/companion-garch-volatility.html`). Move $\alpha_1$, $\beta_1$, $\alpha_0$, the 1-step volatility, the innovation tails ($t_\nu$ or normal) and the horizon, or snap to Tsay's GARCH and IGARCH and the Petrobras fits. Watch the variance term structure, the **half-life of a volatility shock**, $\sigma_\infty$, the kurtosis and the $(\alpha_1, \beta_1)$ plane. Defaults reproduce Tsay's Table 3.1 to ±0.01 percentage points. Linked from L08-06-arch-garch.

Both pages are verified by `companions/check_companions.py` at three control vectors (both ends and the default) plus degenerate cases (unit-root edge, explosive, IGARCH, infinite fourth moment), against their own JavaScript run under `node`.

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### Scope and ceiling

- **Examinable (C):** what the four decks, the eight site pages and Tsay's assigned sections state — the list in the layers table above. The final exam asks for derivations and the reading of printed output, without code (L08-10-formulas §0).
- **Named only (G):** EACF, exact MA likelihood, Brownian-motion limits, KPSS, ARFIMA estimation, long-memory SV, CHARMA/RCA, EGARCH multi-step forecasts, SV samplers beyond Tsay §12.7, the `bayesGARCH` proposal.
- **Not on the class schedule, not developed:** cointegration and error correction (W §18.4; syllabus topic 04, listed in `Map/class-plan.md` as "not on the site schedule"); multivariate GARCH and DCC (Class 9); VAR, impulse responses, BVAR (Class 9).

## Errata found

- **Tsay p. 137** prints $\hat{\alpha}_1 + \hat{\beta}_1 = 0.9772$ for $0.1216 + 0.8511 = 0.9727$ → L08-06-arch-garch §4.4. Tsay p. 125 prints the Intel unconditional sd as 0.1315; $\sqrt{0.0111/0.644} = 0.1313$. Tsay p. 154 prints $E(a_t^4) = 3\exp(2\mu_h^2 + 2\sigma_h^2)$; it is $2\mu_h$ → L08-08-stochastic-volatility §4.
- `ar2-forecast.html`: $\mathrm{Var}(y_t)$ printed with the braces multiplying instead of dividing (its own R code divides and prints 7.12069); the $h = 4$ lines are labelled $E(y_{n+3}\,...)$ → L08-01-ar2-arp-cycles-identification §3.
- `ar-ma.pdf` **slide 18** loses the inequality in "$\phi_1 + \phi_2 \neq 1$". `unitroot.pdf`: the AR(3) rewrite repeats "$\tilde{\phi}_2 \Delta y_{t-2}$" (should start with $\tilde{\phi}_1 \Delta y_{t-1}$); the ACF slide prints "$(k = d)$" for $(k - d) \to$ L08-03-unit-roots-dickey-fuller §6.1, L08-04-long-memory-arfima §4.
- `seasonal.pdf` **slide 10**: in the ARIMA$(1,1,1)\times(1,1,1)_{12}$ expansion the $y_{t-12}$ coefficient should be $(1 + \Phi)$ and the $y_{t-13}$ coefficient $-(1 + \phi)(1 + \Phi)$ → L08-05-seasonal-airline-model §4.1.
- `garchmodeling.html`: the recursion is written with the standardized $\varepsilon$ (it must be the raw residual); the GJR and T-GARCH formulas use an indicator, but `garchFit(leverage=TRUE)` fits the APARCH form, so the printed `gamma1` is APARCH's $\gamma$ (literal reading: persistence 1.078); the legend "GARCH t(3)" is the $\nu = 7$ fit → L08-07-garch-family-asymmetry-tails §3.
- `GARCH11-SVar1.html`: §3 is titled "Gaussian errors" but fits `cond.dist="std"`; the SV table prints a "p-value" of 1.9987 → L08-08-stochastic-volatility §9. `sp500-gaussian-student-SV-models.html` lists SVtl with $\varepsilon \sim N(0,1)$ (should be $t_\nu$) → L08-08-stochastic-volatility §2.
- **Not an error, a trap:** the unit-root deck's ADF $-1.61$ (p 0.45) for GDP and the site page's $-0.37$ (p 0.987) differ because `tseries::adf.test` always adds a trend → L08-03-unit-roots-dickey-fuller §7.

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## Course-wide position

| Class | Relation to Class 8 |
|---|---|
| 1 — review | AR(1) moments, equilibrium law, random walk, AR(p) as regression, VAR(1): the base this class builds on → L01-07; Ljung–Box on squared IBM returns (182.1) is the first ARCH evidence |
| 2 — statistical learning | out-of-sample RMSE (the AirPassengers study) and information criteria as model choice → L02 §3 |
| 3+4 — Bayesian learning | conjugate normal/inverse-gamma updates and Gibbs sampling, used here for M0–M2 and SV; MH for GARCH |
| 5+6 — causal inference | spurious regression is the time-series face of "association is not causation" |
| 7 — more econometric models | Student-$t$ errors and heteroskedasticity-robust inference reappear as $t$-GARCH and ARCH in regressions (W §12.6) |
| 8 | **ARMA, unit roots, long memory, seasonality, ARCH/GARCH, SV, Bayesian GARCH** |

| Class | Relation to Class 8 |
|---|---|
| 9 — multivariate time series | VAR(p), IRFs and structural VARs generalise L08-01/02; factor SV and DCC generalise L08-06/08; TVP-VAR with SV reuses the SV sampler |

**Rules packs:** rules/04 (ARIMA, unit roots, long memory) · rules/05 (GARCH, SV) · scope rules/09.

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### Bibliography actually used in these notes

| Book / source | Where |
|---|---|
| **The Class 8 decks and site pages** (`Aula/class-08/materials/`) — the spine | everywhere |
| **Tsay AFTS 3e** §2.4–2.8 (pp. 40–90), §2.11 (pp. 101–103); §3.1–3.9, 3.12–3.13, 3.16 (pp. 110–166); §12.7 (pp. 636–643) | the C layer of every note |
| **Wooldridge 6e** §10.5e (pp. 336–338), §11.1 (pp. 345–348), §12.6 (pp. 391–396), §18.2–18.3 (pp. 574–580) | notes 02, 03, 05, 06 |
| Box–Jenkins Series G (AirPassengers, printed on `seasonal.pdf`); Tsay's data files `m-deciles08.txt`, `d-gmsp9908.txt` | notes 05, 12 |
| Engle (1982), Bollerslev (1986), Nelson (1991), Glosten–Jagannathan–Runkle (1993), Ding–Granger–Engle (1993), Jacquier–Polson–Rossi (1994), Kastner (2016), Ardia–Hoogerheide (2010), as cited by the pages | notes 06–09 |

**Related project material:** Aula/COURSE.md (the site's Class 8 list) · Map/class-plan.md (goals, readings, practice) · the plan `.agent/plans/notas-L08.md`. Built in parallel by other agents (paths, not links): `Listas/lista-L08-book-exercises.pdf` (solutions to the practice list), `Simulados/quiz-L08-*.md`, `NotebookLM/audio/L08-audio.md`, `Leituras/L08-narrated.txt`.

**Practice (number and printed page only; statements in the books):** T 2.1, 2.2, 2.3 (p. 104), 2.4, 2.5, 2.6 (p. 105); W 18.C13 (p. 604); T 3.3, 3.4 (p. 168), 3.6 (p. 169), 3.11, 3.12 (p. 170), 3.14 (p. 171); W 12.4 (p. 397).

**PDF:** `L08-notes.pdf` (all thirteen notes, rendered by `notes_pdf.py`).

# Tsay §2.4.1–2.4.2 · AR(2) and AR(p): moments, the stationarity triangle, cycles from complex roots, and choosing p

> **Claim. An AR(2) is two AR(1)'s stacked, and its behaviour is read off one quadratic**, $1-\phi_1 x-\phi_2 x^2=0$. **Stationary iff both solutions lie outside the unit circle, which in the** $(\phi_1,\phi_2)$ **plane is the triangle** $\phi_2<1-\phi_1$, $\phi_2<1+\phi_1$, $\phi_2>-1$**.** The ACF obeys the same second-order difference equation as the series (Yule–Walker), so **real roots give a mixture of exponential decays and complex roots give damped waves with average period** $k=2\pi/\cos^{-1}[\phi_1/(2\sqrt{-\phi_2})]$, which is how a quarterly GNP AR(3) delivers a 10.6-quarter business cycle. **The order** $p$ **is chosen by the PACF, which cuts off after lag** $p$ **with standard error** $1/\sqrt{T}$**, or by AIC/BIC, which can disagree** (AR(9) vs AR(1) on the same CRSP returns). Prev: L01-07 · Next: L08-02-forecasting-ma-arma · Map: L08-00-map

What L01-07 already settled and this note uses without repeating: weak stationarity, the ACF and Ljung–Box (L01-07 (1)–(3)), linear series and $\psi$ weights (L01-07 (4)), the AR(1) mean, variance and ACF $\rho_\ell=\phi_1^\ell$ (L01-07 (6)–(11)), conditional least squares = conditional MLE (L01-07 (12)–(13)), and AR(p) as a regression with mean (L01-07 (20)–(21)).

**Notation.**

| Symbol | Meaning | Range |
|---|---|---|
| $r_t$ | the series (Tsay); $y_t$ on the site's pages | |
| $a_t$ | white-noise shock, variance $\sigma_a^2$ ($\sigma^2$ on ar2-forecast.html) | |
| $\phi_0,\phi_1,\dots,\phi_p$ | AR intercept and coefficients | |
| $\mu=E(r_t)$ | stationary mean | |
| $\gamma_\ell,\ \rho_\ell$ | lag-$\ell$ autocovariance and autocorrelation; $\gamma_0=V(r_t)$ | $\rho_\ell\in[-1,1]$ |
| $B$ | back-shift operator, $Br_t=r_{t-1}$, $B\gamma_\ell=\gamma_{\ell-1}$ (Tsay writes $L$ too) | |
| $x_1,x_2$ | solutions of the characteristic equation $1-\phi_1x-\phi_2x^2=0$ | complex in general |
| $\omega_i=1/x_i$ | characteristic roots (Tsay's name for the inverses) | |
| $k$ | average length of the stochastic cycle | periods |
| $\phi_{\ell,\ell}$ | lag-$\ell$ partial autocorrelation (PACF) | |
| $T$ | sample length | |
| $\tilde\sigma_\ell^2$ | MLE of $\sigma_a^2$ in an AR($\ell$) fit | |

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## 1 · C — AR(2): mean and the centred form (slide 18; T pp. 40–41)

The model (T eq. 2.12):

$$r_t=\phi_0+\phi_1 r_{t-1}+\phi_2 r_{t-2}+a_t. \tag{1}$$

Take expectations of (1), with $E(a_t) = 0$ and, under stationarity, $E(r_t) = E(r_{t-1}) = E(r_{t-2}) = \mu$:

$$\mu = \phi_0 + \phi_1\mu + \phi_2\mu$$

$$\mu(1 - \phi_1 - \phi_2) = \phi_0$$

$$\mu = \frac{\phi_0}{1 - \phi_1 - \phi_2}, \qquad \text{provided } \phi_1 + \phi_2 \neq 1. \tag{2}$$

(The slide prints the proviso as "$\phi_1+\phi_2 = 1$": the inequality sign was lost.) Substitute $\phi_0 = (1-\phi_1-\phi_2)\mu$ from (2) into (1):

$$r_t = (1 - \phi_1 - \phi_2)\mu + \phi_1 r_{t-1} + \phi_2 r_{t-2} + a_t$$

$$r_t - \mu = \phi_1(r_{t-1} - \mu) + \phi_2(r_{t-2} - \mu) + a_t. \tag{3}$$

**The intercept is not the mean.** At $\phi_1 + \phi_2 \to 1$ the denominator of (2) vanishes: that is the unit root of L08-03-unit-roots-dickey-fuller.

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## 2 · C — the moment equations and Yule–Walker (slides 18–19; T eq. 2.13, p. 41)

Multiply (3) by $(r_{t-\ell} - \mu)$ and take expectations. For $\ell > 0$, $r_{t-\ell}$ depends only on shocks dated $t - \ell$ and earlier, so $E[(r_{t-\ell} - \mu)a_t] = 0$:

$$E[(r_{t-\ell} - \mu)(r_t - \mu)] = \phi_1 E[(r_{t-\ell} - \mu)(r_{t-1} - \mu)] + \phi_2 E[(r_{t-\ell} - \mu)(r_{t-2} - \mu)] + 0$$

$$\gamma_\ell = \phi_1\gamma_{\ell-1} + \phi_2\gamma_{\ell-2}, \qquad \ell > 0. \tag{4}$$

For $\ell = 0$ the shock term does not vanish: $E[(r_t - \mu)a_t] = E(a_t^2) = \sigma_a^2$ (only the $a_t$ in (3) is correlated with itself), so

$$\gamma_0 = \phi_1\gamma_1 + \phi_2\gamma_2 + \sigma_a^2. \tag{5}$$

Divide (4) by $\gamma_0$ — the **Yule–Walker** (moment) equation of the ACF, T (2.13):

$$\rho_\ell = \phi_1\rho_{\ell-1} + \phi_2\rho_{\ell-2}, \qquad \ell > 0. \tag{6}$$

At $\ell = 1$, using $\rho_0 = 1$ and $\rho_{-1} = \rho_1$ (L01-07 (1)):

$$\rho_1 = \phi_1\rho_0 + \phi_2\rho_{-1} = \phi_1 + \phi_2\rho_1$$

$$\rho_1 = \frac{\phi_1}{1-\phi_2}. \tag{7}$$

At $\ell = 2$, substituting (7) into (6):

$$\rho_2 = \phi_1\rho_1 + \phi_2 = \frac{\phi_1^2}{1-\phi_2} + \phi_2. \tag{8}$$

In operator form (6) is $(1-\phi_1 B - \phi_2 B^2)\rho_\ell = 0$: **the ACF solves the same difference equation as the series without its shock**. That single fact gives the shape of the ACF (§4) and of the forecasts (L08-02-forecasting-ma-arma §1).

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## 3 · C — the variance, step by step (slide 19)

Write (5) with $\gamma_\ell = \rho_\ell\gamma_0$. From (7), $\gamma_1 = \frac{\phi_1}{1-\phi_2}\gamma_0$; from (4) at $\ell = 2$, $\gamma_2 = \phi_1\gamma_1 + \phi_2\gamma_0$. Substitute $\gamma_2$ into (5):

$$\gamma_0 = \phi_1\gamma_1 + \phi_2(\phi_1\gamma_1 + \phi_2\gamma_0) + \sigma_a^2$$

$$\gamma_0 = \phi_1(1+\phi_2)\gamma_1 + \phi_2^2\gamma_0 + \sigma_a^2.$$

Substitute $\gamma_1$ from (7):

$$\gamma_0 = \phi_1(1+\phi_2)\Big(\frac{\phi_1}{1-\phi_2}\gamma_0\Big) + \phi_2^2\gamma_0 + \sigma_a^2$$

$$\gamma_0\Big[1 - \phi_2^2 - \frac{\phi_1^2(1+\phi_2)}{1-\phi_2}\Big] = \sigma_a^2.$$

Multiply both sides by $(1-\phi_2)$ and use $1-\phi_2^2 = (1-\phi_2)(1+\phi_2)$:

$$\gamma_0\big[(1-\phi_2)^2(1+\phi_2) - \phi_1^2(1+\phi_2)\big] = (1-\phi_2)\sigma_a^2$$

$$\boxed{\gamma_0 = \frac{(1-\phi_2)\,\sigma_a^2}{(1+\phi_2)\big[(1-\phi_2)^2 - \phi_1^2\big]}} \tag{9}$$

the slide's "$\gamma_0 = \frac{1-\phi_2}{(1+\phi_2)[(1-\phi_2)^2-\phi_1^2]}\sigma_a^2$". **Erratum:** `ar2-forecast.html` §3 prints $Var(y_t) = (\frac{1-\phi_2}{1+\phi_2})\{(1-\phi_2)^2 - \phi_1^2\}\sigma^2$, with the braces multiplying; its own R line `sqrt(sig^2*(1-phi2)/((1+phi2)*((1-phi2)^2-phi1^2)))` divides, as in (9), and prints 7.12069.

**Worked — the two parameter sets of slides 20–22** ($\sigma_a = 1$):

| | $(\phi_1, \phi_2) = (1.75, -0.80)$ | $(\phi_1, \phi_2) = (0.90, 0.05)$ |
|---|---|---|
| $\rho_1$ by (7) | $\frac{1.75}{1-(-0.80)} = \frac{1.75}{1.80} = 0.9722$ | $\frac{0.90}{1-0.05} = \frac{0.90}{0.95} = 0.9474$ |
| $\rho_2$ by (8) | $1.75(0.9722) + (-0.80) =$ $1.7014 - 0.80 = 0.9014$ | $0.90(0.9474) + 0.05 =$ $0.8526 + 0.05 = 0.9026$ |
| $(1-\phi_2)^2 - \phi_1^2$ | $1.80^2 - 1.75^2 =$ $3.2400 - 3.0625 = 0.1775$ | $0.95^2 - 0.90^2 =$ $0.9025 - 0.8100 = 0.0925$ |
| $\gamma_0$ by (9) | $\frac{1.80}{(0.20)(0.1775)} = \frac{1.80}{0.0355} = 50.704$ | $\frac{0.95}{(1.05)(0.0925)} = \frac{0.95}{0.097125} = 9.7812$ |
| sd $\sqrt{\gamma_0}$ | **7.1207** (the page's 7.12069) | 3.1275 |

The first lag correlations are almost equal (0.97 vs 0.95, 0.90 vs 0.90), yet the variances differ fivefold: the first model sits close to an edge of the triangle (§5), where (9)'s bracket 0.1775 is small.

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## 4 · C — the characteristic equation and its roots (T eq. 2.14, pp. 41–42)

Corresponding to (6) is the polynomial equation (T 2.14)

$$1 - \phi_1 x - \phi_2 x^2 = 0, \tag{10}$$

with solutions (quadratic formula, $\phi_2 \neq 0$)

$$x = \frac{\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}}{-2\phi_2}. \tag{11}$$

Their inverses $\omega_i = 1/x_i$ are the **characteristic roots**. Multiplying (10) by $\omega^2$ with $x = 1/\omega$ shows they solve

$$\omega^2 - \phi_1\omega - \phi_2 = 0, \qquad \omega_1 + \omega_2 = \phi_1, \quad \omega_1\omega_2 = -\phi_2, \tag{12}$$

and the AR polynomial factors as $1 - \phi_1 B - \phi_2 B^2 = (1 - \omega_1 B)(1 - \omega_2 B)$: **an AR(1) operating on top of another AR(1)** (T p. 42).

- **Real roots** ($\phi_1^2 + 4\phi_2 \geq 0$): the solution of (6) is $\rho_\ell = c_1\omega_1^\ell + c_2\omega_2^\ell$, a mixture of two exponential decays.
- **Complex roots** ($\phi_1^2 + 4\phi_2 < 0$): $\omega_{1,2} = Re^{\pm i\theta}$ with, by (12), $R^2 = \omega_1\omega_2 = -\phi_2$ and $2R\cos\theta = \omega_1 + \omega_2 = \phi_1$. So

$$R = \sqrt{-\phi_2}, \qquad \cos\theta = \frac{\phi_1}{2\sqrt{-\phi_2}}, \tag{13}$$

and $\rho_\ell = c\,R^\ell \cos(\theta\ell + \varphi)$: **a damped cosine** with damping $R$ per period and frequency $\theta$ radians per period. One full wave takes $2\pi/\theta$ periods (T p. 42):

$$\boxed{k = \frac{2\pi}{\cos^{-1}\left[\phi_1/(2\sqrt{-\phi_2})\right]}} \tag{14}$$

with the inverse cosine in radians. Writing the complex **solutions** of (10) as $a \pm bi$, $|a \pm bi| = 1/R$ and $\cos\theta = a/\sqrt{a^2+b^2}$, so equivalently $k = 2\pi/\cos^{-1}(a/\sqrt{a^2+b^2})$ (T p. 42) — the form R's `polyroot` output feeds.

> **Complex roots are the business cycle.** Damped sine waves in the ACF, and in the forecasts, are what an economist calls a stochastic cycle (T p. 42).

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## 5 · C — the stationarity triangle (slides 20–21; T p. 45)

**Condition** (T p. 45): both characteristic roots inside the unit circle, $|\omega_i| < 1$; equivalently both solutions $x_i$ of (10) outside it. Translate to $(\phi_1, \phi_2)$ with (12).

*Real roots.* $f(\omega) = \omega^2 - \phi_1\omega - \phi_2$ is an upward parabola. Both roots lie in $(-1,1)$ iff $f(1) > 0$, $f(-1) > 0$ and the vertex $\phi_1/2$ is in $(-1,1)$:

$$f(1) = 1 - \phi_1 - \phi_2 > 0 \iff \phi_2 < 1 - \phi_1, \tag{15}$$

$$f(-1) = 1 + \phi_1 - \phi_2 > 0 \iff \phi_2 < 1 + \phi_1. \tag{16}$$

The product $\omega_1\omega_2 = -\phi_2$ of two numbers in $(-1,1)$ is in $(-1,1)$, so in particular $\phi_2 > -1$.

*Complex roots.* $|\omega_1| = |\omega_2| = R = \sqrt{-\phi_2}$ by (13), so $|\omega_i| < 1 \iff -\phi_2 < 1$:

$$\phi_2 > -1. \tag{17}$$

Together, the stationary region is the **triangle** with vertices $(-2,-1)$, $(2,-1)$, $(0,1)$:

$$\boxed{\phi_2 < 1 - \phi_1, \qquad \phi_2 < 1 + \phi_1, \qquad \phi_2 > -1} \tag{18}$$

Adding (15) and (16) gives $\phi_2 < 1$, so $|\phi_2| < 1$ inside it (and the vertex condition $|\phi_1| < 2$ is implied). The **parabola** $\phi_2 = -\phi_1^2/4$ splits it: complex roots below, real above. The slide's R code draws exactly these lines (`1+phi1`, `-1`, `1-phi1`, the gray polygon `1-abs(phi1)` and `-phi1^2/4`). Note that (18) is also what keeps (9) positive: $(1-\phi_2)^2 - \phi_1^2 = (1-\phi_2-\phi_1)(1-\phi_2+\phi_1) > 0$ by (15)–(16), and $1 + \phi_2 > 0$ by (17).

**Worked — the four points of the figure.**

- $(1.75, -0.80)$**, slide 20.** Discriminant $\phi_1^2 + 4\phi_2 = 3.0625 - 3.2000 = -0.1375 < 0$: **complex** (the point lies just below the parabola, $-1.75^2/4 = -0.7656$). By (13), $R = \sqrt{0.80} = 0.8944$ (solutions of modulus $1/0.8944 = 1.1180$, outside the circle: stationary) and $\cos\theta = 1.75/(2 \times 0.8944) = 1.75/1.7889 = 0.9783$, $\theta = 0.2088$ rad, so by (14) $k = 2\pi/0.2088 =$ **30.09** periods. The ACF is a long, slow wave: it falls from 0.97 to a trough of **−0.1866 at lag 15** — half a cycle, $30.09/2 \approx 15$ — then dies out (damping $0.8944^{30} = 0.035$ per cycle).
- $(0.90, 0.05)$**, slide 20.** Discriminant $0.81 + 0.20 = 1.01 > 0$: **real**. By (11) the solutions are $x = \frac{0.90 \pm \sqrt{1.01}}{-0.10}$, i.e. $\frac{0.90+1.00499}{-0.10} = -19.05$ and $\frac{0.90-1.00499}{-0.10} = 1.0499$. One root is **barely outside the circle** (characteristic root $1/1.0499 = 0.9525$): a near-unit-root process with a slowly decaying, single-signed ACF ($\phi_1 + \phi_2 = 0.95$, close to the edge (15)).
- **GNP factor** $(0.869, -0.274)$ — the worked example below; complex, $k = 10.62$.

- **Tsay Fig. 2.4(b),** $(0.6, -0.4)$ (T p. 42): discriminant $0.36 - 1.6 = -1.24 < 0$; $\cos\theta = 0.6/(2\sqrt{0.4}) = 0.4743$, $\theta = 1.0766$, $k = 5.84$.

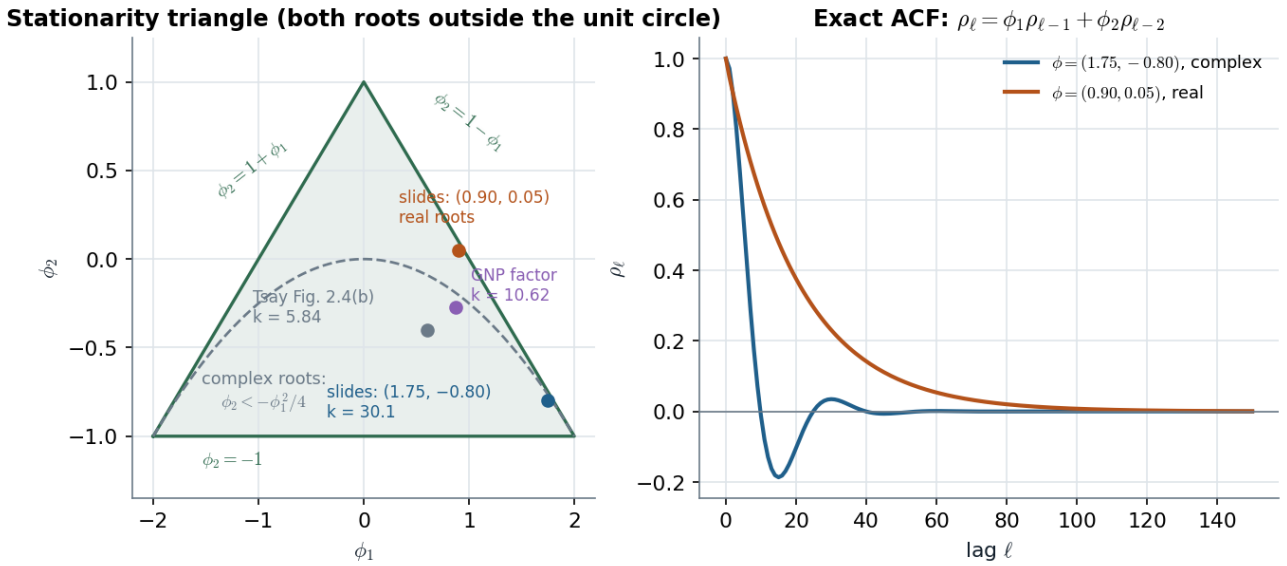



*Figure — region (18) with the parabola $\phi_2 = -\phi_1^2/4$; ACF from the recursion (6)–(7) for slides 20–22's parameter sets.*

On the left, both slide points are inside the triangle, so both processes are stationary, but each hugs a different edge: $(0.90, 0.05)$ is near $\phi_2 = 1 - \phi_1$ (a real root near 1), $(1.75, -0.80)$ is near the same edge from below the parabola (complex roots of modulus 0.89). On the right the near-real-unit-root model gives the long one-signed decay, while the complex model gives the trough at lag 15 and a small second wave near lag 30 — the cycle length (14) is visible in the ACF.

> **Interactive companion.** The AR(2) Triangle (`companions/companion-ar2-dynamics.html`) — move $\phi_1$, $\phi_2$, $\sigma$, $y_n$, $y_{n-1}$ and the horizon $H$; watch the point cross the parabola and the edges of the triangle, the root moduli and cycle length update, the ACF switch between decays and waves, and the forecast fan revert (or explode).

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## 6 · C — worked: US real GNP and the 10.6-quarter cycle (slides 26–29; T Example 2.1, pp. 42–45; ar-ma.html)

Quarterly growth of US real GNP, 1947.II–1991.I, $T = 176$; `ar(gnp, method="mle")` selects order 3 by AIC. Fitted (T eq. 2.15):

$$r_t = 0.0047 + 0.348 r_{t-1} + 0.179 r_{t-2} - 0.142 r_{t-3} + a_t, \qquad \hat{\sigma}_a = 0.0097. \tag{19}$$

**The intercept from R's output.** R's `arima` reports the *mean*, `intercept` $= 0.0077$, with coefficients 0.3480, 0.1793, −0.1423. By (2) generalised (L01-07 (21)):

$$\phi_0 = (1 - \phi_1 - \phi_2 - \phi_3)\mu = (1 - 0.348 - 0.1793 - (-0.1423)) \times 0.0077 = 0.6150 \times 0.0077 = 0.0047355.$$

$$\hat{\sigma}_a = \sqrt{9.427 \times 10^{-5}} = 0.009709.$$

**The characteristic polynomial factors** (T p. 43):

$$1 - 0.348B - 0.179B^2 + 0.142B^3 \approx (1 + 0.521B)(1 - 0.869B + 0.274B^2). \tag{20}$$

Check by multiplying out: $(1+0.521B)(1-0.869B+0.274B^2) = 1+(0.521-0.869)B+(0.274-0.521\times 0.869)B^2 + 0.521 \times 0.274B^3 = 1 - 0.348B - 0.1787B^2 + 0.1428B^3$ ✓ (to rounding).

- The first factor $(1 + 0.521B)$ is a real root $\omega = -0.521$: an alternating exponential decay.
- The second is an AR(2) with $\phi_1 = 0.869$, $\phi_2 = -0.274$. Discriminant $0.869^2 + 4(-0.274) = 0.755161 - 1.096 = -0.341 < 0$: **complex**. By (13)–(14): $\sqrt{0.274} = 0.52345$, $\cos\theta = \frac{0.869}{2\times 0.52345} = \frac{0.869}{1.04690} = 0.83007$, $\theta = \cos^{-1}(0.83007) = 0.59156$ rad,

$$k = \frac{2\pi}{0.59156} = \frac{6.28319}{0.59156} = 10.62 \text{ quarters} \approx 3 \text{ years}.$$

**From R's roots** (ar-ma.html): `polyroot(c(1,-0.3480,-0.1793,0.1423))` gives $1.590253 \pm 1.063882i$ and $-1.920152$, moduli 1.913308, 1.920152 (all $> 1$: stationary). With $a = 1.590253$, $\sqrt{a^2+b^2} = 1.913308$: $a/\sqrt{a^2+b^2} = 0.83115$, $\cos^{-1} = 0.58962$ rad, $k = 6.28319/0.58962 =$ **10.656** — the page's `10.65638`. The difference from 10.62 is the rounding of the coefficients in (20).

> **10.62 quarters is a compromise.** A nonlinear model splitting the economy into regimes finds contractions of about 3 quarters and expansions of about 12; the linear AR(3) averages them into one damped cycle (T p. 44). The same feature appears in many OECD growth rates.

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## 7 · C — AR(p) (slide 30; T p. 46)

$r_t = \phi_0 + \phi_1 r_{t-1} + \cdots + \phi_p r_{t-p} + a_t$ has stationary mean $\phi_0/(1 - \phi_1 - \cdots - \phi_p)$ (L01-07 (21)) and characteristic equation

$$1 - \phi_1 x - \phi_2 x^2 - \cdots - \phi_p x^p = 0. \tag{21}$$

**Stationary iff all solutions of (21) exceed 1 in modulus** (all characteristic roots $< 1$). Multiplying the centred model by $r_{t-\ell} - \mu$ as in §2 gives the AR(p) Yule–Walker equation

$$(1 - \phi_1 B - \cdots - \phi_p B^p)\rho_\ell = 0, \qquad \ell > 0, \tag{22}$$

so the ACF is **a mixture of damped sine–cosine waves (one per complex pair) and exponential decays (one per real root)** — it never cuts off. That is why the ACF cannot identify $p$; the PACF can.

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## 8 · C — the partial autocorrelation function (slides 31–32; T pp. 46–48)

Fit AR models of increasing order by least squares (they are regressions, L01-07 §8):

$$r_t = \phi_{0,1} + \phi_{1,1} r_{t-1} + e_{1t},$$

$$r_t = \phi_{0,2} + \phi_{1,2} r_{t-1} + \phi_{2,2} r_{t-2} + e_{2t},$$

$$r_t = \phi_{0,3} + \phi_{1,3} r_{t-1} + \phi_{2,3} r_{t-2} + \phi_{3,3} r_{t-3} + e_{3t}, \ \ldots \tag{23}$$

**Definition.** The last coefficient of the $j$-th regression, $\hat{\phi}_{j,j}$, is the **lag-$j$ sample PACF**: the added contribution of $r_{t-j}$ over an AR$(j-1)$ — a partial $F$ (here $t$) test logic. For a **stationary Gaussian AR(p)** (T p. 47):

1. $\hat{\phi}_{p,p} \to \phi_p$ as $T \to \infty$;
2. $\hat{\phi}_{\ell,\ell} \to 0$ for all $\ell > p$;
3. $V(\hat{\phi}_{\ell,\ell}) \approx 1/T$ for $\ell > p$.

So **the PACF cuts off at lag** $p$, and $\pm 2/\sqrt{T}$ is the approximate 95% band. Population version: $\phi_{\ell,\ell}$ is a function of $\rho_1, \ldots, \rho_\ell$ (the Durbin–Levinson recursion, used in L08-12-python and the figure of L08-02-forecasting-ma-arma §8). For an AR(2), $\phi_{1,1} = \rho_1$ and $\phi_{2,2} = \phi_2$.

**Worked.**

- CRSP value-weighted monthly returns, 1926–2008, $T = 996$ (T Table 2.1, p. 47): $1/\sqrt{996} = 0.0317 \approx 0.032$, band $\pm 0.063$. PACF: 0.115, −0.030, **−0.102**, 0.033, 0.062, −0.050, 0.031, 0.052, **0.063**, … Lags 1 and 3 clear 0.063, lag 9 sits on it: AR(3) or AR(9) at 5%, AR(3) at 1%.
- GNP (Fig. 2.6, p. 48): $\pm 2/\sqrt{176} = \pm 0.151$; the first three lags are large: AR(3), as in (19).

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## 9 · C — information criteria (slides 33–34; T eq. 2.16, pp. 48–49)

$$\text{AIC} = \frac{-2}{T} \ln(\text{likelihood}) + \frac{2}{T} \times (\text{number of parameters}), \tag{24}$$

goodness of fit plus a penalty, the likelihood at the MLE. For a Gaussian AR$(\ell)$, $-\frac{2}{T} \ln L = \ln \tilde{\sigma}_\ell^2 + \text{const}$ (L01-06's Gaussian log-likelihood at $\hat{\sigma}^2$), so

$$\text{AIC}(\ell) = \ln \tilde{\sigma}_\ell^2 + \frac{2\ell}{T}, \qquad \text{BIC}(\ell) = \ln \tilde{\sigma}_\ell^2 + \frac{\ell \ln T}{T}. \tag{25}$$

**Selection rule** (T p. 49): compute for $\ell = 0, \ldots, P$ and pick the minimum. The penalty per parameter is 2 for AIC and $\ln T$ for BIC; with $T = 996$, $\ln T = 6.90$, so **BIC picks smaller models whenever** $T > e^2 \approx 7.4$.

**Worked — Table 2.1 again.** AIC is minimal at $\ell = 9$ (**−5.849**); BIC at $\ell = 1$ (**−5.833**), with $\ell = 3$ a close second (−5.831). Same data, three answers (PACF: 3 or 9; AIC: 9; BIC: 1). Tsay's verdict: no criterion dominates; substance and simplicity decide. For GNP, R's `ord$aic` (rescaled so the minimum is 0) is 27.847, 2.742, 1.603, **0.000**, 0.323, … for orders 0, 1, 2, **3**, 4: AR(3).

> **Penalty per parameter: 2 (AIC) vs $\ln T$ (BIC).** Same idea as the $C_p$/AIC/BIC of regression in L01-06, same direction of disagreement.

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## 10 · C — fitting and checking (T pp. 49–53)

Conditional least squares from observation $p+1$ (L01-07 §5, §8); residuals $\hat{a}_t$; **the Ljung–Box test on the residuals has $m-g$ degrees of freedom**, $g$ the number of AR/MA coefficients (T p. 51; R and S-Plus use $m$ by default). CRSP AR(3): $Q(12) = 16.35$, $p = 0.060$ on $\chi^2_9$ (R prints 0.1756 on 12 df). Dropping the insignificant lag 2 gives $r_t = 0.0088 + 0.114 r_{t-1} - 0.106 r_{t-3} + a_t$, $\hat{\sigma}_a = 0.0536$, $Q(12) = 16.83$, $p = 0.078$ on $\chi^2_{10}$: adequate. The intercept from R's mean: $(1 - 0.1136 + 0.1063) \times 0.0089 = 0.00884$.

$R^2$ (T §2.4.3, pp. 53–54): $1 - \sum \hat{a}_t^2 / \sum (r_t - \bar{r})^2$ measures fit only for stationary series; **for a unit-root series the $R^2$ of an AR(1) fit tends to 1 whatever the true model**, and it cannot fall when lags are added (L01-06's warning again).

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## 11 · U — what the slides' simulation shows (slides 23–25)

Slides 23–25 simulate each model (`acf1.hedi`: burn-in of $n$ draws, keep the next $n$) and overlay the sample ACF on the theoretical one at $n = 10\,000$, 1 000, 500 and 200. At $n = 10\,000$ the two agree; as $n$ falls the sample ACF wanders around the theoretical curve with growing error, most visibly at the long lags of the near-unit-root $(0.90, 0.05)$ model, whose slow decay needs long samples to be estimated. **Sample ACFs at high lags are noisy, and for persistent series biased toward zero**; read them with the $\pm 2/\sqrt{T}$ band.

## 12 · G — EACF (T §2.6.3, pp. 66–68)

When the model is a mixed ARMA, neither ACF nor PACF cuts off (L08-02-forecasting-ma-arma §6). Tsay and Tiao's **extended ACF** is a two-way table whose triangle of O's has its upper-left vertex at $(p, q)$ (Table 2.4 for ARMA(1,1)); X means $|\text{EACF}| \geq 2/\sqrt{T}$. 3M monthly log returns, $T = 755$: $2/\sqrt{755} = 0.073$, vertex at $(0, 0)$ — white noise. Named only.

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## Links

- Neighbours: L01-07 (AR(1), equilibrium law, VAR(1)) · L08-02-forecasting-ma-arma (forecasts, MA, ARMA, identification figure).
- Unit: L08-00-map · L08-10-formulas · L08-11-pitfalls · L08-12-python · L08-03-unit-roots-dickey-fuller (the edge $\phi_1 + \phi_2 = 1$).
- Class 1: L01-06 (AIC/BIC in regression). Class 9 (plain forward link): the VAR(p) has the matrix version of (21): stationarity iff all roots of $\det(I - \Phi_1 x - \cdots - \Phi_p x^p) = 0$ lie outside the unit circle.
- Rules: rules/04 (ARIMA, unit roots).
- **Practice (number and printed page only; statements in Tsay):** T 2.1 (p. 104), T 2.2 (p. 104), T 2.3 (p. 104), T 2.4 (p. 105). Full solutions are being built at `Listas/lista-L08-book-exercises.pdf`.

# Tsay §2.4.4–2.6 · Forecasting, MA and ARMA: conditional means, $\psi$ weights, and what the ACF and PACF identify

**Claim. The MSE-optimal forecast is the conditional mean, and its error is the sum of the shocks still to come, weighted by the $\psi$ weights: $e_h(\ell) = a_{h+\ell} + \psi_1 a_{h+\ell-1} + \cdots + \psi_{\ell-1} a_{h+1}$, so $V[e_h(\ell)] = \sigma_a^2 \sum_{i<\ell} \psi_i^2$ rises to $\gamma_0$ while the point forecast reverts to $\mu$. For the deck's AR(2) $(1.75, -0.80)$ the forecast sd goes 1, 2.02, 3.03, 3.97, ... → 7.12. An MA(q) is a finite-memory model: always stationary, its ACF cuts off after lag $q$, its forecasts hit the mean after $q$ steps, and it is identified only up to invertibility ($|\rho_1| \le \frac{1}{2}$ for an MA(1)). An ARMA(1,1) has the AR(1)'s stationarity condition and an AR(1)-like ACF that starts decaying one lag late; neither its ACF nor its PACF cuts off.** Prev: L08-01-ar2-arp-cycles-identification · Next: L08-03-unit-roots-dickey-fuller · Map: L08-00-map

**Notation.**

| Symbol | Meaning | Range |
|---|---|---|
| $h$, $\ell$ | forecast origin, forecast horizon | $\ell \ge 1$ |
| $F_h$ | information available at $h$ (inclusive) | |
| $\hat{r}_h(\ell)$ | $\ell$-step-ahead forecast at origin $h$ | |
| $e_h(\ell) = r_{h+\ell} - \hat{r}_h(\ell)$ | forecast error | |
| $\phi_i$, $\theta_j$ | AR and MA coefficients, **Tsay's sign**: $r_t = c_0 + a_t - \theta_1 a_{t-1} - ...$ | |
| $c_0$ | MA constant ($= E(r_t)$) | |
| $\psi_i$ | MA($\infty$) weights / impulse response, $\psi_0 = 1$ | |
| $\pi_i$ | AR($\infty$) weights | |
| $\phi(B)$, $\theta(B)$ | AR and MA polynomials $1 - \sum \phi_i B^i$, $1 - \sum \theta_j B^j$ | |
| $y_n, y_{n-1}$ | last two observations on ar2-forecast.html | |

**Sign convention.** Tsay writes the MA part with minus signs, $r_t = c_0 + a_t - \theta_1 a_{t-1}$. R's `arima`, statsmodels and Wooldridge (11.1) write plus signs, $r_t = c_0 + a_t + \theta_1 a_{t-1}$. A printed R/statsmodels `ma1 = -0.39` is Tsay's $\theta_1 = 0.39$. Every formula below uses Tsay's sign.

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## 1 · C — the forecast is the conditional mean (slide 38; T p. 54)

The forecast minimises expected squared error given $F_h$ over functions $g$ of $F_h$:

$$E\{[r_{h+\ell} - \hat{r}_h(\ell)]^2 \mid F_h\} \le \min_g E[(r_{h+\ell} - g)^2 \mid F_h]. \tag{1}$$

For any $g \in F_h$, adding and subtracting $m = E(r_{h+\ell} \mid F_h)$, the cross term has conditional mean zero:

$$E[(r_{h+\ell} - g)^2 \mid F_h] = V(r_{h+\ell} \mid F_h) + (m - g)^2,$$

minimised at $g = m$. Hence

$$\hat{r}_h(\ell) = E(r_{h+\ell} \mid F_h). \tag{2}$$

Forecasts with *estimated* parameters are conditional forecasts: they ignore parameter uncertainty, which Bayesian forecasting by MCMC would include (T p. 55; Classes 3+4, L08-09-bayesian-garch).

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## 2 · C — AR(p) forecasts and mean reversion (slides 39–42; T pp. 54–56)

**One step.** From $r_{h+1} = \phi_0 + \sum_{i=1}^{p} \phi_i r_{h+1-i} + a_{h+1}$ and (2), every $r$ on the right is in $F_h$ and $E(a_{h+1} \mid F_h) = 0$:

$$\hat{r}_h(1) = \phi_0 + \sum_{i=1}^{p} \phi_i r_{h+1-i}, \qquad e_h(1) = a_{h+1}, \qquad V[e_h(1)] = \sigma_a^2. \tag{3}$$

With normal shocks the 95% interval is $\hat{r}_h(1) \pm 1.96\sigma_a$.

**Two steps.** $r_{h+2} = \phi_0 + \phi_1 r_{h+1} + \phi_2 r_h + \cdots + a_{h+2}$; replace the unknown $r_{h+1}$ by its forecast:

$$\hat{r}_h(2) = \phi_0 + \phi_1 \hat{r}_h(1) + \phi_2 r_h + \cdots + \phi_p r_{h+2-p}. \tag{4}$$

Subtract (4) from the model:

$$e_h(2) = \phi_1[r_{h+1} - \hat{r}_h(1)] + a_{h+2} = a_{h+2} + \phi_1 a_{h+1}, \qquad V[e_h(2)] = (1+\phi_1^2)\sigma_a^2 \geq V[e_h(1)]. \tag{5}$$

$\ell$ **steps** (T p. 56): recursively,

$$\hat{r}_h(\ell) = \phi_0 + \sum_{i=1}^{p} \phi_i \hat{r}_h(\ell - i), \qquad \hat{r}_h(i) = r_{h+i} \text{ if } i \leq 0. \tag{6}$$

The point forecasts obey the deterministic part of the model — the same difference equation as the ACF (L08-01-ar2-arp-cycles-identification (6), (22)). **Mean reversion:** for a stationary AR(p), $\hat{r}_h(\ell) \to E(r_t)$ and $V[e_h(\ell)] \to V(r_t)$ as $\ell \to \infty$. For an AR(1), $\hat{x}_h(\ell) = \phi_1^\ell x_h$ for the mean-adjusted series, and the half-life is $\ell = \ln 0.5 / \ln|\phi_1|$ (L01-07 (17)).

**Worked — Tsay Table 2.2 (p. 57).** AR(3) for CRSP value-weighted returns re-estimated on 984 months, $r_t = 0.0098 + 0.1024 r_{t-1} - 0.0201 r_{t-2} - 0.1090 r_{t-3} + a_t$, $\hat{\sigma}_a = 0.054$. Forecasts at origin December 2007: 0.0076, 0.0161, 0.0118, 0.0099, 0.0089, 0.0093, then 0.0095–0.0097; standard errors 0.0534, 0.0537, 0.0537, then 0.0540 from step 4. The subsample mean and sd are 0.0095 and 0.0540: weak dependence means **reversion within a few months**. Only October 2008 ($-0.1847$) leaves the 95% band.

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## 3 · C — worked: exact h-step forecast distributions of an AR(2) (ar2-forecast.html §§1–4)

Model $y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \epsilon_t$, $\epsilon_t \sim N(0, \sigma^2)$, $\theta = (\phi_1, \phi_2, \sigma) = (1.75, -0.80, 1)$; 300 observations simulated with `set.seed(1233)` after a 300-draw burn-in; last two values $y_n = -5.487150$, $y_{n-1} = -3.508837$.

$h = 1$**.** By (3) with $\phi_0 = 0$:

$$E(y_{n+1} \mid y_{1:n}, \theta) = \phi_1 y_n + \phi_2 y_{n-1}, \qquad V = \sigma^2. \tag{7}$$

$h = 2$**.** Substitute the model for $y_{n+1}$ into $y_{n+2} = \phi_1 y_{n+1} + \phi_2 y_n + \epsilon_{n+2}$:

$$y_{n+2} = \phi_1(\phi_1 y_n + \phi_2 y_{n-1} + \epsilon_{n+1}) + \phi_2 y_n + \epsilon_{n+2}$$

$$y_{n+2} = (\phi_1^2 + \phi_2) y_n + \phi_1\phi_2 y_{n-1} + \phi_1\epsilon_{n+1} + \epsilon_{n+2}, \tag{8}$$

so $E = (\phi_1^2 + \phi_2) y_n + \phi_1\phi_2 y_{n-1}$ and $V = (1 + \phi_1^2)\sigma^2$.

$h = 3$**.** Substitute (8) and $y_{n+1}$ into $y_{n+3} = \phi_1 y_{n+2} + \phi_2 y_{n+1} + \epsilon_{n+3}$ and collect:

$$E(y_{n+3} \mid \cdot) = (\phi_1^3 + 2\phi_1\phi_2) y_n + (\phi_1^2\phi_2 + \phi_2^2) y_{n-1}, \qquad V = [1 + \phi_1^2 + (\phi_1^2 + \phi_2)^2]\sigma^2. \tag{9}$$

$h = 4$**.** One more substitution:

$$E(y_{n+4} \mid \cdot) = (\phi_1^4 + 3\phi_1^2\phi_2 + \phi_2^2) y_n + (\phi_1^3\phi_2 + 2\phi_1\phi_2^2) y_{n-1}, \quad V = \left[1 + \phi_1^2 + (\phi_1^2 + \phi_2)^2 + (\phi_1^3 + 2\phi_1\phi_2)^2\right]\sigma^2. \tag{10}$$

(The page labels these $h = 4$ lines "$E(y_{n+3}\,...)$" and prints "$2\phi\phi_2^2$" for $2\phi_1\phi_2^2$; its R code is right.) The coefficients on $y_n$ are the $\psi$ weights $\psi_1 = \phi_1$, $\psi_2 = \phi_1^2 + \phi_2$, $\psi_3 = \phi_1^3 + 2\phi_1\phi_2$, $\psi_4 = \phi_1^4 + 3\phi_1^2\phi_2 + \phi_2^2$, generated by $\psi_j = \phi_1\psi_{j-1} + \phi_2\psi_{j-2}$; **the variances are** $\sigma^2 \sum_{j<h} \psi_j^2$, which is (21) below.

**Numbers** ($\phi_1 = 1.75$, $\phi_2 = -0.80$, $\sigma = 1$):

| $h$ | $\psi$ coefficients | $E(y_{n+h} \mid y_{1:n})$ | $V$ |
|---|---|---|---|
| 1 | 1.75; −0.80 | $1.75(-5.487150) + (-0.80)(-3.508837) = -9.602513 + 2.807070 =$ **−6.7954** | 1 |
| 2 | $3.0625 - 0.80 = 2.2625$; $1.75(-0.80) = -1.40$ | $2.2625(-5.487150) + (-1.40)(-3.508837) = -12.414677 + 4.912372 =$ **−7.5023** | $1 + 1.75^2 =$ **4.0625** |
| 3 | $5.359375 - 2.80 = 2.559375$; $-2.45 + 0.64 = -1.81$ | $2.559375(-5.487150) + (-1.81)(-3.508837) = -14.043677 + 6.350995 =$ **−7.6927** | $4.0625 + 2.2625^2 =$ **9.1814** |

| $h$ | $\psi$ coefficients | $E(y_{n+h} \mid y_{1:n})$ | $V$ |
|---|---|---|---|
| 4 | $9.378906 - 7.35 + 0.64 = 2.668906$; $-4.2875 + 2.24 = -2.0475$ | $2.668906(-5.487150) + (-2.0475)(-3.508837) = -14.644688 + 7.184344 =$ **$-7.4603$** | $9.1814 + 2.559375^2 =$ **15.7318** |

The mean keeps *falling* for three steps before turning: the model's momentum ($\psi_j$ rising to 2.67) outweighs reversion at first. All four distributions are normal because each $y_{n+h}$ is a fixed number plus a linear combination of iid normal shocks.

**The limit.** The AR(2) is stationary (L08-01-ar2-arp-cycles-identification §5), so $E \to 0$ and $V \to \gamma_0 = 50.704$ by L08-01-ar2-arp-cycles-identification (9): limiting sd **7.12069**. After 50 steps the exact sd is already 7.1206.

**Monte Carlo check (page §4).** $M = 10\,000$ paths simulated 50 steps forward from $(y_{n-1}, y_n)$: at $h = 50$ the sample mean is **0.0297** and sd **7.1259**, against 0 and 7.1207; the MC histograms at $h = 1, \ldots, 4$ sit on the exact normals (7)–(10). The 5%, 50%, 95% quantile lines converge to $0 \pm 1.645 \times 7.12069 = \pm 11.71$.

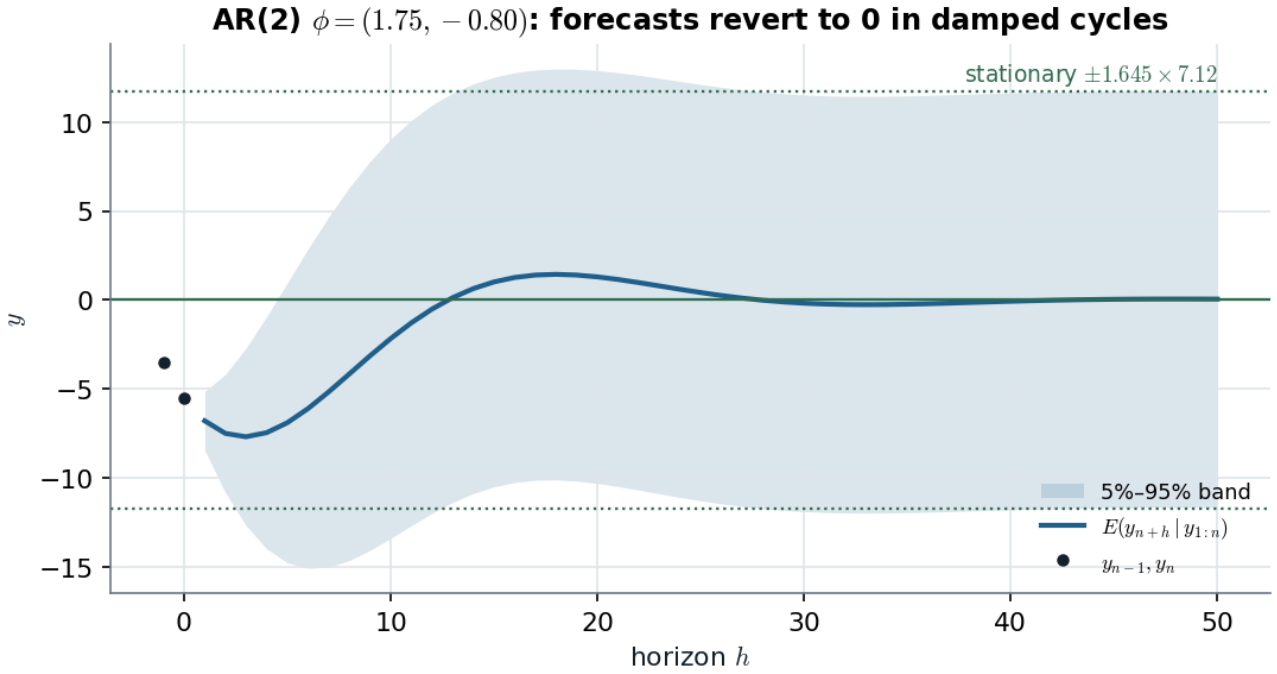



*Figure — exact mean recursion (6) and band $E \pm 1.645\sigma(\sum_{j<h} \psi_j^2)^{1/2}$ for ar2-forecast.html's model and origin; dotted lines: $\pm 1.645\sqrt{\gamma_0}$.*

The mean overshoots zero and returns — the complex roots' 30-period cycle seen in the forecasts rather than in the ACF. The band reaches its stationary width within about 15 steps: by then the forecast carries no information beyond the unconditional distribution.

> **Interactive companion.** The AR(2) Triangle (`companions/companion-ar2-dynamics.html`) — set $y_n$, $y_{n-1}$ and the horizon $H$, then move $\phi_1, \phi_2$ toward an edge of the triangle: the band stops converging (unit root) or the mean explodes; the read-outs reproduce the table above at the defaults.

## 4 · C — MA(1) as a constrained AR(∞), and MA(q) (slides 43–46; T eqs. 2.17–2.22, pp. 57–59)

An AR of infinite order, $r_t = \phi_0 + \phi_1 r_{t-1} + \phi_2 r_{t-2} + \cdots + a_t$, has too many parameters. Constrain them to one, $\phi_i = -\theta_1^i$ (T 2.17):

$$r_t = \phi_0 - \theta_1 r_{t-1} - \theta_1^2 r_{t-2} - \theta_1^3 r_{t-3} - \cdots + a_t, \tag{11}$$

stationary only if $|\theta_1| < 1$. Rewrite (11) and lag it one period (T 2.18–2.19):

$$r_t + \theta_1 r_{t-1} + \theta_1^2 r_{t-2} + \cdots = \phi_0 + a_t, \tag{12}$$

$$r_{t-1} + \theta_1 r_{t-2} + \theta_1^2 r_{t-3} + \cdots = \phi_0 + a_{t-1}. \tag{13}$$

Multiply (13) by $\theta_1$ and subtract it from (12): every $r_{t-j}$, $j \geq 1$, cancels:

$$r_t = \phi_0(1 - \theta_1) + a_t - \theta_1 a_{t-1}. \tag{14}$$

That is the **MA(1)**, $r_t = c_0 + (1 - \theta_1 B)a_t$ (T 2.20); in general (T 2.22)

$$r_t = c_0 + a_t - \theta_1 a_{t-1} - \cdots - \theta_q a_{t-q} = c_0 + (1 - \theta_1 B - \cdots - \theta_q B^q)a_t. \tag{15}$$

**Moments.** A finite linear combination of white noise has time-invariant first two moments, so **every MA(q) is weakly stationary** (no condition on the $\theta$'s), with

$$E(r_t) = c_0, \qquad V(r_t) = (1 + \theta_1^2 + \cdots + \theta_q^2)\sigma_a^2. \tag{16}$$

Wooldridge's example of a weakly dependent but correlated series is the MA(1) $x_t = e_t + \alpha_1 e_{t-1}$ (W (11.1), §11.1b): observations two or more periods apart are independent, so the LLN and CLT apply.

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## 5 · C — ACF of MA models, the $\frac{1}{2}$ bound, invertibility (slides 47–48; T pp. 59–60)

**MA(1)**, $c_0 = 0$. Multiply $r_t = a_t - \theta_1 a_{t-1}$ by $r_{t-\ell}$ and take expectations:

$$\gamma_1 = E[(a_{t-1} - \theta_1 a_{t-2})(a_t - \theta_1 a_{t-1})] = -\theta_1 \sigma_a^2, \qquad \gamma_\ell = 0 \ (\ell > 1).$$

Divide by $\gamma_0 = (1 + \theta_1^2)\sigma_a^2$ from (16):

$$\rho_1 = \frac{-\theta_1}{1 + \theta_1^2}, \qquad \rho_\ell = 0 \ \ (\ell > 1). \tag{17}$$

**MA(2)**: $\gamma_1 = E[(a_{t-1} - \theta_1 a_{t-2} - \theta_2 a_{t-3})(a_t - \theta_1 a_{t-1} - \theta_2 a_{t-2})] = (-\theta_1 + \theta_1\theta_2)\sigma_a^2$ and $\gamma_2 = -\theta_2\sigma_a^2$, so

$$\rho_1 = \frac{-\theta_1 + \theta_1\theta_2}{1+\theta_1^2+\theta_2^2}, \qquad \rho_2 = \frac{-\theta_2}{1+\theta_1^2+\theta_2^2}, \qquad \rho_\ell = 0 \ \ (\ell > 2). \tag{18}$$

**The ACF of an MA(q) cuts off after lag** $q$: a "finite-memory" model, linearly related only to its first $q$ lags.

**The $\frac{1}{2}$ bound.** Since $1+\theta_1^2 \geq 2|\theta_1|$, (17) gives $|\rho_1| \leq \frac{1}{2}$, with equality at $\theta_1 = \pm 1$. A sample lag-1 autocorrelation of 0.7 with nothing beyond is not an MA(1).

**Invertibility.** (17) is unchanged when $\theta_1 \to 1/\theta_1$: $\theta_1 = 0.5$ and $\theta_1 = 2$ both give $\rho_1 = -0.5/1.25 = -2/2.5 = -0.4$. To pick one, write $a_t = r_t + \theta_1 a_{t-1}$ and substitute repeatedly:

$$a_t = r_t + \theta_1 r_{t-1} + \theta_1^2 r_{t-2} + \theta_1^3 r_{t-3} + ... \tag{19}$$

The present shock depends less on remote returns only if $|\theta_1| < 1$: the **invertible** MA(1) (T p. 60). $|\theta_1| = 1$ is noninvertible.

**Worked.** $\theta_1 = 0.6$ (the figure's MA(1)): $\rho_1 = -0.6/1.36 = -0.4412$, all higher ACFs zero.

---

## 6 · C/G — estimation (slide 49; T pp. 61–62)

MLE, with the first shocks handled two ways. **Conditional likelihood**: set $a_t = 0$ for $t \leq 0$, then $a_1 = r_1 - c_0$, $a_2 = r_2 - c_0 + \theta_1 a_1$, … recursively, and the likelihood is a product of normal densities of these $a_t$. **Exact likelihood** (G): treat the initial shocks as extra parameters (or integrate them out). Preferred near noninvertibility ($|\theta_1| \to 1$), where the conditional recursion forgets its start slowly; the two coincide in large samples.

**Worked — Tsay's MA(9), CRSP equal-weighted monthly returns 1926–2008.** The sample ACF is significant at lags 1, 3, 9 (MA lags are identified from the ACF). Conditional MLE (T 2.23): $r_t = 0.012 + a_t + 0.189a_{t-1} - 0.121a_{t-3} + 0.122a_{t-9}$, $\hat{\sigma}_a = 0.0714$ (printed in the plus-sign form). Residual $Q(12) = 17.5$: $p = 0.041$ on $12-3 = 9$ df, 0.132 on 12. Exact MLE (T 2.24): 0.191, −0.120, 0.123, same $\hat{\sigma}_a$ — negligible difference here.

---

## 7 · C — MA forecasts (slides 50–51; T pp. 62–63)

**MA(1).** $r_{h+1} = c_0 + a_{h+1} - \theta_1 a_h$, and $a_h \in F_h$ (the residual, or by (19)):

$$\hat{r}_h(1) = c_0 - \theta_1 a_h, \quad e_h(1) = a_{h+1}; \qquad \hat{r}_h(2) = c_0, \quad e_h(2) = a_{h+2} - \theta_1 a_{h+1}, \quad V[e_h(2)] = (1+\theta_1^2)\sigma_a^2. \tag{20}$$

So $\hat{r}_h(\ell) = c_0$ for all $\ell \geq 2$ and the 2-step error variance is already $V(r_t)$: **mean reversion takes one period. MA(2):** $\hat{r}_h(1) = c_0 - \theta_1 a_h - \theta_2 a_{h-1}$, $\hat{r}_h(2) = c_0 - \theta_2 a_h$, $\hat{r}_h(\ell) = c_0$ for $\ell > 2$. In general an MA(q) forecast reaches the mean after $q$ steps.

**Worked — Tsay Table 2.3 (p. 63).** The MA(9) re-estimated on 986 months: the 10-step forecast is **0.0128**, the subsample mean; forecast standard errors 0.0712 → 0.0729 → 0.0734, approaching the subsample sd 0.0736.

---

## 8 · C — ARMA(1,1) and the identification table (T §2.6.1–2.6.2, pp. 64–66; slide 52)

$$r_t - \phi_1 r_{t-1} = \phi_0 + a_t - \theta_1 a_{t-1}, \qquad \phi_1 \neq \theta_1 \tag{21}$$

(T 2.25; with $\phi_1 = \theta_1$ the factors cancel and $r_t$ is white noise). **Mean:** $E(r_t) = \phi_0/(1-\phi_1)$, as for the AR(1). Set $\phi_0 = 0$. Multiplying (21) by $a_t$ (T 2.26):

$$E(r_t a_t) = E(a_t^2) - \theta_1 E(a_t a_{t-1}) = \sigma_a^2. \tag{22}$$

**Variance.** Take the variance of $r_t = \phi_1 r_{t-1} + a_t - \theta_1 a_{t-1}$; $r_{t-1}$ is uncorrelated with $a_t$, and $\mathrm{Cov}(r_{t-1}, a_{t-1}) = \sigma_a^2$ by (22) at $t-1$:

$$V(r_t) = \phi_1^2 V(r_{t-1}) + \sigma_a^2 + \theta_1^2 \sigma_a^2 - 2\phi_1\theta_1\sigma_a^2.$$

Under stationarity $V(r_t) = V(r_{t-1}) = \gamma_0$:

$$\gamma_0 = \frac{(1 - 2\phi_1\theta_1 + \theta_1^2)\,\sigma_a^2}{1 - \phi_1^2}, \tag{23}$$

positive only if $\phi_1^2 < 1$: **the AR(1) stationarity condition**, unaffected by the MA part.

**Autocovariances.** Multiply (21) by $r_{t-\ell}$ and take expectations. At $\ell = 1$, $E(a_t r_{t-1}) = 0$ and $E(a_{t-1} r_{t-1}) = \sigma_a^2$:

$$\gamma_1 - \phi_1\gamma_0 = -\theta_1\sigma_a^2; \qquad \gamma_\ell - \phi_1\gamma_{\ell-1} = 0, \quad \ell > 1 \quad \text{(T 2.27)}. \tag{24}$$

Divide by $\gamma_0$ and substitute $\sigma_a^2/\gamma_0$ from (23):

$$\rho_1 = \phi_1 - \frac{\theta_1\sigma_a^2}{\gamma_0} = \phi_1 - \frac{\theta_1(1-\phi_1^2)}{1 - 2\phi_1\theta_1 + \theta_1^2}, \qquad \rho_\ell = \phi_1\rho_{\ell-1} \quad (\ell > 1). \tag{25}$$

**Worked, $(\phi_1, \theta_1) = (0.8, 0.4)$.** $1 - 2(0.8)(0.4) + 0.4^2 = 1 - 0.64 + 0.16 = 0.52$; $\gamma_0 = 0.52/(1-0.64) = 0.52/0.36 = 1.4444\,\sigma_a^2$; $\rho_1 = 0.8 - \frac{0.4(0.36)}{0.52} = 0.8 - 0.2769 = \mathbf{0.5231}$; $\rho_2 = 0.8(0.5231) = 0.4185$. Compared with an AR(1) with $\phi_1 = 0.8$ ($\rho_1 = 0.8$): lower start, same decay rate from lag 2.

**General ARMA(p, q)** (T 2.28): $\phi(B) r_t = \phi_0 + \theta(B) a_t$, with no common factors; weakly stationary iff all characteristic roots of $\phi(B)$ are inside the unit circle; mean $\phi_0/(1 - \phi_1 - \cdots - \phi_p)$. ACF and PACF are not informative about $(p, q)$; Tsay uses the EACF (L08-01-ar2-arp-cycles-identification §12) or AIC/BIC over a grid. Residual $Q(m)$ has $m - g$ df, $g$ = number of AR+MA coefficients (T p. 68).

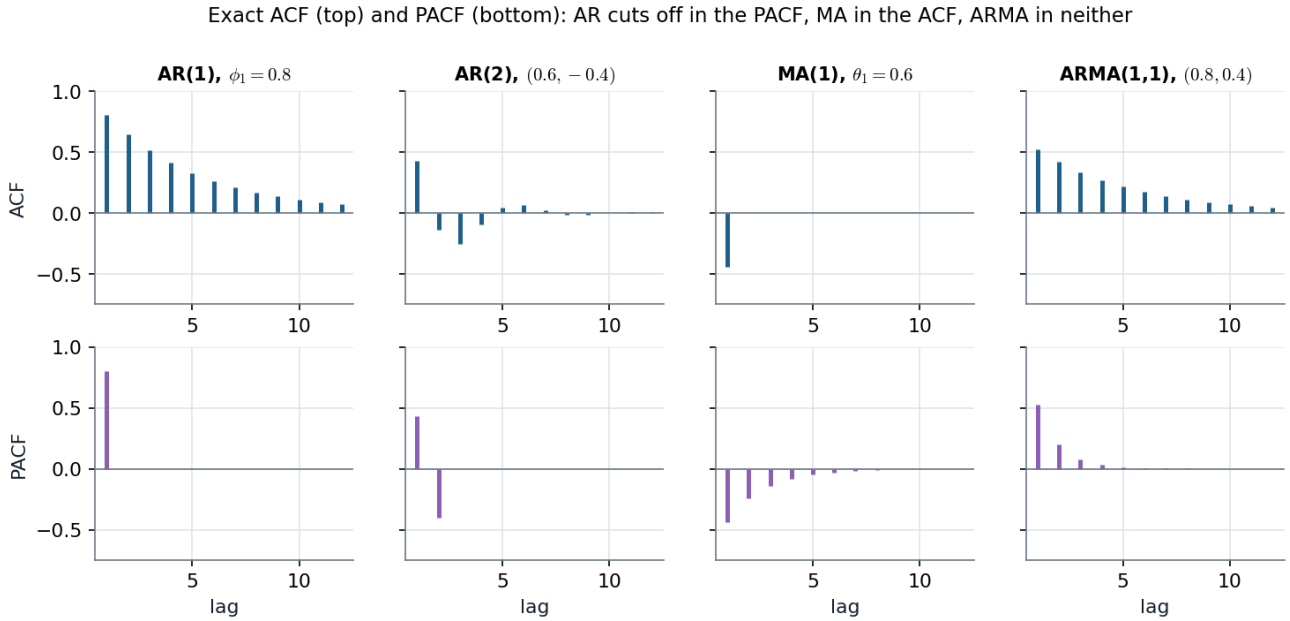



*Figure — exact ACF (from the $\psi$ weights, L01-07 (4)) and PACF (Durbin–Levinson on that ACF) for four models; Tsay's MA sign.*

Read the columns as the identification table (slide 52; T pp. 63, 66):

| Model | ACF | PACF |
|---|---|---|
| AR(p) | tails off (decays, waves if complex roots) | **cuts off after lag** $p$ |
| MA(q) | **cuts off after lag** $q$ | tails off |
| ARMA(p, q) | tails off from lag $q$ | tails off from lag $p$ |

The AR(2) PACF is exactly $\rho_1 = 0.6/1.4 = 0.4286$ then $\phi_2 = -0.4$ then zeros; the MA(1)'s single ACF spike is $-0.4412$ from (17); the ARMA(1,1) shows 0.5231 and then the 0.8 decay of (25), with a PACF that never stops.

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## 9 · C — three representations; the forecast-error variance (T §2.6.5, pp. 69–71)

By long division of the two polynomials (T 2.29–2.30):

$$\frac{\theta(B)}{\phi(B)} = 1 + \psi_1 B + \psi_2 B^2 + \cdots \equiv \psi(B), \qquad \frac{\phi(B)}{\theta(B)} = 1 - \pi_1 B - \pi_2 B^2 - \cdots \equiv \pi(B). \tag{26}$$

For ARMA(1,1): $\psi_i = \phi_1^{i-1}(\phi_1 - \theta_1)$ and $\pi_i = \theta_1^{i-1}(\phi_1 - \theta_1)$, $i \geq 1$; with $(0.8, 0.4)$, $\psi = 1, 0.4, 0.32, 0.256, \ldots$

**AR representation** (T 2.31): $r_t = \frac{\phi_0}{1-\theta_1-\cdots-\theta_q} + \pi_1 r_{t-1} + \pi_2 r_{t-2} + \cdots + a_t$. **Invertible** iff $\pi_i \to 0$; sufficient: all zeros of $\theta(B)$ outside the unit circle (MA(1): $|\theta_1| < 1$, as in (19)).

**MA representation** (T 2.32): $r_t = \mu + a_t + \psi_1 a_{t-1} + \psi_2 a_{t-2} + \ldots$, the $\psi_i$ being the **impulse response** of the series to a shock. At origin $h$ the shocks $a_h, a_{h-1}, \ldots$ are known, so (T 2.33)

$$\hat{r}_h(\ell) = \mu + \psi_\ell a_h + \psi_{\ell+1} a_{h-1} + \dots, \qquad e_h(\ell) = a_{h+\ell} + \psi_1 a_{h+\ell-1} + \cdots + \psi_{\ell-1} a_{h+1}, \tag{27}$$

$$\boxed{V[e_h(\ell)] = (1 + \psi_1^2 + \cdots + \psi_{\ell-1}^2)\sigma_a^2} \quad \text{(T 2.34)} \tag{28}$$

non-decreasing in $\ell$. **Mean reversion, proved:** stationarity means $\psi_i \to 0$, so by (27) $\hat{r}_h(\ell) \to \mu$, and by (28) $V[e_h(\ell)] \to (1 + \sum_{i\geq 1} \psi_i^2)\sigma_a^2 = V(r_t)$ (L01-07 (4)). The ar2-forecast variances of §3 are (28) with the AR(2) $\psi$'s; for an MA(q), $\psi_i = -\theta_i$ up to $q$ and 0 after, which is (20).

> **The $\psi$ weights do three jobs:** they give the ACF (L01-07 (4)), the impulse response, and the forecast-error variance (28). A unit root makes them stop decaying — the subject of L08-03-unit-roots-dickey-fuller.

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## 10 · U — summary of slide 52

MA: order from the ACF, always stationary. AR: order from the PACF, stationary iff all characteristic roots are inside the unit circle. Stationary series: forecasts converge to the mean, forecast-error variances to the variance. "Carefully read Section 2.6 of Tsay (2010) about ARMA models" — §§8–9 above.

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## Links

- Neighbours: L08-01-ar2-arp-cycles-identification (Yule–Walker, triangle, PACF, AIC/BIC) · L08-03-unit-roots-dickey-fuller ($\psi_i = 1$ for all $i$).
- Unit: L08-00-map · L08-10-formulas · L08-11-pitfalls · L08-12-python · L08-06-arch-garch (GARCH as an ARMA for $a_t^2$).
- Class 1: L01-07 (linear series, AR(1), half-life). Class 9 (plain forward link): the VAR's MA representation and impulse responses generalise (26)–(28) to matrices.
- Rules: rules/04.
- **Practice (number and printed page only; statements in Tsay):** T 2.1 (p. 104), T 2.2 (p. 104), T 2.3 (p. 104), T 2.4 (p. 105). Full solutions are being built at `Listas/lista-L08-book-exercises.pdf`.

## §2.7 · Unit roots — shocks that never die, forecasts that never revert, and a $t$ ratio that is not normal

> **Claim. A random walk is the AR(1) at $\phi_1 = 1$: every shock is permanent ($\psi_i = 1$), the point forecast never moves from today's value, and the forecast-error variance grows like $\ell\sigma^2$.** Adding a drift $\mu$ makes the constant a *slope*, not a mean; a trend-stationary series has the same straight-line mean but a *bounded* variance. **Whether a series has a unit root is tested with the Dickey–Fuller $t$ ratio, whose null distribution is skewed left and has its own critical values ($-2.86$ at 5% with a constant, $-3.41$ with a trend), not $-1.645$.** Using normal critical values rejects a true unit root almost half the time. US GDP and the S&P 500 do not reject; their differences do. Regressing one independent random walk on another "finds" a relationship most of the time. Prev: L08-02-forecasting-ma-arma · Next: L08-04-long-memory-arfima · Map: L08-00-map

Builds on L01-07 §7, where the random walk $y_t = y_0 + e_1 + \cdots + e_t$, its variance $t\sigma^2$, the non-reverting forecast and $\text{Corr}(y_t, y_{t+h}) = \sqrt{t/(t+h)}$ were derived (L01-07 (18)–(19)). This note does not repeat them; it uses them.

**Notation.**

| Symbol | Meaning | Range |
|---|---|---|
| $p_t$ | log price (Tsay) or any $I(1)$ candidate | |
| $a_t$, $e_t$ | white-noise shock (Tsay: $a_t$ in models, $e_t$ in test regressions), variance $\sigma^2$ | |
| $\mu$ | drift: $E(p_t - p_{t-1})$ in a random walk with drift | |
| $h$, $\ell$ | forecast origin, horizon | $\ell \geq 1$ |
| $\hat{p}_h(\ell)$, $e_h(\ell)$ | $\ell$-step forecast and its error | |
| $\psi_i$ | MA($\infty$) weights | |
| $\phi_0, \phi_1$ | constant and slope in the DF regressions (2.38)–(2.39) | |
| $\rho$ | AR(1) coefficient in Wooldridge's (18.17); $\theta = \rho - 1$ | |
| $c = \beta - 1$ | coefficient of $x_{t-1}$ in the $\Delta$ form of the ADF regression (Tsay); $u$ in the figure | $c \leq 0$ |
| $\beta$ | **two uses:** Tsay (2.40) levels coefficient ($H_0 : \beta = 1$); the slides' $\Delta$-form $\beta = -\phi(1)$ ($H_0 : \beta = 0$). Stated each time | |
| $\phi(B) = 1 - \phi_1 B - \cdots - \phi_p B^p$ | AR polynomial; $\phi(1) = 1 - \sum \phi_i$ | |
| $c_t$ | deterministic term: 0, $\omega_0$, or $\omega_0 + \omega_1 t$ | |
| $I(d)$ | integrated of order $d$: stationary after $d$ differences | |

## 1 · C — the random walk as a forecasting model (slides 4–12; T §2.7.1, pp. 72–73)

The random walk (T eq. 2.35):

$$p_t = p_{t-1} + a_t, \qquad p_0 \text{ given}, \quad a_t \text{ white noise.} \tag{1}$$

It is the AR(1) of L01-07 (5) with $\phi_0 = 0$, $\phi_1 = 1$, which violates $|\phi_1| < 1$: "a unit-root nonstationary time series" (T p. 72).

**Point forecasts.** At origin $h$, (1) at $t = h + 1$ gives $p_{h+1} = p_h + a_{h+1}$; take $E(\cdot \mid F_h)$ with $E(a_{h+1} \mid F_h) = 0$:

$$\hat{p}_h(1) = p_h. \tag{2}$$

For two steps, write $p_{h+2} = p_{h+1} + a_{h+2}$ and condition:

$$\hat{p}_h(2) = E(p_{h+1} \mid F_h) + E(a_{h+2} \mid F_h) = \hat{p}_h(1) + 0 = p_h,$$

and by induction

$$\hat{p}_h(\ell) = p_h \quad \text{for all } \ell \geq 1. \tag{3}$$

**The process is not mean reverting**: compare L01-07 (19), where a stationary AR(1) forecast decays as $\phi^\ell$ toward $\mu$.

**MA representation and forecast errors.** Substituting (1) into itself, as in L01-07 (8) with $\phi_1 = 1$:

$$p_t = a_t + a_{t-1} + a_{t-2} + \cdots, \qquad \psi_i = 1 \text{ for all } i. \tag{4}$$

The forecast error is the sum of the shocks not yet seen. From (1) iterated $\ell$ times, $p_{h+\ell} = p_h + a_{h+1} + \cdots + a_{h+\ell}$; subtract (3):

$$e_h(\ell) = p_{h+\ell} - \hat{p}_h(\ell) = a_{h+\ell} + a_{h+\ell-1} + \cdots + a_{h+1}, \tag{5}$$

$$V[e_h(\ell)] = \ell\,\sigma^2 \;\longrightarrow\; \infty \quad (\ell \to \infty). \tag{6}$$

(6) is T eq. (2.34) with all $\psi_i = 1$: $\sigma^2 \sum_{i=0}^{\ell-1} \psi_i^2 = \sigma^2 \ell$. "The usefulness of point forecast $\hat{p}_h(\ell)$ diminishes as $\ell$ increases, which again implies that the model is not predictable" (slide 9; T p. 73).

**Memory.** $\psi_i = 1$ in (4) means a shock from any date enters today's level with weight one: "in economics, the shocks are said to have a permanent effect on the series" (T p. 73). The sample ACF of a random walk approaches 1 at every fixed lag as $T$ grows (slide 11 shows it at lag 200 from 100,000 simulated steps). This is what makes the ACF useless for identifying anything once the series is $I(1)$.

> **Stationary: shocks fade at rate $\psi_i \to 0$, forecasts revert, the forecast variance converges to $\gamma_0$. Unit root: $\psi_i = 1$, forecasts are flat at $p_h$, the variance grows without bound.**

**U — the slides' coin walk (slides 5–6, `unitroot.html`).** With $a_t \in \{-1, +1\}$ equally likely ($\sigma^2 = 1$), the four panels show the first 50, 100, 200 and 1000 steps of one path. Each zoom-out looks like a new "trend": by (6) the typical distance from the start after 1000 steps is $\sqrt{1000} \approx 32$, and nothing pulls the walk back to 0.

---

## 2 · C — random walk with drift: the constant becomes a slope (slides 13–15; T §2.7.2, pp. 73–75)

$$p_t = \mu + p_{t-1} + a_t \qquad \text{(T eq. 2.36)}. \tag{7}$$

Repeated substitution, one line at a time:

$$p_1 = \mu + p_0 + a_1,$$

$$p_2 = \mu + p_1 + a_2 = \mu + (\mu + p_0 + a_1) + a_2 = 2\mu + p_0 + a_2 + a_1,$$

$$p_3 = \mu + p_2 + a_3 = \mu + (2\mu + p_0 + a_2 + a_1) + a_3 = 3\mu + p_0 + a_3 + a_2 + a_1,$$

$$p_t = p_0 + \mu t + \sum_{i=1}^{t} a_i. \tag{8}$$

So, conditional on $p_0$,

$$E(p_t) = p_0 + \mu t, \qquad V(p_t) = t\sigma^2, \qquad \mathrm{sd}(p_t) = \sigma\sqrt{t}. \tag{9}$$

**The mean grows like $t$, the sd like $\sqrt{t}$**: the trend eventually dominates the noise. They are equal when $\mu t = \sigma\sqrt{t}$, i.e. at

$$t^* = \left(\frac{\sigma}{\mu}\right)^2. \tag{10}$$

**Worked — 3M monthly log returns (T eq. 2.37, p. 74; slides 14–15).** $r_t = 0.0103 + a_t$, $\hat{\sigma} = 0.0637$, $T = 755$ months. The standard error of the mean is $0.0637/\sqrt{755} = 0.0637/27.48 = 0.002318$ (Tsay rounds it to 0.0023), so

$$t = \frac{0.0103}{0.002318} = 4.44,$$

the drift is significant at any usual level. By (10), $t^* = (0.0637/0.0103)^2 = (6.184)^2 = 38.2$ months: after about three years the drift is larger than one sd of the accumulated noise. The slides simulate $n = 10{,}000$ steps with $a_t \sim N(0, 0.0637^2)$, with and without $\mu = 0.0103$: at $t = 10{,}000$ the drift has added $0.0103 \times 10{,}000 = 103$ to the log price, against a noise sd of $0.0637 \times \sqrt{10{,}000} = 6.37$. In Tsay's Figure 2.10 the log price of 3M follows the line $0.0103\,t$ closely for that reason.

**Trend-stationary alternative (slide 17; T §2.7.3, p. 75).**

$$p_t = \beta_0 + \beta_1 t + r_t, \qquad r_t \text{ stationary (e.g. AR}(p)\text{)}. \tag{11}$$

| | Mean | Variance | A shock at $t$ |
|---|---|---|---|
| RW with drift (7) | $p_0 + \mu t$ | $t\sigma^2$, grows | permanent ($\psi_i = 1$) |
| Trend-stationary (11) | $\beta_0 + \beta_1 t$ | $V(r_t)$, finite and constant | dies out ($\psi_i \to 0$ in $r_t$) |

Both have a straight-line mean; only the variance tells them apart. Remove the trend from (11) by regression and $r_t$ is stationary; remove a trend from (8) and $\sum a_i$ is still a random walk. **Differencing** is the right transformation for (7), **detrending** for (11).

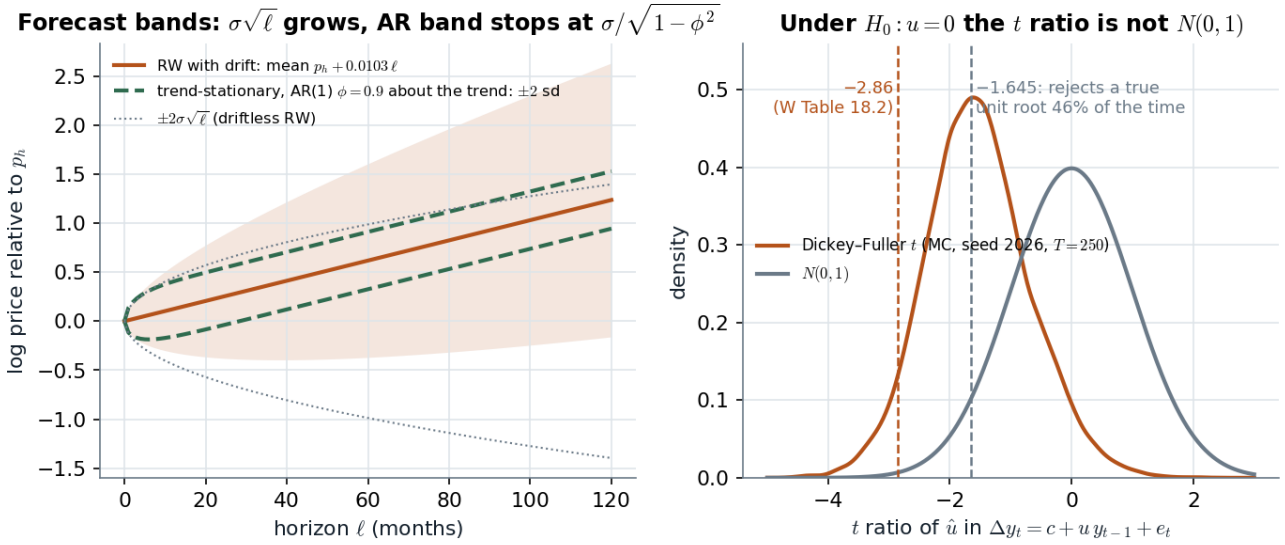



*Figure — left: exact $\pm 2$ sd forecast bands from (9) and from an AR(1) about the trend, L01-07 (15), with the 3M calibration $\mu = 0.0103$, $\sigma = 0.0637$ (T eq. 2.37). Right: the DF $t$ ratio of §5, simulated (seed 2026, $T = 250$, 20,000 replications), against $N(0,1)$; W Table 18.2 (p. 575).*

On the left the two models share the mean line, but the random-walk band keeps opening ($2 \times 0.0637\sqrt{120} = 1.40$ at ten years) while the trend-stationary band stops at $2 \times 0.0637/\sqrt{1 - 0.81} = 0.29$ after a few half-lives. Ten years of data can be consistent with both, which is why the test of §5 has low power against $\phi$ near 1. The right panel is read in §5.

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## 3 · C — what the constant term means (slide 16; T p. 74)

| Model | The constant is | Formula |
|---|---|---|
| MA($q$), T eq. (2.22) | the **mean** | $E(r_t) = c_0$ |
| stationary AR($p$) or ARMA($p, q$) | **related to the mean** | $\mu = \phi_0/(1 - \phi_1 - \cdots - \phi_p)$ (L01-07 (21)) |
| random walk with drift (7) | the **time slope** | $E(p_t) = p_0 + \mu t$ by (9) |

The third line follows from the second: as $\sum \phi_i \to 1$ the denominator vanishes and the "mean" ceases to exist; the constant becomes the slope instead. Tsay adds the contrast with regression: in $r_t = \phi_0 + \phi_1 r_{t-1} + a_t$ the coefficient must satisfy $|\phi_1| \leq 1$ for the model to be meaningful, while in $y_t = \beta_0 + \beta_1 x_t + a_t$ the slope can be any real number (T p. 74).

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## 4 · C — ARIMA($p$,1,$q$) and differencing (slides 18–19; T §2.7.4, pp. 75–76)

Allowing the AR polynomial of an ARMA to have a root at 1 gives an ARIMA model. **Definition (T p. 76):** $y_t$ is ARIMA($p$,1,$q$) if

$$c_t = y_t - y_{t-1} = (1-B)y_t \text{ is a stationary and invertible ARMA}(p,q). \tag{12}$$

Write the stationary ARMA for $c_t$ as $\phi(B)c_t = \theta_0 + \theta(B)a_t$ and substitute $c_t = (1-B)y_t$:

$$\phi(B)(1-B)\, y_t = \theta_0 + \theta(B)a_t. \tag{13}$$

The AR polynomial of $y_t$ is $\phi(B)(1-B)$, which vanishes at $B=1$: a unit root. Like (4), the $\psi$ weights of $y_t$ do not decay, so ARIMA processes have "strong memory" (slide 18). The canonical case: log prices are ARIMA, **log returns** $r_t = \ln P_t - \ln P_{t-1}$ are stationary.

**Double unit roots.** If $y_t$ and $c_t$ are both unit-root nonstationary but $s_t = c_t - c_{t-1} = (1-B)^2 y_t = y_t - 2y_{t-1} + y_{t-2}$ is stationary, $y_t$ is $I(2)$, and ARIMA($p$,2,$q$) if $s_t$ is ARMA($p,q$). A nonzero mean of $s_t$ then implies a **quadratic** time trend in $y_t$ (one more level of integration than (8)). Tsay's candidate is the US GDP deflator (T p. 76; that is Exercise 2.15, not on our list).

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## 5 · C — the Dickey–Fuller test (slides 20–23; T §2.7.5, pp. 76–77; W §18.2, pp. 574–576)

### 5.1 · The regressions and the hypothesis

$$p_t = \phi_1 p_{t-1} + e_t \qquad \text{(T eq. 2.38)}, \tag{14}$$

$$p_t = \phi_0 + \phi_1 p_{t-1} + e_t \qquad \text{(T eq. 2.39)}, \tag{15}$$

$$H_0 : \phi_1 = 1 \quad \text{vs} \quad H_a : \phi_1 < 1. \tag{16}$$

The alternative is one-sided: $\phi_1 > 1$ would be explosive. Wooldridge writes (15) as $y_t = \alpha + \rho y_{t-1} + e_t$ (18.17) and subtracts $y_{t-1}$ from both sides, with $\theta = \rho - 1$:

$$\Delta y_t = \alpha + \theta y_{t-1} + e_t \qquad \text{(W 18.21)}, \qquad H_0 : \theta = 0 \text{ vs } H_1 : \theta < 0. \tag{17}$$

Testing $\rho = 1$ in levels and $\theta = 0$ in differences is the same test: the two regressions have the same residuals, and $\widehat{\theta} = \widehat{\rho} - 1$ with the same standard error. **Note what is null:** the unit root ($I(1)$). "We do not take the null to be $I(0)$ … because $\{y_t\}$ is $I(0)$ for any value of $\rho$ strictly between $-1$ and 1" (W p. 575).

### 5.2 · The $t$ ratio, derived

For (14) with $p_0 = 0$, least squares gives

$$\hat{\phi}_1 = \frac{\sum_{t=1}^{T} p_{t-1} p_t}{\sum_{t=1}^{T} p_{t-1}^2}, \qquad \hat{\sigma}_e^2 = \frac{\sum_{t=1}^{T} (p_t - \hat{\phi}_1 p_{t-1})^2}{T-1}. \tag{18}$$

Subtract 1 and put it over the common denominator:

$$\hat{\phi}_1 - 1 = \frac{\sum p_{t-1} p_t - \sum p_{t-1}^2}{\sum p_{t-1}^2} = \frac{\sum p_{t-1}(p_t - p_{t-1})}{\sum p_{t-1}^2}.$$

Under $H_0$, (14) says $p_t - p_{t-1} = e_t$; substituting in place:

$$\hat{\phi}_1 - 1 = \frac{\sum p_{t-1}\,(e_t)}{\sum p_{t-1}^2}. \tag{19}$$

The OLS standard error is $\mathrm{std}(\hat{\phi}_1) = \hat{\sigma}_e / \sqrt{\sum p_{t-1}^2}$. Dividing (19) by it:

$$\text{DF } t \text{ ratio} = \frac{\hat{\phi}_1 - 1}{\mathrm{std}(\hat{\phi}_1)} = \frac{\sum_{t=1}^{T} p_{t-1} e_t}{\hat{\sigma}_e \sqrt{\sum_{t=1}^{T} p_{t-1}^2}} \qquad \text{(T p. 77)}. \tag{20}$$

### 5.3 · Why (20) is not $N(0,1)$

**W's argument (p. 575).** In (17), under $H_0$ the regressor $y_{t-1}$ is $I(1)$, "and so the usual central limit theorem that underlies the asymptotic standard normal distribution for the $t$ statistic does not apply." The sums in (20) are dominated by the late, large values of the walk ($\sum p_{t-1}^2$ is of order $T^2$, not $T$).

**U — one exact line that shows the skew.** Square (14) under $H_0$: $p_t^2 = (p_{t-1} + e_t)^2 = p_{t-1}^2 + 2p_{t-1}e_t + e_t^2$. Sum over $t = 1, \ldots, T$; the $p_{t-1}^2$ terms telescope, with $p_0 = 0$:

$$p_T^2 = 2\sum_{t=1}^{T} p_{t-1} e_t + \sum_{t=1}^{T} e_t^2 \quad \Longrightarrow \quad \sum_{t=1}^{T} p_{t-1} e_t = \tfrac{1}{2}\Big(p_T^2 - \sum_{t=1}^{T} e_t^2\Big). \tag{21}$$

Divide by $T\sigma^2$. By L01-07 (18), $p_T \sim N(0, T\sigma^2)$ for Gaussian shocks, so $p_T^2/(T\sigma^2) \sim \chi_1^2$; and $\sum e_t^2/(T\sigma^2) \to 1$. The numerator of (20) therefore behaves like $\frac{1}{2}(\chi_1^2 - 1)$, which is **negative whenever $\chi_1^2 < 1$, probability 0.683**. Under the null, $\hat{\phi}_1 < 1$ about two times in three: the estimate is biased toward stationarity, and the $t$ ratio is shifted left. **G:** the limit is a functional of standard Brownian motion $W$, $t \Rightarrow \frac{1}{2}[W(1)^2 - 1] \big/ \sqrt{\int_0^1 W(r)^2 dr}$ (Chan and Wei 1988; Phillips 1987; T p. 77); named, not examinable.

### 5.4 · Three cases (T p. 77)

| Regression | True $\phi_0$ | Null law of the $t$ ratio |
|---|---|---|
| (14), no constant | 0 | DF (Brownian functional), critical values by simulation |

| Regression | True $\phi_0$ | Null law of the $t$ ratio |
|---|---|---|
| (15), constant | 0 (driftless walk) | **another** nonstandard law (W Table 18.2) |
| (15), constant | $\neq 0$ (walk with drift) | asymptotically **normal**, but "large sample sizes are needed" |

The third case is why the deterministic term has to match the alternative being considered, §6.

### 5.5 · Critical values (W Tables 18.2–18.3, pp. 575–577)

| Significance | 1% | 2.5% | 5% | 10% |
|---|---|---|---|---|
| constant, no trend (W Table 18.2) | −3.43 | −3.12 | −2.86 | −2.57 |
| constant + trend (W Table 18.3) | −3.96 | −3.66 | −3.41 | −3.12 |
| our simulation, constant, $T = 250$ | −3.45 | −3.16 | −2.87 | −2.58 |

Reject $H_0$ when $t < c$. "If we use the standard normal critical value to test for a unit root, we would reject $H_0$ much more often than 5% of the time when $H_0$ is true" (W p. 575). **Worked (the figure, right panel):** in the seeded simulation, $P(t < -1.645 \mid H_0) = 0.465$. Using the normal 5% value is a **46.5% test**. A trend in the regression pushes the critical values further left because "detrending a unit root process tends to make it look more like an I(0) process" (W p. 577).

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## 6 · C — the augmented DF test (slides 24–27; T eq. 2.40, p. 77; W eqs. 18.23–18.25)

### 6.1 · From AR($p$) to the $\Delta$ form

**AR(3), derived step by step** (slide 25). Start from

$$y_t = \phi_0 + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \phi_3 y_{t-3} + e_t. \tag{22}$$

Express the far lags through $y_{t-1}$ and differences, $\Delta y_s = y_s - y_{s-1}$:

$$y_{t-2} = y_{t-1} - \Delta y_{t-1}, \qquad y_{t-3} = y_{t-2} - \Delta y_{t-2} = y_{t-1} - \Delta y_{t-1} - \Delta y_{t-2}. \tag{23}$$

Substitute (23) into (22):

$$y_t = \phi_0 + \phi_1 y_{t-1} + \phi_2 (y_{t-1} - \Delta y_{t-1}) + \phi_3 (y_{t-1} - \Delta y_{t-1} - \Delta y_{t-2}) + e_t.$$

Collect terms:

$$y_t = \phi_0 + (\phi_1 + \phi_2 + \phi_3)\, y_{t-1} - (\phi_2 + \phi_3)\, \Delta y_{t-1} - \phi_3\, \Delta y_{t-2} + e_t.$$

Subtract $y_{t-1}$ from both sides:

$$\Delta y_t = \phi_0 + \beta\, y_{t-1} + \tilde{\phi}_1 \Delta y_{t-1} + \tilde{\phi}_2 \Delta y_{t-2} + e_t, \tag{24}$$

$$\beta = \phi_1 + \phi_2 + \phi_3 - 1, \qquad \tilde{\phi}_1 = -(\phi_2 + \phi_3), \qquad \tilde{\phi}_2 = -\phi_3. \tag{25}$$

(The slide prints the right-hand side as "$\tilde{\phi}_2 \Delta y_{t-2} + \tilde{\phi}_2 \Delta y_{t-2}$"; the first term is $\tilde{\phi}_1 \Delta y_{t-1}$, as (24) shows.) **General AR($p$)** (slide 24): $\Delta y_t = \phi_0 + \beta y_{t-1} + \sum_{i=1}^{p-1} \tilde{\phi}_i \Delta y_{t-i} + e_t$ with

$$\beta = -\phi(1) = \phi_1 + \cdots + \phi_p - 1. \tag{26}$$

**A (single) unit root means** $\phi(1) = 0$, i.e. 1 solves the characteristic equation $1 - \phi_1 x - \cdots - \phi_p x^p = 0$ (L08-01), i.e. $\beta = 0$, i.e.

$$\phi_1 + \phi_2 + \cdots + \phi_p = 1. \tag{27}$$

### 6.2 · The ADF regression and statistic

Tsay writes the regression in levels:

$$x_t = c_t + \beta x_{t-1} + \sum_{i=1}^{p-1} \phi_i \Delta x_{t-i} + e_t, \qquad H_0 : \beta = 1 \text{ vs } H_a : \beta < 1 \qquad \text{(T eq. 2.40)}, \tag{28}$$

$$\text{ADF} = \frac{\hat{\beta} - 1}{\text{std}(\hat{\beta})}, \qquad c_t \in \{0,\ \omega_0,\ \omega_0 + \omega_1 t\}. \tag{29}$$

Equivalently $\Delta x_t = c_t + c\, x_{t-1} + \sum \phi_i \Delta x_{t-i} + e_t$ with $c = \beta - 1$ and $H_0 : c = 0$ vs $c < 0$. **Here $\beta$ in (28) is the levels coefficient (null 1); the $\beta$ of (26) is the $\Delta$-form coefficient (null 0).** The slides use both names; the statistic is the same.

The lagged differences "clean up any serial correlation in $\Delta y_t$" (W p. 576). Too few lags: wrong size even asymptotically; too many: lower power. The $t$ statistics on the $\Delta y_{t-i}$ terms do have the usual asymptotic $t$ law, so standard tests can choose the lag length (W p. 577).

**Deterministic term.** Use $c_t = \omega_0 + \omega_1 t$ when the series trends (alternative: trend-stationary (11)). Without the trend, "if we carry out the usual DF or augmented DF test on a trending but I(0) series, we will probably have little power for rejecting a unit root" (W p. 577). Then the critical values of W Table 18.3 apply.

### 6.3 · Worked — Wooldridge's three examples

**T-bill rate (W Example 18.2, p. 576, eq. 18.22).** $\widehat{\Delta r3_t} = 0.625 - 0.091\, r3_{t-1}$, standard errors (0.261) and (0.037), $n = 123$. So $\hat{\rho} = 1 + \hat{\theta} = 1 - 0.091 = 0.909$ and

$$t = \frac{-0.091}{0.037} = -2.46 > -2.57,$$

fail to reject even at 10%. **Failing to reject is not accepting.** A standard $t$ test of $\rho = 0.9$ (valid, since $y_t$ is $I(0)$ under that null) gives $(0.909 - 0.9)/0.037 = 0.24$: no evidence against $\rho = 0.9$ either, and "it makes no sense to accept $\rho = 1$ and $\rho = .9$" (W p. 576). With $\rho = 0.9$, observations ten periods apart have correlation $0.9^{10} = 0.35$, not nearly one.

**US inflation (W Example 18.3, p. 577).** $\widehat{\Delta inf}_t = 1.36 - 0.310\, inf_{t-1} + 0.138\, \Delta inf_{t-1}$, s.e. $(0.517)(0.103)(0.126)$, $n = 47$:

$$t = \frac{-0.310}{0.103} = -3.01 < -2.86 \quad \Rightarrow \quad \text{reject at } 5\%, \qquad \hat{\rho} = 0.690.$$

Dropping the insignificant lag gives $\hat{\theta} = -0.335$, $t = -3.13$: slightly stronger.

**US log GDP with a trend (W Example 18.4, p. 578, eq. 18.26).** $\widehat{gGDP}_t = 1.65 + 0.0059\, t - 0.210 \log(GDP_{t-1}) + 0.264\, gGDP_{t-1}$, s.e. $(0.67)(0.0027)(0.087)(0.165)$, $n = 35$:

$$t = \frac{-0.210}{0.087} = -2.41 > -3.12 \text{ (10\%, Table 18.3)} \quad \Rightarrow \quad \text{do not reject,}$$

although $\hat{\rho} = 1 - 0.21 = 0.79$. With $n = 35$, a near-unit root cannot be told from a unit root. Without the trend, $\hat{\theta} = -0.023$, $t = -1.92$: "the estimate of $\rho$ is much closer to one, but this is misleading due to the omitted time trend" (W p. 578).

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## 7 · C — the decks' examples: US GDP and the S&P 500 (slides 28–31; T Example 2.2, pp. 78–81; `unitroot.html`)

**US quarterly log GDP, 1947.I–2008.IV.** The level trends upward with sample ACF near 1; the difference (growth) varies around a fixed mean. With $p = 10$ lags (chosen from the PACF of the differences), Tsay reports the ADF $-1.701$**, p-value 0.4297** (S-Plus, constant), with coefficient on the lagged level $-0.0008$ (s.e. 0.0005), so

$$\hat{\beta} = 1 + \hat{c} = 1 - 0.0008 = 0.9992 \qquad \text{(T p. 78)}.$$

The R run in Tsay's book (`fUnitRoots`) prints $-1.6109$, p 0.4569; the slides round it to "$-1.61$ with a p-value 0.45". The site page runs `tseries::adf.test(gdp, k=10)` and prints **Dickey–Fuller** $= -0.37047$**, p-value 0.9872**. The difference is the regression, not the data: `adf.test` always includes a constant **and a linear trend**, so its statistic is compared with the trend critical values (W Table 18.3). Three numbers, one conclusion: **the unit root is not rejected.**

**S&P 500 log index, 3 Jan 1950 – 16 Apr 2008, 14,462 days.** To test "random walk with drift" against trend stationarity, $c_t = \omega_0 + \omega_1 t$ and $p = 15$: ADF $-1.998$**, p 0.602** (T p. 79). The constant (0.0019, $t = 2.39$) is significant, the trend (t ratio 1.85, p 0.064) only at 10%: "the log series of the S&P 500 index contains a unit root and a positive drift, but there is no strong evidence of a time trend" (T p. 80). The R value in the book is $-1.9946$ (p 0.5807). The site page picks $k = 2$ by `ar(diff(sp5))` (AR order 2, coefficients 0.0721 and $-0.0387$, mean daily return 0.000299) and prints `adf.test(sp5, k=2)` $= -2.0179$, p 0.5708; the PDF deck's $k = 10$ run gives $-1.9482$, p 0.6004. **On the differences**, `adf.test(diff(sp5), k=10)` gives $-37.537$ with "p-value = 0.01": `tseries` truncates p-values at 0.01, so read it as $p \le 0.01$. Log prices are $I(1)$, log returns $I(0)$.

**Read an ADF output with three questions: which deterministic terms, how many lags, which table.** The three GDP statistics ($-1.70$, $-1.61$, $-0.37$) differ because the regressions differ.

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## 8 · C — spurious regression (W §18.3, pp. 578–580)

Let $x_t = x_{t-1} + a_t$ and $y_t = y_{t-1} + e_t$ (W 18.27–18.28) with independent iid shocks and $x_0 = y_0 = 0$: the two series are independent. Regress $y_t$ on $x_t$ (W 18.29) and test $\beta_1 = 0$ at 5%. Granger and Newbold (1974) found frequent "significance"; in Davidson and MacKinnon's simulation the usual $t$ test rejects **66.2% of the time at $n = 50$ and 84.7% at $n = 250$** (W p. 579): it gets worse with more data.

**Why.** Under $H_0 : \beta_1 = 0$ the model $y_t = \beta_0 + \beta_1 x_t + u_t$ (W 18.30) can hold only with $\beta_0 = 0$ and

$$u_t = y_t = \sum_{j=1}^{t} e_j, \tag{30}$$

a **random walk error**: the errors are not even weakly dependent, so neither the $t$ statistic (which diverges as $n \to \infty$) nor $R^2$ (which converges to a random variable, not to a population value) has its usual behaviour (W pp. 579–580). Adding a time trend does not fix it. The safe course is to difference $I(1)$ variables first (§4); then $\Delta y_t$ on $\Delta x_t$ is a regression of white noise on white noise.

**G — named only:** when some combination $y_t - bx_t$ is $I(0)$ the two $I(1)$ series are *cointegrated* and the levels regression is meaningful (W §18.4). Cointegration is in the syllabus table but not on the class schedule (Map/class-plan, "Syllabus items not on the site schedule").

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## 9 · U — reading it as an economist

A unit root in log GDP means a recession's output loss is never made up: the level forecast is today's level plus drift, (3) with (8). A trend-stationary GDP returns to its old trend line. The tests cannot reliably choose between them on post-war data (W Example 18.4: $\hat{\rho} = 0.79$, not rejected), which is why the decks state conclusions as "cannot be rejected" and work with growth rates. For asset prices, the unit root is the statement that returns are unpredictable in level terms, which is where the volatility models of L08-06-arch-garch start: the *returns* are serially uncorrelated, their *squares* are not.

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## Links

- Neighbours: L08-02-forecasting-ma-arma (the $\psi$-weight forecast variance (2.34) that (6) specialises) · L08-04-long-memory-arfima (memory between $\psi_i \to 0$ geometrically and $\psi_i = 1$).
- Unit: L08-00-map · L08-10-formulas · L08-11-pitfalls · L08-12-python (the DF simulation and `adfuller` on simulated series).
- Class 1: L01-07 §6–§7 (equilibrium law, random walk variance $t\sigma^2$, persistence $\sqrt{t/(t+h)}$).
- Rules: rules/04 (ARIMA, unit roots).
- **Practice (number and printed page only):** W 18.C13 (p. 604). Worked solutions: `Listas/lista-L08-book-exercises.pdf`.

## §2.11 · Long memory — an ACF that decays like a power of the lag, between ARMA and the unit root

> **Claim. The fractionally differenced process $(1-B)^d x_t = a_t$ with $-0.5 < d < 0.5$ is stationary, yet its autocorrelations decay hyperbolically, $\rho_k \sim c\,k^{2d-1}$, instead of geometrically like every stationary ARMA.** Its whole ACF follows from one ratio, $\rho_k/\rho_{k-1} = (k-1+d)/(k-d)$, starting at $\rho_1 = d/(1-d)$. With $d = 0.4$ the correlation at lag 100 is still 0.27, where an AR(1) with the same $\rho_1$ has $10^{-18}$. **Long memory sits between the two cases of L08-03-unit-roots-dickey-fuller:** shocks fade (unlike a random walk) but slowly (unlike ARMA). Absolute returns are the financial example. Prev: L08-03-unit-roots-dickey-fuller · Next: L08-05-seasonal-airline-model · Map: L08-00-map

**Notation.**

| Symbol | Meaning | Range |
|---|---|---|
| $x_t$ | the series | |
| $a_t$ | white noise, variance $\sigma_a^2$ | |
| $d$ | fractional differencing parameter | $-0.5 < d < 0.5$ |
| $B$ | back-shift operator, $Bx_t = x_{t-1}$ | |
| $\binom{d}{k}$ | generalised binomial coefficient $d(d-1)\cdots(d-k+1)/k!$ | |
| $\psi_k$, $\pi_k$ | MA($\infty$) and AR($\infty$) weights | |
| $\rho_k$, $\phi_{k,k}$ | ACF and PACF at lag $k$ | |
| $f(\omega)$ | spectral density at frequency $\omega \in [0, 2\pi]$ | |
| $x! = \Gamma(x+1)$ | factorial of a non-integer, through the gamma function | |

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### 1 · U — where long memory comes from (slides 33–34)

The idea began in hydrology: Hurst (1951) found that the rescaled range of the Nile's levels grew "anomalously fast". Mandelbrot showed that, if stationarity is kept, new mathematics is needed to explain the Hurst effect, and defined long memory (Mandelbrot and Van Ness 1968; Mandelbrot and Wallis 1968). The slides cite Palma (2007) and Beran et al. (2013) for the theory.

The empirical signature (T p. 102): a sample ACF that is **not large but decays very slowly**. In Tsay's Figure 2.22 the ACF of *absolute* daily returns of the CRSP value- and equal-weighted indexes, 1970–2008, is small yet still significant at the 5% level after 300 lags, while the returns themselves are nearly uncorrelated. Ding, Granger and Engle (1993) documented the same for $|r_t|$; it is the reason long-memory volatility models exist (§6).

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### 2 · C — fractional differencing: the operator (slide 32; T eq. 2.52, p. 101; binomial expansion p. 102)

$$(1-B)^d x_t = a_t, \qquad -0.5 < d < 0.5. \tag{1}$$

For non-integer $d$ the operator is defined by the binomial series (T p. 102):

$$(1-B)^d = \sum_{k=0}^{\infty} \binom{d}{k} (-B)^k = \sum_{k=0}^{\infty} (-1)^k \frac{d(d-1)\cdots(d-k+1)}{k!} B^k. \tag{2}$$

**The first terms, one at a time.**

$k = 0$: 1.

$k = 1$: $(-1)^1 \dfrac{d}{1!} B = -dB.$

$k = 2$: $(-1)^2 \dfrac{d(d-1)}{2!} B^2 = \dfrac{d(d-1)}{2} B^2 = -\dfrac{d(1-d)}{2} B^2.$

$k = 3$: $(-1)^3 \dfrac{d(d-1)(d-2)}{3!} B^3 = -\dfrac{d(1-d)(2-d)}{6} B^3$, using $(d-1)(d-2) = (1-d)(2-d)$.

So

$$(1-B)^d = 1 - dB - \frac{d(1-d)}{2} B^2 - \frac{d(1-d)(2-d)}{6} B^3 - \cdots \tag{3}$$

For $d = 1$ the series stops after two terms and (3) is the ordinary difference $1 - B$; for $0 < d < 1$ **every** past value enters, with slowly shrinking weights. That is the source of the long memory.

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## 3 · C — the two representations (slide 35; T properties 1–2, p. 101)

**Property 1 (MA form).** If $d < 0.5$, $x_t$ is weakly stationary and

$$x_t = a_t + \sum_{k=1}^{\infty} \psi_k a_{t-k}, \qquad \psi_k = \frac{d(1+d)\cdots(k-1+d)}{k!} = \frac{(k+d-1)!}{k!\,(d-1)!}. \tag{4}$$

These are the coefficients of $(1-B)^{-d}$, the inverse of (2): $\psi_1 = d$, $\psi_2 = d(1+d)/2$. For $d = 0.25$: $\psi_1 = 0.25$, $\psi_2 = 0.25 \times 1.25/2 = 0.156$.

**Property 2 (AR form).** If $d > -0.5$, $x_t$ is invertible. The coefficients of (2) are, written as products,

$$\pi_k = (-1)^k \binom{d}{k} = \frac{-d(1-d)\cdots(k-1-d)}{k!} = \frac{(k-d-1)!}{k!\,(-d-1)!}, \qquad \pi_0 = 1, \tag{5}$$

(Tsay's $\pi_k$, p. 101), so that (1) reads $\sum_{k \geq 0} \pi_k x_{t-k} = a_t$, i.e.

$$x_t = a_t - \sum_{k=1}^{\infty} \pi_k x_{t-k} = a_t + d\, x_{t-1} + \frac{d(1-d)}{2} x_{t-2} + \cdots \tag{6}$$

using $\pi_1 = -d$ and $\pi_2 = -d(1-d)/2$ from (3). (Check of the factorial form at $k = 1$: $(-d)!/(-d-1)! = \Gamma(1-d)/\Gamma(-d) = -d$.)

**Range of $d$.** $\sum \psi_k^2 < \infty$ (finite variance) needs $d < 0.5$; $\sum \pi_k^2 < \infty$ needs $d > -0.5$. Both hold on $(-0.5, 0.5)$, the range in (1).

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## 4 · C — the ACF, PACF and spectrum (slides 36–37; T properties 3–5, p. 102)

**Property 3 (ACF).** For $-0.5 < d < 0.5$,

$$\rho_k = \frac{d(1+d)\cdots(k-1+d)}{(1-d)(2-d)\cdots(k-d)}, \qquad k = 1, 2, \ldots \tag{7}$$

(The slide prints the last denominator factor as "$(k = d)$"; it is $(k - d)$.) Two consequences, each read straight off (7):

$k = 1$, one factor above and below:

$$\rho_1 = \frac{d}{1-d}. \tag{8}$$

Dividing (7) at $k$ by (7) at $k-1$, all factors but the last cancel:

$$\frac{\rho_k}{\rho_{k-1}} = \frac{k-1+d}{k-d}. \tag{9}$$

**Asymptotics.** Write the products in (7) with gamma functions, $d(1+d)\cdots(k-1+d) = \Gamma(k+d)/\Gamma(d)$ and $(1-d)\cdots(k-d) = \Gamma(k+1-d)/\Gamma(1-d)$:

$$\rho_k = \frac{\Gamma(1-d)}{\Gamma(d)} \cdot \frac{\Gamma(k+d)}{\Gamma(k+1-d)}. \tag{10}$$

**G — the ratio of gamma functions.** For large $k$, $\Gamma(k+a)/\Gamma(k+b) \approx k^{a-b}$ (Stirling). With $a = d$, $b = 1-d$, $a - b = 2d - 1$:

$$\rho_k \approx \frac{(-d)!}{(d-1)!}\, k^{2d-1} \qquad (k \to \infty), \tag{11}$$

since $\Gamma(1-d) = (-d)!$ and $\Gamma(d) = (d-1)!$. **This is the definition of long memory: polynomial decay,** $\log \rho_k \approx$ **const** $+ (2d-1)\log k$**.** For $0 < d < 0.5$ the exponent lies in $(-1, 0)$, so $\sum_k \rho_k$ diverges even though $\rho_k \to 0$.

**Property 4 (PACF):** $\phi_{k,k} = d/(k-d)$. At $k = 1$ it equals (8), as any PACF must; then it decays like $1/k$: it does not cut off, so neither the ACF nor the PACF identifies an ARFIMA by a cut-off (compare L08-02-forecasting-ma-arma).

**Property 5 (spectrum, named):** $f(\omega) \sim \omega^{-2d}$ as $\omega \to 0$ (T eq. 2.53). For $d > 0$ the spectrum **diverges at frequency zero**, while every stationary ARMA spectrum is bounded on $[0, 2\pi]$. This is what the log-periodogram estimator of $d$ exploits.

### 4.1 · Worked — the ACF from (8)–(9)

$d = 0.25$**.** By (8), $\rho_1 = 0.25/0.75 = 1/3 = 0.3333$. By (9) with $k = 2$:

$$\rho_2 = \rho_1 \cdot \frac{1 + 0.25}{2 - 0.25} = \frac{1}{3} \cdot \frac{1.25}{1.75} = 0.2381.$$

With $k = 3$:

$$\rho_3 = \rho_2 \cdot \frac{2 + 0.25}{3 - 0.25} = 0.2381 \cdot \frac{2.25}{2.75} = 0.1948.$$

PACF: $\phi_{2,2} = 0.25/1.75 = 0.1429$. At $k = 100$, (11) gives $\frac{\Gamma(0.75)}{\Gamma(0.25)} 100^{-0.5} = \frac{1.2254}{3.6256} \times 0.1 = 0.0338$, equal to the exact product (7) to four decimals.

$d = 0.4$. $\rho_1 = 0.4/0.6 = 2/3$. An AR(1) with the same first autocorrelation has $\phi = 2/3$ and $\rho_{100} = (2/3)^{100} = 2.5 \times 10^{-18}$. The ARFIMA has, from (11),

$$\rho_{100} \approx \frac{\Gamma(0.6)}{\Gamma(0.4)} 100^{-0.2} = \frac{1.4892}{2.2182} \times 0.3981 = 0.6714 \times 0.3981 = 0.267,$$

again matching (7) (0.2673). Same lag-1 dependence; at lag 100 one is still 0.27, the other is zero for every practical purpose.

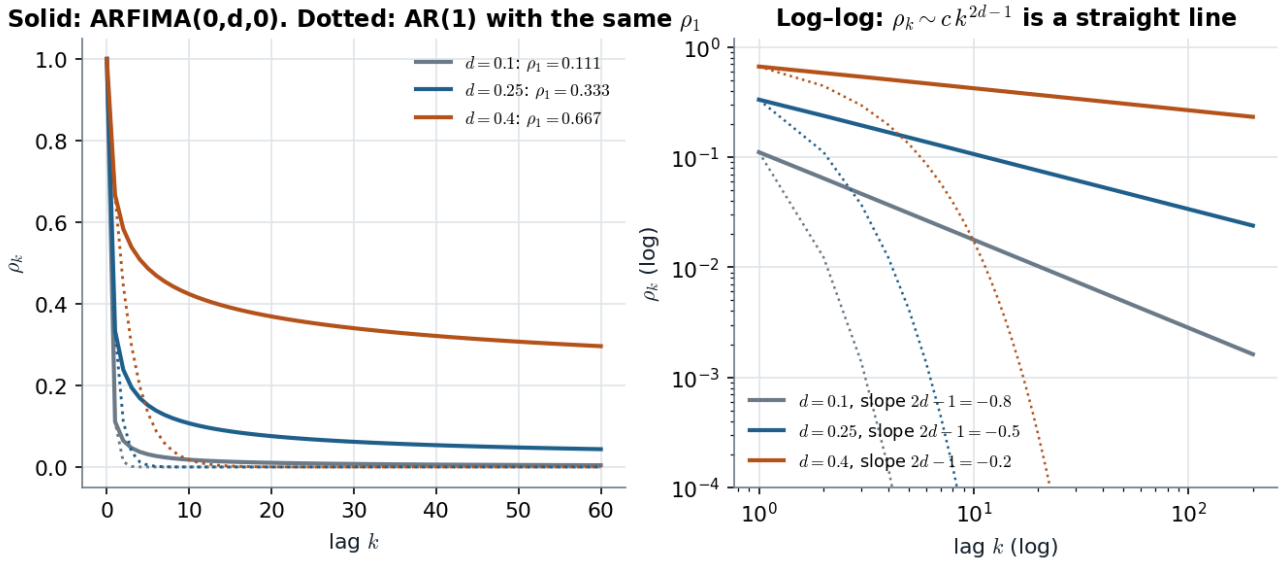



*Figure — exact ACF (7) of ARFIMA(0,d,0) for $d = 0.1, 0.25, 0.4$ against an AR(1) with the same $\rho_1 = d/(1-d)$ from (8); right panel log–log, slope $2d - 1$ from (11). T p. 102.*

On the linear scale the two families start at the same height, and the difference is in the tail: the dotted AR(1) curves are gone by lag 20 even for $\phi = 2/3$. On the log–log scale the ARFIMA correlations are straight lines with slopes $-0.8$, $-0.5$ and $-0.2$, exactly $2d - 1$, while a geometric decay $\phi^k$ is a curve that bends down ever faster. **A straight line on the log–log ACF plot is the visual test for long memory.**

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## 5 · C — ARFIMA$(p, d, q)$ and the three kinds of memory (slide 38; T p. 102)

If the fractionally differenced series $(1-B)^d x_t$ follows an ARMA$(p, q)$, $x_t$ is **ARFIMA$(p, d, q)$**: $\phi(B)(1-B)^d x_t = \theta(B) a_t$. It generalises ARIMA (L08-03-unit-roots-dickey-fuller (13)) from integer to non-integer $d$; the ARMA part shapes the short lags, $d$ the long ones.

| | $\psi_k$ (shock at $t-k$) | ACF at lag $k$ | Example |
|---|---|---|---|
| stationary ARMA ($d=0$) | geometric, $\propto \lvert\lambda\rvert^k$ | geometric | returns |
| long memory ($0<d<0.5$) | $\psi_k \approx k^{d-1}/(d-1)!$, hyperbolic | $\rho_k \sim ck^{2d-1}$ | $\lvert r_t\rvert$, volatility |
| unit root ($d=1$) | $\psi_k = 1$ | sample ACF $\to 1$ | log prices |

($\psi_k \approx k^{d-1}/(d-1)!$ follows from (4) by the same gamma-ratio step as (11).) Estimation of $d$ in the pure model (1): **maximum likelihood**, or a **regression on the logged periodogram at low frequencies** (T p. 102), which uses property 5. Bayesian treatments (Koop, Ley, Osiewalski and Steel 1997; Pai and Ravishanker 1998; Graves et al. 2015, …) are listed on slide 38; named only.

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## 6 · G — long memory in volatility (T §3.13, p. 154)

The long-memory stochastic volatility model (T eq. 3.41) puts the fractional difference in the log-volatility: $a_t = \sigma_t \epsilon_t$, $\sigma_t = \sigma \exp(u_t/2)$, $(1-B)^d u_t = \eta_t$, $0 < d < 0.5$. Then $\ln a_t^2$ is a Gaussian long-memory signal plus non-Gaussian noise. Bollerslev and Jubinski (1999) and Ray and Tsay (2000) find a median $d$ of about 0.38 for the log squared daily returns of S&P 500 companies. Named here; the short-memory SV model is L08-08-stochastic-volatility.

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## Links

- Neighbours: L08-03-unit-roots-dickey-fuller ($d=1$: the unit root) · L08-05-seasonal-airline-model.
- Unit: L08-00-map · L08-10-formulas · L08-11-pitfalls · L08-12-python (the ACF recursion (9) checked against (10)) · L08-02-forecasting-ma-arma (geometric ACF and the identification table) · L08-06-arch-garch (why $\lvert r_t\rvert$ is autocorrelated).
- Class 1: L01-07 (the linear-process ACF from $\psi$ weights, L01-07 (4)).
- Rules: rules/04 (long memory, ARFIMA).
- **Practice (number and printed page only):** T 2.5 (p. 105). Worked solutions: `Listas/lista-L08-book-exercises.pdf`.

## §2.8 · Seasonal models — difference twice, model the remainder with two MA terms, and know when the seasonality is deterministic

**Claim. The airline model $(1-B^s)(1-B)x_t = (1-\theta B)(1-\Theta B^s)a_t$ removes trend and seasonality by regular and seasonal differencing and describes what is left with two MA parameters.** Its differenced series has nonzero autocorrelations only at lags 1, $s-1$, $s$, $s+1$, and the two side lags are the *product* $\rho_1\rho_s$: the regular and seasonal dynamics are orthogonal. It is the ARIMA$(0,1,1)\times(0,1,1)_s$ special case of Box–Jenkins SARIMA. **When the fitted seasonal MA coefficient reaches 1 it cancels the seasonal difference and the seasonality is deterministic**: dummies then do the same job (Decile 1's January effect). For the AirPassengers series, BIC picks the airline model, statsmodels reproduces the deck's estimates, and out of sample it forecasts 1958–60 with RMSE 36.6 against 63.3 for a trend-plus-dummies regression. Prev: L08-04-long-memory-arfima · Next: L08-06-arch-garch · Map: L08-00-map

**Notation.**

| Symbol | Meaning | Range |
|---|---|---|
| $x_t$ | the (usually log) series | |
| $s$ | periodicity: 4 quarterly, 12 monthly | $s \geq 3$ in §3 |
| $\Delta = 1-B$, $\Delta_s = 1-B^s$ | regular and seasonal difference | |
| $w_t = (1-B^s)(1-B)x_t$ | the doubly differenced series | |
| $\theta$, $\Theta$ | regular and seasonal MA coefficients (Tsay's sign: $1-\theta B$) | $\lvert\theta\rvert, \lvert\Theta\rvert < 1$ |
| $a_t$ | white noise, variance $\sigma_a^2$ | |
| $\gamma_\ell$, $\rho_\ell$ | autocovariance and ACF of $w_t$ | |
| $\phi(B), \theta(B)$; $\Phi_P(B^s), \Theta_Q(B^s)$ | regular and seasonal AR and MA polynomials | |

**Sign convention.** R's `arima`/`Arima` and statsmodels print MA coefficients with a plus sign, $1+\theta B$. A printed `ma1 = -0.3864` is Tsay's $\theta = 0.3864$. AR coefficients print with the same sign in all three.

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### 1 · C — seasonal series and seasonal adjustment (slides 3–4; T §2.8, pp. 81–82; W §10.5e, pp. 336–337)

A seasonal series repeats a pattern every $s$ periods: Tsay's Figure 2.13 shows the quarterly earnings per share of Johnson & Johnson, 1960–1980, growing exponentially with a strong period-4 pattern and rising variability. **Seasonal adjustment** removes the seasonal component before inference; most US government series (GDP growth, unemployment) are published adjusted (T p. 82; W p. 336). "In other applications such as forecasting, seasonality is as important as other characteristics of the data and must be handled accordingly" (slide 4): the course models it.

**Logs first** (T pp. 82–83). The log turns exponential growth into a linear trend and stabilises the variance: the widening seasonal swings of the J&J earnings (and of AirPassengers, §6) become swings of constant amplitude. In R this is `lambda = 0` (Box–Cox with $\lambda = 0$).

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## 2 · C — seasonal differencing (slide 5; T §2.8.1, pp. 83–84)

With $x_t$ the log earnings, the ACF of $x_t$ decays slowly (trend), and the ACF of $\Delta x_t$ spikes at multiples of 4 (seasonality). Following Box, Jenkins and Reinsel, take both differences:

$$\Delta_s y_t = (1 - B^s) y_t = y_t - y_{t-s}, \tag{1}$$

$$\Delta_s \Delta x_t = (1 - B^s)(1 - B) x_t. \tag{2}$$

Multiply the operators in (2), $(1 - B^s)(1 - B) = 1 - B - B^s + B^{s+1}$, and apply to $x_t$:

$$\Delta_s \Delta x_t = x_t - x_{t-1} - x_{t-s} + x_{t-s-1}. \tag{3}$$

For quarterly data, $\Delta_4 \Delta x_t = x_t - x_{t-1} - x_{t-4} + x_{t-5}$ (T p. 83): this year's quarterly change minus last year's change in the same quarter. Its ACF for J&J has "a significant negative ACF at lag 1 and a marginal negative correlation at lag 4" (T p. 84), the pattern the next model is built for.

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## 3 · C — the airline model and its ACF (slides 6–8; T §2.8.2, eq. 2.41, pp. 84–85)

$$(1 - B^s)(1 - B) x_t = (1 - \theta B)(1 - \Theta B^s) a_t, \qquad |\theta| < 1,\ |\Theta| < 1. \tag{4}$$

The AR side is just the two differences; all the modelling is in the MA side. Expanding the right-hand side, with $w_t$ as in the notation:

$$w_t = (1 - \theta B - \Theta B^s + \theta\Theta B^{s+1}) a_t = a_t - \theta a_{t-1} - \Theta a_{t-s} + \theta\Theta a_{t-s-1}. \tag{5}$$

$w_t$ is an MA$(s+1)$ with only four nonzero weights, so $E(w_t) = 0$.

**Variance.** The four shocks in (5) are uncorrelated:

$$\gamma_0 = \sigma_a^2 (1 + \theta^2 + \Theta^2 + \theta^2\Theta^2) = (1 + \theta^2)(1 + \Theta^2)\, \sigma_a^2. \tag{6}$$

**Autocovariances.** For each lag write $w_{t-\ell}$ from (5) and keep only the shocks it shares with $w_t$ (assume $s \geq 3$, so that lags 1, $s-1$, $s$, $s+1$ are distinct).

Lag 1: $w_{t-1} = a_{t-1} - \theta a_{t-2} - \Theta a_{t-s-1} + \theta\Theta a_{t-s-2}$. Shared shocks: $a_{t-1}$ (weights $-\theta$ in $w_t$, 1 in $w_{t-1}$) and $a_{t-s-1}$ (weights $\theta\Theta$ and $-\Theta$):

$$\gamma_1 = \sigma_a^2 \big[(-\theta)(1) + (\theta\Theta)(-\Theta)\big] = -\theta(1 + \Theta^2)\, \sigma_a^2. \tag{7}$$

Lag $s-1$: $w_{t-s+1} = a_{t-s+1} - \theta a_{t-s} - \Theta a_{t-2s+1} + \theta\Theta a_{t-2s}$. Shared: $a_{t-s}$ (weights $-\Theta$ and $-\theta$):

$$\gamma_{s-1} = \sigma_a^2 (-\Theta)(-\theta) = \theta\Theta\, \sigma_a^2. \tag{8}$$

Lag $s$: $w_{t-s} = a_{t-s} - \theta a_{t-s-1} - \Theta a_{t-2s} + \theta\Theta a_{t-2s-1}$. Shared: $a_{t-s}$ (weights $-\Theta$ and 1) and $a_{t-s-1}$ (weights $\theta\Theta$ and $-\theta$):

$$\gamma_s = \sigma_a^2[(-\Theta)(1) + (\theta\Theta)(-\theta)] = -\Theta(1+\theta^2)\,\sigma_a^2. \tag{9}$$

Lag $s+1$: $w_{t-s-1} = a_{t-s-1} - \theta a_{t-s-2} - \ldots$. Shared: $a_{t-s-1}$ (weights $\theta\Theta$ and 1):

$$\gamma_{s+1} = \theta\Theta\,\sigma_a^2. \tag{10}$$

Every other lag shares no shock: $\gamma_\ell = 0$ for $\ell > 0$, $\ell \notin \{1, s-1, s, s+1\}$.

**ACF.** Divide (7)–(10) by (6):

$$\rho_1 = \frac{-\theta(1+\Theta^2)}{(1+\theta^2)(1+\Theta^2)} = \frac{-\theta}{1+\theta^2}, \qquad \rho_s = \frac{-\Theta}{1+\Theta^2}, \tag{11}$$

$$\rho_{s-1} = \rho_{s+1} = \frac{\theta\Theta}{(1+\theta^2)(1+\Theta^2)} = \rho_1\rho_s. \tag{12}$$

**Why "multiplicative"** (T p. 85). The MA(1) $y_t = (1-\theta B)a_t$ has $\rho_1(y) = -\theta/(1+\theta^2)$ and nothing else; the MA($s$) $z_t = (1-\Theta B^s)a_t$ has $\rho_s(z) = -\Theta/(1+\Theta^2)$ and nothing else. By (11)–(12), $\rho_1 = \rho_1(y)$, $\rho_s = \rho_s(z)$ and $\rho_{s\pm1} = \rho_1(y)\rho_s(z)$: the side lags are the interaction of the regular and seasonal dependence. "A multiplicative seasonal model says that the dynamics of the regular and seasonal components of the series are approximately orthogonal" (slide 8).

**Non-multiplicative alternative** (T eq. 2.42, p. 85):

$$w_t = (1-\theta B - \Theta B^s)a_t = a_t - \theta a_{t-1} - \Theta a_{t-s}. \tag{13}$$

Repeat the bookkeeping: $w_{t-s+1}$ contains $-\theta a_{t-s}$, which meets $-\Theta a_{t-s}$ in $w_t$, so $\gamma_{s-1} = \theta\Theta\sigma_a^2 \neq 0$; but $w_{t-s-1} = a_{t-s-1} - \theta a_{t-s-2} - \Theta a_{t-2s-1}$ shares no shock with $w_t$, so

$$\rho_{s+1} = 0 \quad \text{in (13).} \tag{14}$$

Both models have two parameters; the multiplicative one has more nonzero autocorrelations (lags $1, s-1, s, s+1$ against $1, s-1, s$), hence it is the more parsimonious description of a symmetric pattern around lag $s$ (T p. 85).

### 3.1 · Worked — the two fitted airline models

**Johnson & Johnson (T Example 2.3, p. 85; slide 11).** Exact likelihood on the log earnings:

$$(1-B)(1-B^4)x_t = (1-0.678B)(1-0.314B^4)a_t, \qquad \hat{\sigma}_a = 0.089,$$

standard errors 0.080 and 0.101; residual $Q(12) = 10.0$, p 0.44: adequate. By (11)–(12), with $1+\theta^2 = 1+0.678^2 = 1.4597$ and $1+\Theta^2 = 1+0.314^2 = 1.0986$:

$$\rho_1 = \frac{-0.678}{1.4597} = -0.4645, \qquad \rho_4 = \frac{-0.314}{1.0986} = -0.2858, \qquad \rho_3 = \rho_5 = (-0.4645)(-0.2858) = 0.1328,$$

and by (6) $\gamma_0 = 1.4597 \times 1.0986 \times 0.089^2 = 1.6036 \times 0.007921 = 0.0127$.

**AirPassengers (§6, page fit on 1949–1957, $s = 12$).** $\theta = 0.3864$, $\Theta = 0.5885$: $1 + \theta^2 = 1.1493$, $1 + \Theta^2 = 1.3463$,

$$\rho_1 = \frac{-0.3864}{1.1493} = -0.3362, \qquad \rho_{12} = \frac{-0.5885}{1.3463} = -0.4371, \qquad \rho_{11} = \rho_{13} = (-0.3362)(-0.4371) = 0.1470.$$

With $\hat{\sigma}_a^2 = 0.001416$, (6) gives $\gamma_0 = 1.5473 \times 0.001416 = 0.00219$.

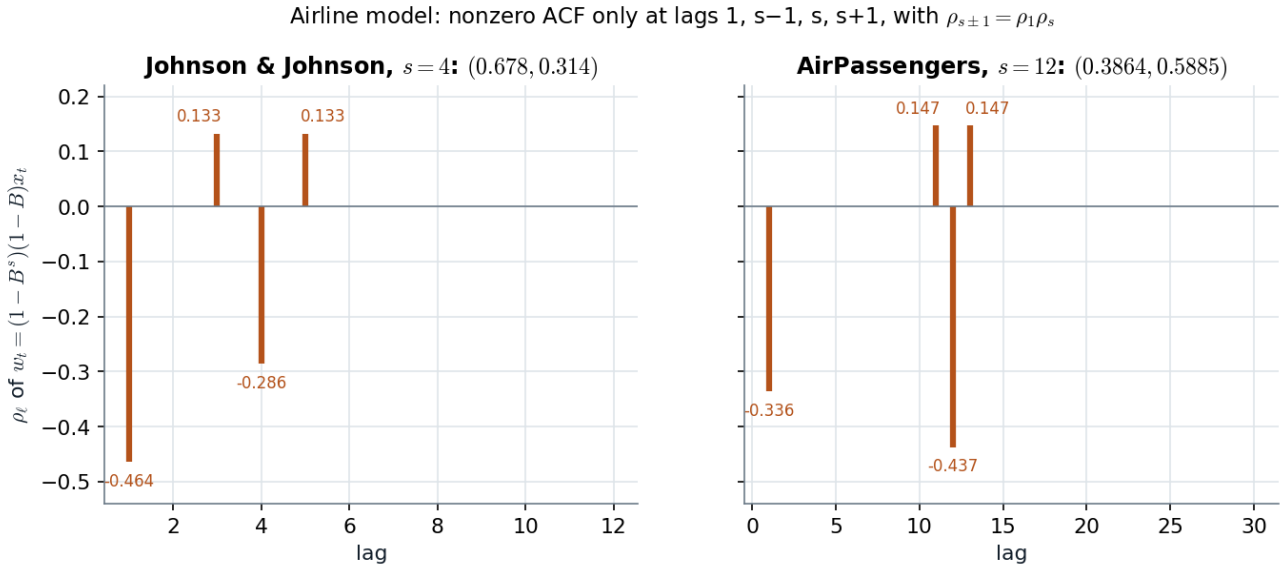



*Figure — exact ACF (11)–(12) of $w_t$ for the two fitted airline models: J&J $(\theta, \Theta) = (0.678, 0.314)$, $s = 4$ (T p. 85); AirPassengers $(0.3864, 0.5885)$, $s = 12$ (`airpassengers-seasonalmodels.html`).*

The theoretical ACF has exactly four spikes. The two negative ones, at lags 1 and $s$, are the regular and seasonal MA(1) effects; the two small positive ones flank lag $s$ with equal height $\rho_1\rho_s$. Reading a sample ACF of $\Delta_s\Delta x_t$: spikes at 1 and $s$ with small symmetric shoulders at $s \pm 1$ point to the airline model; a shoulder only at $s - 1$ points to (13).

---

## 4 · C — SARIMA, and how constrained it is (slides 9–10)

The airline model is the special case of the seasonal ARIMA of Box and Jenkins (1970):

$$\Phi_P(B^s)\,\phi(B)\,(1 - B^s)^D(1 - B)^d\,x_t = \delta + \Theta_Q(B^s)\,\theta(B)\,a_t, \tag{15}$$

written ARIMA$(p, d, q) \times (P, D, Q)_s$, with $\phi, \theta$ of degrees $p, q$ in $B$ and $\Phi_P, \Theta_Q$ of degrees $P, Q$ in $B^s$. **The airline model is ARIMA$(0, 1, 1) \times (0, 1, 1)_s$.**

### 4.1 · Worked — ARIMA$(1, 1, 1) \times (1, 1, 1)_{12}$ is a constrained ARMA(26,13)

$$(1 - \phi L)(1 - \Phi L^{12})\,\Delta\Delta_{12}\,y_t = (1 - \theta L)(1 - \Theta L^{12})\,\epsilon_t. \tag{16}$$

Multiply the four AR factors in pairs. Regular pair: $(1-\phi L)(1-L) = 1-(1+\phi)L+\phi L^2$. Seasonal pair: $(1-\Phi L^{12})(1-L^{12}) = 1-(1+\Phi)L^{12}+\Phi L^{24}$. Their product, term by term:

$$\big[1-(1+\phi)L+\phi L^2\big]\big[1-(1+\Phi)L^{12}+\Phi L^{24}\big]$$

$$= 1-(1+\phi)L+\phi L^2-(1+\Phi)L^{12}+(1+\phi)(1+\Phi)L^{13}-\phi(1+\Phi)L^{14}+\Phi L^{24}-(1+\phi)\Phi L^{25}+\phi\Phi L^{26}. \quad (17)$$

The MA side is $(1-\theta L)(1-\Theta L^{12}) = 1-\theta L-\Theta L^{12}+\theta\Theta L^{13}$. Moving every lagged $y$ to the right:

$$\begin{aligned} y_t = {} & (1+\phi)y_{t-1} - \phi y_{t-2} + (1+\Phi)y_{t-12} \\ & - (1+\phi)(1+\Phi)y_{t-13} + (1+\Phi)\phi y_{t-14} \\ & - \Phi y_{t-24} + (1+\phi)\Phi y_{t-25} - \phi\Phi y_{t-26} \\ & + \epsilon_t - \theta\epsilon_{t-1} - \Theta\epsilon_{t-12} + \theta\Theta\epsilon_{t-13}. \end{aligned} \quad (18)$$

An ARMA(26,13) whose 26 AR and 13 MA coefficients are functions of four parameters, "with severe parameter constraints" (slide 10). **Slide typos:** the slide prints the $y_{t-12}$ coefficient as $(1-\phi+\Phi)$ and the $y_{t-13}$ coefficient as $-(1+\phi\Phi+\Phi)$; (17) gives $(1+\Phi)$ and $-(1+\phi)(1+\Phi) = -(1+\phi+\Phi+\phi\Phi)$. Check: setting $L=1$ in (17) must give 0 (the polynomial contains $(1-L)$), and $1-(1+\phi)+\phi-(1+\Phi)+(1+\phi)(1+\Phi)-\phi(1+\Phi)+\Phi-(1+\phi)\Phi+\phi\Phi = 0$ ✓. The other coefficients on the slide agree with (18).

---

## 5 · C — deterministic seasonality: when $\Theta \to 1$ (slides 12–14; T pp. 86–90; W §10.5e)

**Why $\Theta = 1$ means deterministic** (T p. 86). With $\Theta = 1$, (4) is $(1-B^s)(1-B)x_t = (1-\theta B)(1-B^s)a_t$. Move everything to one side and factor $(1-B^s)$:

$$(1-B^s)\big[(1-B)x_t - (1-\theta B)a_t\big] = 0.$$

The bracket $S_t$ satisfies $S_t - S_{t-s} = 0$, i.e. it is a fixed period-$s$ pattern set by the initial conditions:

$$(1-B)x_t = (1-\theta B)a_t + S_t, \qquad S_t = S_{t-s}. \quad (19)$$

A fixed seasonal pattern is exactly what seasonal dummies estimate, so "the same forecasts are obtained by using either dummy variables or a multiplicative seasonal model when the seasonal pattern is deterministic. Yet use of dummy variables can lead to inferior forecasts if the seasonal pattern is not deterministic" (T p. 86). Tsay recommends the **exact** likelihood, which can reach $\Theta$ near 1 where the conditional likelihood cannot.

**Worked — CRSP Decile 1 monthly returns, 1970–2008, 468 months (T Example 2.4, pp. 86–90; slides 12–14).** The time plot shows no seasonality but the ACF is significant at lags 1, 12, 24, 36. The model $(1-\phi_1 B)(1-\phi_{12}B^{12})R_t = (1-\theta_{12}B^{12})a_t$ gives:

| Method | $\hat{\phi}_1$ | $\hat{\phi}_{12}$ (seasonal AR) | $\hat{\theta}_{12}$ (seasonal MA) | $\tilde{\sigma}_a$ |
|---|---|---|---|---|
| conditional likelihood | 0.18 | 0.87 | 0.74 | 0.069 |
| exact likelihood | 0.188 | **0.951** | **0.997** | 0.063 |

With the exact method the seasonal AR and MA factors nearly cancel, $(1-0.951B^{12}) \approx (1-0.997B^{12})$, and $\hat{\theta}_{12} \approx 1$: by (19) the seasonality looks deterministic. (R's exact `arima` gives `ar1` 0.1787, `sar1` 0.9886, `sma1` −0.9127, i.e. $\theta_{12} = 0.9127$; "slightly differ from those of SCA", T p. 89.) The confirming regression on a January dummy (T p. 88):

$$\hat{R}_t = 0.0029 + 0.1253\, Jan_t, \qquad \text{s.e. } 0.0033 \text{ and } 0.0115, \qquad R^2 = 0.2016.$$

The January slope has $t = 0.1253/0.0115 = 10.9$ (R prints 10.848): the average January return is $0.0029 + 0.1253 = 0.1282$, about 12.8%, against 0.29% in other months. The residual ACF loses its lags 12, 24, 36. **The seasonality of Decile 1 is the January effect**, and one dummy captures it.

**Seasonal dummies in regression (W §10.5e, pp. 336–338).** With monthly data, include eleven month dummies (one month as base) in the regression, alone or with a trend and other regressors; the dummy coefficients are the average seasonal differences from the base month. This is the "deterministic trend + cycle" model of the AirPassengers page.

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## 6 · C — the AirPassengers study (slides 15–28; `airpassengers-seasonalmodels.html`)

**Data** (slide 16; Box, Jenkins and Reinsel's Series G): monthly international airline passengers in thousands, 1949–1960, $n = 144$, from 112 in Jan 1949 to 432 in Dec 1960 (sum 40,363). Seasonal swings grow with the level, so the analysis is on $\log y_t$ (`lambda = 0`); the page plots $y_t - y_{t-1}$, $y_t - y_{t-12}$ and $y_t - y_{t-1} - y_{t-12} + y_{t-13}$, which is (3), before and after the log.

**Design (page).** Training 1949.1–1957.12 ($n_1 = 108$, nine years), testing 1958.1–1960.12 ($n_2 = 36$). Two competitors:

1. *Deterministic trend + cycle*: OLS of $\log y_t$ (and of $y_t$) on an intercept, $t$, and 11 month dummies (W §10.5e).
2. *Seasonal ARIMA*: all sixteen ARIMA$(i,1,j) \times (k,1,l)_{12}$ with $i, j, k, l \in \{0, 1\}$, chosen by BIC.

**BIC choice.** The page's loop returns `0 1 0 1 -336.9304`: **ARIMA**$(0,1,1) \times (0,1,1)_{12}$**, the airline model, BIC** −336.93. Fitted (page):

$$(1-B)(1-B^{12}) \log y_t = (1 - 0.3864B)(1 - 0.5885B^{12})a_t, \qquad \text{s.e. } 0.1097,\ 0.0927, \quad \hat{\sigma}_a^2 = 0.001416,$$

log-likelihood 175.3, AIC −344.59, BIC −336.93. The printed `ma1 = -0.3864`, `sma1 = -0.5885` are $-\theta$, $-\Theta$.

**Reproduction in Python** (statsmodels `SARIMAX` on the logs, exact likelihood; our computation, in L08-12-python):

| Sample | Deck / page (ma1, sma1) | statsmodels | log-lik, AIC, BIC (statsmodels) |
|---|---|---|---|
| to 1956.12 ($n = 96$, slide 19) | $-0.3941$, $-0.6129$; BIC $-284.27$ | $-0.3944$, $-0.6133$ | 148.76, $-291.52$, $-284.26$ |
| to 1957.12 ($n = 108$, page) | $-0.3864$, $-0.5885$; BIC $-336.93$ | $-0.3876$, $-0.5886$ | 175.29, $-344.59$, $-336.92$ |
| full ($n = 144$, slide 26) | $-0.4018$, $-0.5569$; AIC $-483.4$ | $-0.4017$, $-0.5571$ | 244.70, $-483.39$, $-474.77$ |

The log-likelihoods and criteria agree to the printed precision; the coefficients to about 0.001, the difference between R's and statsmodels' optimisers. In statsmodels the BIC-best of the sixteen on $n = 108$ is also (0,1,1)(0,1,1), with (1,1,0)(0,1,1) second at $-335.85$.

**Forecasts (slides 20–21).** `forecast(air.model, h=48)` from the 1956 fit, and refits ending 1953–1956 with forecasts to 1960: the airline forecasts reproduce the growing seasonal pattern, with 80% and 95% bands that widen with the horizon (the two unit roots, L08-03-unit-roots-dickey-fuller (6)).

**RMSE($h$) study (slides 22–24, 27).** For every origin $t = 48, \dots, n-12$, refit fifteen models (the sixteen minus $(1,1,1) \times (1,1,1)$) on observations $1, \dots, t$ and forecast $y_t(h)$, $h = 1, \dots, 12$:

$$\mathrm{RMSE}(h) = \sqrt{\frac{1}{n-11-t_0} \sum_{t=t_0}^{n-12} \left(y_{t+h} - y_t(h)\right)^2}, \qquad t_0 = 48, \tag{20}$$

plotted in levels and relative to model 6, the airline model ($100(\mathrm{RMSE}_i/\mathrm{RMSE}_6 - 1)$). The models without a regular or seasonal MA term do clearly worse at all horizons. **The full-sample fits of the leading models (slide 26):**

| Model | Coefficients (R sign) | $\hat{\sigma}^2$ | log-lik | AIC | BIC |
|---|---|---|---|---|---|
| (0,1,1)(0,1,1) | ma1 $-0.4018$ (0.0896), sma1 $-0.5569$ (0.0731) | 0.001371 | 244.70 | $-483.40$ | $-474.77$ |
| (0,1,1)(1,1,1) | ma1 $-0.4143$, sar1 $-0.1116$ (0.1547), sma1 $-0.4817$ | 0.001375 | 244.96 | $-481.91$ | $-470.41$ |
| (1,1,0)(1,1,1) | ar1 $-0.3451$, sar1 $-0.0760$ (0.1548), sma1 $-0.5108$ | 0.001399 | 243.86 | $-479.73$ | $-468.23$ |
| (1,1,0)(0,1,1) | ar1 $-0.3395$ (0.0822), sma1 $-0.5619$ (0.0748) | 0.001391 | 243.74 | $-481.49$ | $-472.86$ |

The seasonal AR terms are insignificant ($t = -0.72$ and $-0.49$), and the extra parameter only lowers the BIC. The "two best models" plotted on slide 28 are (0,1,1)(0,1,1) and (1,1,0)(0,1,1): an MA(1) or an

AR(1) for the regular part, both with the seasonal MA. A regular MA(1) with $\theta = 0.40$ and an AR(1) with $\phi = -0.34$ have close ACFs at lag 1 ($-0.35$ and $-0.34$), which is why they forecast alike.

**Out of sample, 1958–1960 (our computation).** Fitting on $n_1 = 108$ and forecasting 36 months by exponentiating the log forecasts: RMSE **36.6** thousand passengers for the airline model against **63.3** for the log trend + 11 dummies regression. The deterministic model forces the same seasonal shape and trend slope estimated on 1949–57 onto 1958–60; the airline model lets both drift, which is (19) read in reverse.

> **Seasonality is stochastic until the data say otherwise.** Use the exact likelihood; if $\widehat{\Theta} \to 1$ (or a seasonal AR and MA cancel), switch to dummies.

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## Links

- Neighbours: L08-04-long-memory-arfima · L08-06-arch-garch.
- Unit: L08-00-map · L08-10-formulas · L08-11-pitfalls · L08-12-python (the SARIMAX reproduction, the RMSE comparison, the Decile 1 January regression) · L08-02-forecasting-ma-arma (MA ACF, forecasts) · L08-03-unit-roots-dickey-fuller (differencing, ARIMA).
- Class 1: L01-07 (ACF of a linear process from its $\psi$ weights, L01-07 (4)).
- Rules: rules/04.
- **Practice (number and printed page only):** T 2.6 (p. 105). Worked solutions: `Listas/lista-L08-book-exercises.pdf`.

## 3.1–3.6 · ARCH and GARCH: a variance that is forecastable, a kurtosis that is not normal

**Claim. Returns can be serially uncorrelated and still dependent: their conditional variance moves.** ARCH writes that variance as a linear function of past squared shocks, $\sigma_t^2 = \alpha_0 + \sum \alpha_i a_{t-i}^2$. That one equation produces volatility clusters and, even with Gaussian innovations, **unconditional kurtosis above 3**: $3(1-\alpha_1^2)/(1-3\alpha_1^2)$ for ARCH(1). **GARCH adds the lagged variance and is an ARMA model for $a_t^2$**: $\sigma_\infty^2 = \alpha_0/(1-\alpha_1-\beta_1)$, and forecasts decay to it at rate $\alpha_1+\beta_1$. The empirical regularity is $\alpha_1+\beta_1$ close to 1 (0.9727 for Tsay's S&P 500, 0.980 for Petrobras), so a volatility shock has a half-life of months or weeks. At $\alpha_1+\beta_1 = 1$ (IGARCH) the variance has a unit root, the unconditional variance does not exist, and RiskMetrics' exponential smoothing is the special case $\alpha_0 = 0$. Prev: L08-05-seasonal-airline-model · Next: L08-07-garch-family-asymmetry-tails · Map: L08-00-map

**Notation.**

| Symbol | Meaning | Range |
|---|---|---|
| $r_t$ | log return at $t$ (garch-sv-petrobras.html uses simple returns) | |
| $F_{t-1}$ | information at $t-1$ | |
| $\mu_t = E(r_t \mid F_{t-1})$ | conditional mean (mean equation) | |
| $a_t = r_t - \mu_t$ | shock, innovation | |
| $\sigma_t^2 = V(r_t \mid F_{t-1})$ | conditional variance (volatility equation); $\sigma_t$ the volatility | $> 0$ |
| $\varepsilon_t = a_t/\sigma_t$ | standardized innovation, iid, mean 0, variance 1 | |
| $\alpha_0, \alpha_i, \beta_j$ | constant, ARCH and GARCH coefficients | $\alpha_0 > 0$; $\alpha_i, \beta_j \geq 0$ |
| $m, s$ | ARCH and GARCH orders | |
| $\eta_t = a_t^2 - \sigma_t^2$ | "variance surprise", martingale difference | |
| $\sigma_\infty^2$ | unconditional variance $E(a_t^2)$ | |
| $\sigma_h^2(\ell)$ | $\ell$-step variance forecast at origin $h$ | |
| $K$ | excess kurtosis of $\varepsilon_t$: $E(\varepsilon_t^4) = K+3$ | $K = 0$ normal |
| $\nu$ | degrees of freedom of the standardized Student-$t$ | $\nu > 2$ |

**Mapping to the site pages.** garchmodeling.html writes $y_t = \sigma_t \varepsilon_t$ and $\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$: its $\omega$ is Tsay's $\alpha_0$, and the $\varepsilon_{t-1}$ inside the recursion must be read as the **raw residual** $a_{t-1} = y_{t-1}$ (fGarch filters the residuals, not the standardized ones). Taken literally, $\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2$ with $\varepsilon$ standardized would make $\sigma_t^2$ independent of the level of past volatility. Some authors write $h_t$ for $\sigma_t^2$ (T p. 117; bayes-garch.pdf does).

## 1 · U — what volatility does (T §3.1, pp. 110–111)

Volatility is **not observed**: one return per day carries no variance by itself. Implied volatility (VIX) rests on an option-pricing model. Tsay lists four regularities that any model must reproduce:

1. **clusters**: high-volatility periods and calm periods;
2. **continuity**: volatility evolves smoothly, jumps are rare;
3. **mean reversion within a range**: volatility does not diverge, so it is "often stationary";
4. **leverage**: a big price drop raises volatility more than a big rise of the same size.

ARCH/GARCH capture 1–3. Fact 4 needs the asymmetric models of L08-07-garch-family-asymmetry-tails. The motivating evidence is already in L01: IBM monthly returns pass Ljung–Box at lag 30, while their squares give $Q(30) = 182.12 \to$ L01-07 §2.

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## 2 · C — the structure of a volatility model (T §3.2–3.3, pp. 111–115)

The conditional mean and variance of $r_t$ given $F_{t-1}$ are

$$\mu_t = E(r_t \mid F_{t-1}), \qquad \sigma_t^2 = V(r_t \mid F_{t-1}) = E[(r_t - \mu_t)^2 \mid F_{t-1}]. \tag{1}$$

The **mean equation** is a simple stationary ARMA with regressors (T eq. 3.3):

$$r_t = \mu_t + a_t, \qquad \mu_t = \sum_{i=1}^{p} \phi_i y_{t-i} - \sum_{i=1}^{q} \theta_i a_{t-i}, \qquad y_t = r_t - \phi_0 - \sum_{i=1}^{k} \beta_i x_{it}. \tag{2}$$

Because $\mu_t$ is known at $t-1$, substituting (2) into (1) gives (T eq. 3.4)

$$\sigma_t^2 = V(\mu_t + a_t \mid F_{t-1}) = V(a_t \mid F_{t-1}). \tag{3}$$

**A volatility model is an equation for (3).** GARCH makes $\sigma_t^2$ an exact function of $F_{t-1}$. Stochastic volatility adds its own shock (L08-08-stochastic-volatility).

**Four steps** (T p. 113): (i) fit the mean equation and remove linear dependence; (ii) test the residuals for ARCH effects; (iii) if present, specify a volatility model and estimate mean and volatility **jointly**; (iv) check and refine.

### 2.1 · C — testing for ARCH effects (T §3.3.1, p. 114)

**McLeod–Li:** apply Ljung–Box $Q(m)$ (L01-07 (3)) to the series $a_t^2$.

**Engle's LM test:** run the auxiliary regression

$$a_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \cdots + \alpha_m a_{t-m}^2 + e_t, \qquad t = m+1, \ldots, T, \tag{4}$$

and test $H_0 : \alpha_1 = \cdots = \alpha_m = 0$ with

$$F = \frac{(SSR_0 - SSR_1)/m}{SSR_1/(T-2m-1)}, \qquad SSR_0 = \sum_{t=m+1}^{T} (a_t^2 - \bar{\omega})^2, \quad \bar{\omega} = \frac{1}{T}\sum_{t=1}^{T} a_t^2, \quad SSR_1 = \sum_{t=m+1}^{T} \hat{e}_t^2. \tag{5}$$

Tsay states that (5) is asymptotically $\chi^2_m$. Strictly it is $m \cdot F$ (equivalently $TR^2$ of (4)) that has the $\chi^2_m$ limit; the software prints that statistic against $\chi^2_m$.

**Worked — Intel monthly log returns 1973–2008, $T = 432$ (T pp. 114–115).** Returns: $Q(12) = 18.26$, $p = 0.108$, no serial correlation, so $a_t = r_t - \bar{r}$. Squares: $Q(12) = 89.85$, $p = 5.3 \times 10^{-14}$. LM statistic 53.62 on $\chi^2_{12}$, $p \approx 0$. **Uncorrelated but not independent.**

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## 3 · C — the ARCH model (T §3.4, pp. 115–123)

An **ARCH($m$)** model (Engle, 1982; T eq. 3.5):

$$a_t = \sigma_t \varepsilon_t, \qquad \sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \cdots + \alpha_m a_{t-m}^2, \qquad \varepsilon_t \text{ iid}, \ E\varepsilon_t = 0, \ V\varepsilon_t = 1, \tag{6}$$

with $\alpha_0 > 0$ and $\alpha_i \geq 0$, which keep $\sigma_t^2 > 0$. A large $a_{t-1}^2$ makes $\sigma_t^2$ large, which makes a large $|a_t|$ **more likely** but not certain: that is the cluster.

### 3.1 · Moments of ARCH(1) (T pp. 117–118)

Take $m = 1$: $\sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2$.

**Mean.** By iterated expectations, with $\sigma_t$ known at $t-1$ and $E\varepsilon_t = 0$:

$$E(a_t) = E[E(a_t \mid F_{t-1})] = E[\sigma_t E(\varepsilon_t)] = 0. \tag{7}$$

**Variance.** $E(a_t^2 \mid F_{t-1}) = \sigma_t^2 E(\varepsilon_t^2) = \sigma_t^2$, so

$$V(a_t) = E(a_t^2) = E[E(a_t^2 \mid F_{t-1})] = E(\alpha_0 + \alpha_1 a_{t-1}^2) = \alpha_0 + \alpha_1 E(a_{t-1}^2).$$

Under stationarity $E(a_t^2) = E(a_{t-1}^2) = V(a_t)$:

$$V(a_t) = \alpha_0 + \alpha_1 V(a_t) \iff V(a_t) = \frac{\alpha_0}{1 - \alpha_1}, \qquad 0 \leq \alpha_1 < 1. \tag{8}$$

**Fourth moment.** With Gaussian $\varepsilon_t$, $E(\varepsilon_t^4) = 3$:

$$E(a_t^4 \mid F_{t-1}) = \sigma_t^4 E(\varepsilon_t^4) = 3\sigma_t^4 = 3(\alpha_0 + \alpha_1 a_{t-1}^2)^2. \tag{9}$$

Expand the square and take expectations:

$$E(a_t^4) = 3E(\alpha_0^2 + 2\alpha_0\alpha_1 a_{t-1}^2 + \alpha_1^2 a_{t-1}^4) = 3\big[\alpha_0^2 + 2\alpha_0\alpha_1 V(a_t) + \alpha_1^2 E(a_{t-1}^4)\big].$$

Under fourth-order stationarity $m_4 = E(a_t^4) = E(a_{t-1}^4)$; substitute (8) for $V(a_t)$:

$$m_4 = 3\Big[\alpha_0^2 + 2\alpha_0\alpha_1\Big(\frac{\alpha_0}{1 - \alpha_1}\Big) + \alpha_1^2 m_4\Big]. \tag{10}$$

Collect the $m_4$ terms:

$$m_4(1-3\alpha_1^2) = 3\alpha_0^2\Big(1+\frac{2\alpha_1}{1-\alpha_1}\Big) = 3\alpha_0^2\,\frac{1+\alpha_1}{1-\alpha_1}$$

$$m_4 = \frac{3\alpha_0^2(1+\alpha_1)}{(1-\alpha_1)(1-3\alpha_1^2)}. \tag{11}$$

Since $m_4 > 0$, it exists only if $1-3\alpha_1^2 > 0$, i.e. $\alpha_1^2 < 1/3$ ($\alpha_1 < 0.577$). **Kurtosis**, dividing (11) by the square of (8):

$$\frac{m_4}{[V(a_t)]^2} = \frac{3\alpha_0^2(1+\alpha_1)}{(1-\alpha_1)(1-3\alpha_1^2)}\cdot\frac{(1-\alpha_1)^2}{\alpha_0^2} = \frac{3(1+\alpha_1)(1-\alpha_1)}{1-3\alpha_1^2} = 3\,\frac{1-\alpha_1^2}{1-3\alpha_1^2} > 3. \tag{12}$$

**Gaussian conditionally, fat-tailed unconditionally.** A mixture of normals with random variances has heavier tails than a normal, so ARCH "produces outliers" more often than Gaussian white noise (T p. 118). Every $\alpha_1 \in (0, 1/\sqrt{3})$ gives kurtosis above 3.

### 3.2 · Weaknesses (T p. 119)

(1) Positive and negative shocks enter through $a_{t-1}^2$, so they have the **same** effect (no leverage). (2) It is restrictive: for ARCH(1) with a finite fourth moment, $\alpha_1^2 < 1/3$, which limits the kurtosis it can generate. (3) It describes, it does not explain. (4) It responds slowly to large isolated shocks and tends to **over-predict** volatility afterwards.

### 3.3 · Order, likelihood, checking, forecasts (T pp. 119–123)

**Order.** Define $\eta_t = a_t^2 - \sigma_t^2$. Substituting $\sigma_t^2 = a_t^2 - \eta_t$ into (6):

$$a_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \cdots + \alpha_m a_{t-m}^2 + \eta_t, \tag{13}$$

an AR($m$) for $a_t^2$ with uncorrelated (but not iid) errors. So **the PACF of $a_t^2$ suggests** $m$ (T p. 120), as the PACF suggested $p$ in L08-01-ar2-arp-cycles-identification.

**Gaussian conditional likelihood.** The joint density factors into one-step conditionals; dropping the density of the first $m$ shocks (T p. 120):

$$f(a_{m+1},\dots,a_T \mid \alpha, a_1,\dots,a_m) = \prod_{t=m+1}^{T}\frac{1}{\sqrt{2\pi\sigma_t^2}}\exp\Big(-\frac{a_t^2}{2\sigma_t^2}\Big), \tag{14}$$

$$\ell(\alpha) = -\frac{1}{2}\sum_{t=m+1}^{T}\Big[\ln(2\pi) + \ln\sigma_t^2 + \frac{a_t^2}{\sigma_t^2}\Big], \tag{15}$$

where $\sigma_t^2$ is computed recursively from (6). There is no closed form: (15) is maximised numerically (L08-12-python does it with `scipy.optimize`).

**Standardized Student-$t$ (T eq. 3.7).** If $x_\nu \sim t_\nu$, then $V(x_\nu) = \nu/(\nu-2)$ for $\nu > 2$, so $\varepsilon_t = x_\nu/\sqrt{\nu/(\nu-2)}$ has variance 1. Change variables $x = \varepsilon\sqrt{\nu/(\nu-2)}$ in the $t_\nu$ density, using $x^2/\nu = \varepsilon^2/(\nu-2)$ and the Jacobian $\sqrt{\nu/(\nu-2)}$:

$$f(\varepsilon \mid \nu) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi}\,\Gamma(\frac{\nu}{2})}\Big(1+\frac{\varepsilon^2}{\nu-2}\Big)^{-\frac{\nu+1}{2}}\sqrt{\frac{\nu}{\nu-2}} = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2})\sqrt{(\nu-2)\pi}}\Big(1+\frac{\varepsilon^2}{\nu-2}\Big)^{-\frac{\nu+1}{2}}. \tag{16}$$

**The rescaling matters**: it keeps $\sigma_t^2$ the conditional *variance*, so the volatility equation means the same thing under both distributions. With $a_t = \sigma_t\varepsilon_t$, $f(a_t \mid F_{t-1}) = f(a_t/\sigma_t \mid \nu)/\sigma_t$, and for fixed $\nu$ the log-likelihood is, up to a constant (T eq. 3.8),

$$\ell(\alpha) = -\sum_{t=m+1}^{T}\Big[\frac{\nu+1}{2}\ln\Big(1+\frac{a_t^2}{(\nu-2)\sigma_t^2}\Big) + \frac{1}{2}\ln\sigma_t^2\Big]. \tag{17}$$

Estimating $\nu$ jointly adds $(T-m)\big[\ln\Gamma(\frac{\nu+1}{2}) - \ln\Gamma(\frac{\nu}{2}) - \frac{1}{2}\ln((\nu-2)\pi)\big]$. Prespecified values are usually between 4 and 8. Skew-$t$ (T eq. 3.9, Fernández–Steel) and GED (T eq. 3.10) are the other options; they are read in L08-07-garch-family-asymmetry-tails §5.

**Checking.** If the model is right, $\tilde{a}_t = a_t/\hat{\sigma}_t$ is iid: Ljung–Box on $\tilde{a}_t$ checks the mean equation, on $\tilde{a}_t^2$ the volatility equation, and skewness, kurtosis and a QQ plot check the distribution (T pp. 122–123).

**Forecasts** (T eq. 3.11). At origin $h$, $\sigma_h^2(1) = \alpha_0 + \alpha_1 a_h^2 + \cdots + \alpha_m a_{h+1-m}^2$, and in general

$$\sigma_h^2(\ell) = \alpha_0 + \sum_{i=1}^{m}\alpha_i\,\sigma_h^2(\ell-i), \qquad \sigma_h^2(\ell-i) = a_{h+\ell-i}^2 \text{ if } \ell-i \le 0, \tag{18}$$

the AR forecast recursion of L08-02-forecasting-ma-arma applied to (13).

### 3.4 · Worked — Intel ARCH (T Ex. 3.1, pp. 123–126)

ARCH(3), Gaussian: $\sigma_t^2 = 0.0106 + 0.2131a_{t-1}^2 + 0.0770a_{t-2}^2 + 0.0599a_{t-3}^2$, with $\alpha_2, \alpha_3$ insignificant ($t = 1.60$ and 0.87). Dropping them (T eq. 3.12):

$$r_t = 0.0126 + a_t, \qquad \sigma_t^2 = 0.0111 + 0.3560\,a_{t-1}^2, \tag{19}$$

with standard errors 0.0053, 0.0010, 0.0761. Checks: $Q(10) = 12.64$ ($p = 0.24$) for $\tilde{a}_t$, $Q(10) = 14.75$ ($p = 0.14$) for $\tilde{a}_t^2$. Adequate.

- Mean return 1.26% a month.
- $\hat{\alpha}_1^2 = 0.356^2 = 0.1267 < 1/3$: the fourth moment exists.
- Unconditional sd by (8): $\sqrt{0.0111/(1-0.356)} = \sqrt{0.0111/0.644} = \sqrt{0.017236} = 0.1313$. Tsay prints "$\approx 0.1315$" from unrounded estimates.
- Kurtosis by (12): $3(1-0.126736)/(1-3\times0.126736) = 3(0.873264)/0.619792 = 4.227$.

$t$ **innovations** (T eq. 3.13): $r_t = 0.0169 + a_t$, $\sigma_t^2 = 0.0120 + 0.2845a_{t-1}^2$, $\hat{\nu} = 6.01$ (se 1.50). Unconditional sd $\sqrt{0.0120/0.7155} = 0.1295$. **The heavy-tailed $\varepsilon_t$ takes over part of the kurtosis, and $\hat{\alpha}_1$ falls from 0.356 to 0.2845.** Here $Q(12) = 35.42$ ($p = 0.0004$) for $\tilde{a}_t^2$, driven by a lag-12 ACF of 0.188, which points to GARCH.

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## 4 · C — GARCH (T §3.5, pp. 131–138)

ARCH often needs many lags (an ARCH(9) for the S&P 500 monthly series, T p. 131). Bollerslev's (1986) **GARCH($m, s$)** (T eq. 3.14):

$$a_t = \sigma_t \varepsilon_t, \qquad \sigma_t^2 = \alpha_0 + \sum_{i=1}^{m} \alpha_i a_{t-i}^2 + \sum_{j=1}^{s} \beta_j \sigma_{t-j}^2, \qquad \sum_{i=1}^{\max(m,s)} (\alpha_i + \beta_i) < 1. \tag{20}$$

### 4.1 · GARCH is ARMA for $a_t^2$

$\eta_t = a_t^2 - \sigma_t^2$ is a martingale difference: $E(\eta_t \mid F_{t-1}) = \sigma_t^2 E(\varepsilon_t^2 - 1) = 0$, so $E\eta_t = 0$ and $\mathrm{Cov}(\eta_t, \eta_{t-j}) = 0$ for $j \geq 1$. It is not iid. Substitute $\sigma_{t-i}^2 = a_{t-i}^2 - \eta_{t-i}$ $(i = 0, \ldots, s)$ into (20):

$$a_t^2 - \eta_t = \alpha_0 + \sum_{i=1}^{m} \alpha_i a_{t-i}^2 + \sum_{j=1}^{s} \beta_j (a_{t-j}^2 - \eta_{t-j}),$$

$$a_t^2 = \alpha_0 + \sum_{i=1}^{\max(m,s)} (\alpha_i + \beta_i) a_{t-i}^2 + \eta_t - \sum_{j=1}^{s} \beta_j \eta_{t-j}. \tag{21}$$

That is T eq. (3.15), an ARMA$(\max(m, s), s)$ for $a_t^2$: **AR coefficients $\alpha_i + \beta_i$, MA coefficients $\beta_j$.** The ARMA mean formula of L08-02-forecasting-ma-arma gives

$$\sigma_\infty^2 = E(a_t^2) = \frac{\alpha_0}{1 - \sum_{i=1}^{\max(m,s)} (\alpha_i + \beta_i)}. \tag{22}$$

### 4.2 · GARCH(1,1): kurtosis (T §3.16, pp. 165–166)

$$\sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \qquad 0 \leq \alpha_1, \beta_1 \leq 1, \quad \alpha_1 + \beta_1 < 1. \tag{23}$$

Let $E(\varepsilon_t^4) = K + 3$, $v = E(\sigma_t^2) = \alpha_0/(1 - \alpha_1 - \beta_1)$ by (22), and $s_4 = E(\sigma_t^4)$. Then $E(a_t^4) = (K + 3)s_4$ and, since $\sigma_{t-1}$ is known before $\varepsilon_{t-1}$ is drawn, $E(\sigma_{t-1}^2 a_{t-1}^2) = E(\sigma_{t-1}^4 \varepsilon_{t-1}^2) = s_4$. Square (23):

$$\sigma_t^4 = \alpha_0^2 + \alpha_1^2 a_{t-1}^4 + \beta_1^2 \sigma_{t-1}^4 + 2\alpha_0\alpha_1 a_{t-1}^2 + 2\alpha_0\beta_1 \sigma_{t-1}^2 + 2\alpha_1\beta_1 \sigma_{t-1}^2 a_{t-1}^2.$$

Take expectations, stationarity giving $E\sigma_t^4 = E\sigma_{t-1}^4 = s_4$:

$$s_4 = \alpha_0^2 + \alpha_1^2 (K + 3) s_4 + \beta_1^2 s_4 + 2\alpha_0(\alpha_1 + \beta_1) v + 2\alpha_1\beta_1 s_4.$$

The $s_4$ coefficients add to $(K + 3)\alpha_1^2 + \beta_1^2 + 2\alpha_1\beta_1 = (\alpha_1 + \beta_1)^2 + (K + 2)\alpha_1^2$. Write $\pi = \alpha_1 + \beta_1$:

$$s_4[1 - \pi^2 - (K + 2)\alpha_1^2] = \alpha_0^2 + 2\alpha_0\pi\Big(\frac{\alpha_0}{1 - \pi}\Big) = \alpha_0^2 \, \frac{1 + \pi}{1 - \pi},$$

$$s_4 = \frac{\alpha_0^2 (1 + \pi)}{(1 - \pi)[1 - \pi^2 - (K + 2)\alpha_1^2]}. \tag{24}$$

Kurtosis of $a_t$, dividing $(K + 3)s_4$ by $v^2 = \alpha_0^2/(1 - \pi)^2$:

$$\frac{E(a_t^4)}{[E(a_t^2)]^2} = \frac{(K+3)(1+\pi)(1-\pi)}{1-\pi^2-(K+2)\alpha_1^2} = \frac{(K+3)(1-\pi^2)}{1-\pi^2-(K+2)\alpha_1^2}.$$

With Gaussian $\varepsilon_t$ ($K=0$) the excess kurtosis is

$$K_a^{(g)} = \frac{3(1-\pi^2)}{1-\pi^2-2\alpha_1^2} - 3 = \frac{6\alpha_1^2}{1-2\alpha_1^2-(\alpha_1+\beta_1)^2}, \tag{25}$$

finite iff $1-2\alpha_1^2-(\alpha_1+\beta_1)^2 > 0$ (T p. 132). **$\alpha_1$, not $\beta_1$, creates the fat tails**: with $\alpha_1 = 0$, $K_a^{(g)} = 0$. With $\beta_1 = 0$, (25) collapses to (12): $6\alpha_1^2/(1-3\alpha_1^2)$ is the excess of $3(1-\alpha_1^2)/(1-3\alpha_1^2)$.

### 4.3 · GARCH(1,1): forecasts (T pp. 133)

Write $a_t^2 = \sigma_t^2\varepsilon_t^2$ in (23), one period ahead:

$$\sigma_{t+1}^2 = \alpha_0 + (\alpha_1+\beta_1)\sigma_t^2 + \alpha_1\sigma_t^2(\varepsilon_t^2 - 1). \tag{26}$$

At origin $h$, $\sigma_h^2(1) = \alpha_0 + \alpha_1 a_h^2 + \beta_1\sigma_h^2$ is known. Set $t = h+1$ in (26) and take $E(\cdot \mid F_h)$, using $E(\varepsilon_{h+1}^2 - 1 \mid F_h) = 0$; repeat:

$$\sigma_h^2(\ell) = \alpha_0 + (\alpha_1+\beta_1)\,\sigma_h^2(\ell-1), \qquad \ell > 1, \tag{27}$$

an AR(1) recursion with coefficient $\pi = \alpha_1+\beta_1$ (T eq. 3.17). Iterate (27) $\ell-1$ times and sum the geometric series:

$$\sigma_h^2(\ell) = \alpha_0\big[1+\pi+\cdots+\pi^{\ell-2}\big] + \pi^{\ell-1}\sigma_h^2(1) = \frac{\alpha_0(1-\pi^{\ell-1})}{1-\pi} + \pi^{\ell-1}\sigma_h^2(1).$$

Substitute $\alpha_0/(1-\pi) = \sigma_\infty^2$ from (22):

$$\sigma_h^2(\ell) = \sigma_\infty^2 + \pi^{\ell-1}\big[\sigma_h^2(1) - \sigma_\infty^2\big] \xrightarrow{\ell\to\infty} \sigma_\infty^2 \quad (\pi < 1). \tag{28}$$

**Half-life of a volatility shock**: the horizon at which the gap $\sigma_h^2(\ell) - \sigma_\infty^2$ has halved, $\pi^\ell = \frac{1}{2}$:

$$\ell_{1/2} = \frac{\ln 0.5}{\ln(\alpha_1+\beta_1)}, \tag{29}$$

the AR(1) half-life of L01-07 (17) applied to (27).

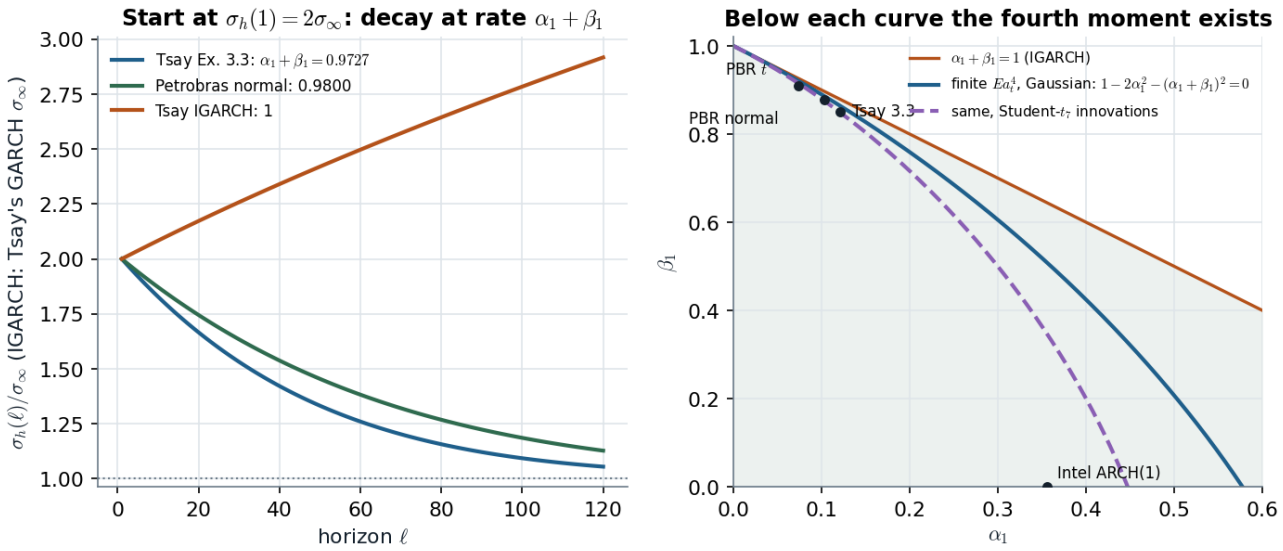



*Figure — left: (28) with $\sigma_h(1) = 2\sigma_\infty$, so $\sigma_h^2(\ell)/\sigma_\infty^2 = 1 + 3\pi^{\ell-1}$, and (32) for the IGARCH fit; right: $\alpha_1 + \beta_1 = 1$, the boundary of (25), and the $t_7$ boundary $1 - 2\alpha_1^2(\nu-1)/(\nu-4) - (\alpha_1+\beta_1)^2 = 0$ (T p. 166). Source: T Ex. 3.3 (pp. 136–141), garchmodeling.html.*

The term structure is (28) exactly: at $\ell = 2$ the Tsay curve is $\sqrt{1 + 3(0.9727)} = 1.98$, and it takes about 25 months to close half the gap in variance. All three empirical points sit **just inside** the Gaussian kurtosis boundary and close to the IGARCH line. That is the typical picture: high persistence, and a fourth moment that barely exists. Under $t_7$ innovations the admissible region shrinks, because the innovations bring their own kurtosis.

> **Interactive companion.** The Volatility Dial (`companions/companion-garch-volatility.html`) — move $\alpha_1$, $\beta_1$, $\alpha_0$, the 1-step volatility $\sigma_h(1)$, the innovation tails ($\nu$ or Normal) and the horizon $L$. Watch the variance term structure close on $\sigma_\infty$, the **half-life** read-out, and the point in the $(\alpha_1, \beta_1)$ plane against the IGARCH line and the kurtosis boundaries. The defaults reproduce Tsay's Table 3.1.

### 4.4 · Worked — S&P 500 monthly excess returns 1926–1991, $T = 792$ (T Ex. 3.3, pp. 134–139)

The PACF of $r_t^2$ shows strong dependence. An AR(3) mean with GARCH(1,1) gives insignificant AR terms, so the refined model is (T eq. 3.19)

$$r_t = 0.0076 + a_t, \qquad \sigma_t^2 = 0.000086 + 0.1216\, a_{t-1}^2 + 0.8511\, \sigma_{t-1}^2, \tag{30}$$

with standard errors 0.0015, 0.000024, 0.0197, 0.0190. Checks: $Q(12) = 11.99$ ($p = 0.45$) for $\tilde{a}_t$, $Q(12) = 13.11$ ($p = 0.36$) for $\tilde{a}_t^2$.

- **Persistence**: $\hat{\alpha}_1 + \hat{\beta}_1 = 0.1216 + 0.8511 = 0.9727$. *Tsay prints "0.9772" on p. 137, a transposition of the last two digits.*
- **Unconditional variance** by (22): $0.000086/(1 - 0.9727) = 0.000086/0.0273 = 0.003150$, sd $\sqrt{0.003150} = 0.0561$. Tsay rounds to 0.00314, whose square root 0.0560 is the "$\infty$" entry of Table 3.1.
- **Half-life** by (29): $\ln 0.5 / \ln 0.9727 = (-0.69315)/(-0.02768) = 25.04$ months.
- **Kurtosis** by (25): $2\alpha_1^2 = 2(0.1216)^2 = 0.02957$, $\pi^2 = 0.9727^2 = 0.94615$, denominator $1 - 0.02957 - 0.94615 = 0.02428$. So $K_a^{(g)} = 6(0.014787)/0.02428 = 0.08872/0.02428 = 3.654$ and the

kurtosis is 6.654. **Close to the boundary: a small change in $\alpha_1$ moves the denominator a lot.**

- **Forecasts** (Table 3.1, origin $h = 792$, December 1991): volatility 0.0536, 0.0537, 0.0537, 0.0538, 0.0538 for $\ell = 1, \ldots, 5$, and 0.0560 at $\infty$. Because $\sigma_h(1)$ starts **below** $\sigma_\infty$, (28) rises toward it, slowly ($\pi^4 = 0.895$).

$t$ **innovations.** With $\nu = 5$ fixed (T eq. 3.20): $\sigma_t^2 = 0.00012 + 0.1121a_{t-1}^2 + 0.8432\sigma_{t-1}^2$, persistence 0.9553 ("$\approx 0.95$, close to 1", hence "essentially IGARCH" in Tsay's words). With $\nu$ estimated (T eq. 3.21): $\hat{\nu} = 7.02$ (se 1.78), the same volatility equation, and $\nu = 5$ is not rejected: $(7.02 - 5)/1.78 = 1.13$.

**Two caveats (T pp. 139–140).** (i) **Forecast evaluation**: $a_{h+1}^2$ is an unbiased but very noisy proxy of $\sigma_{h+1}^2$ (one draw of a random variable cannot estimate its variance), so a low $R^2$ of $a_{h+\ell}^2$ on $\sigma_h^2(\ell)$ is expected and does not condemn the model. (ii) **Two-pass estimation**: fit the mean, then fit (21) as an ARMA(1,1) to $\hat{a}_t^2$. Tsay's SCA fit $a_t^2 = 0.00014 + 0.9583a_{t-1}^2 + \eta_t - 0.8456\eta_{t-1}$ gives $\widehat{\beta}_1 = 0.8456$ and $\widehat{\alpha}_1 = 0.9583 - 0.8456 = 0.1127$, close to (30). It is an approximation with unexplored properties.

### 4.5 · Worked — Petrobras daily log returns, August 2000 – February 2021 (garchmodeling.html, $n = 5167$)

No mean term (`include.mean=FALSE`), Gaussian $\varepsilon$.

| Model | $\widehat{\alpha}_0$ ($\omega$) | $\widehat{\alpha}_1$ | $\widehat{\beta}_1$ | log-lik |
|---|---|---|---|---|
| ARCH(1) | 0.00070027 | 0.36979 | — | 10672.32 |
| GARCH(1,1) | 0.00002255 | 0.10317 | 0.87681 | 11151.72 |

- ARCH(1): unconditional sd by (8), $\sqrt{0.00070027/0.63021} = \sqrt{0.0011112} = 0.0333$ (3.3% a day); kurtosis by (12), $3(1 - 0.13674)/(1 - 0.41023) = 4.391$.
- GARCH(1,1): persistence $0.10317 + 0.87681 = 0.97997$. Unconditional variance $0.00002255/0.020027 = 0.0011259$, sd **0.03356**. Half-life $\ln 0.5 / \ln 0.97997 = (-0.69315)/(-0.020231) =$ **34.3 trading days**. Excess kurtosis by (25): $2\alpha_1^2 = 0.021287$, $\pi^2 = 0.960347$, denominator 0.018367, so $6(0.010643)/0.018367 =$ **3.48**.
- One extra parameter raises the log-likelihood by 479.4: LR $= 2(11151.72 - 10672.32) = 958.8$. ARCH(1) is nested at $\beta_1 = 0$, a boundary point, so $\chi_1^2$ is only a rough guide here; a statistic this large settles the question anyway.

**The 2025 update (garch-sv-petrobras.html, simple returns to November 2025, $n = 6363$):** ARCH(1) $\widehat{\alpha}_1 = 0.33114$ (uncond. sd $\sqrt{0.00066851/0.66886} = 0.0316$); GARCH normal ($1.8817 \times 10^{-5}$, 0.081406, 0.89841), persistence 0.97982, sd 0.0305; GARCH-$t$ ($1.1404 \times 10^{-5}$, 0.059208, 0.92773), $\hat{\nu} = 6.23$, persistence **0.98694**, sd 0.0295. The $t$ fit again has a smaller $\alpha_1$ and a larger $\beta_1$ than the Gaussian one.

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## 5 · C — IGARCH and RiskMetrics (T §3.6, pp. 140–142)

If the AR polynomial of (21) has a unit root, $\alpha_1 + \beta_1 = 1$, the model is **integrated** GARCH:

$$a_t = \sigma_t \varepsilon_t, \qquad \sigma_t^2 = \alpha_0 + \beta_1 \sigma_{t-1}^2 + (1 - \beta_1) a_{t-1}^2, \qquad 0 < \beta_1 < 1. \tag{31}$$

Tsay's S&P 500 IGARCH: $r_t = 0.0067 + a_t$, $\sigma_t^2 = 0.000119 + 0.8059\sigma_{t-1}^2 + 0.1941 a_{t-1}^2$. The coefficients are close to (30), but **the unconditional variance (22) no longer exists** (denominator 0), which is hard to defend for an excess return. Tsay suggests occasional level shifts in volatility as a cause.

**Forecasts.** Put $\alpha_1 + \beta_1 = 1$ in (27): $\sigma_h^2(\ell) = \alpha_0 + \sigma_h^2(\ell - 1)$. Iterating:

$$\sigma_h^2(\ell) = \sigma_h^2(1) + (\ell - 1)\alpha_0, \qquad \ell \geq 1, \tag{32}$$

(T eq. 3.22): a straight line with slope $\alpha_0$. **A shock to $\sigma_h^2(1)$ is permanent**, as a shock to a random walk is (L08-03-unit-roots-dickey-fuller). The variance process is a martingale; it can be strictly but not weakly stationary (Nelson, 1990).

**RiskMetrics** is (31) with $\alpha_0 = 0$: $\sigma_t^2 = (1 - \beta_1)a_{t-1}^2 + \beta_1 \sigma_{t-1}^2$. Substitute the same equation for $\sigma_{t-1}^2$:

$$\sigma_t^2 = (1 - \beta_1)a_{t-1}^2 + \beta_1\big[(1 - \beta_1)a_{t-2}^2 + \beta_1 \sigma_{t-2}^2\big] = (1 - \beta_1)a_{t-1}^2 + (1 - \beta_1)\beta_1 a_{t-2}^2 + \beta_1^2 \sigma_{t-2}^2,$$

and repeat:

$$\sigma_t^2 = (1 - \beta_1)(a_{t-1}^2 + \beta_1 a_{t-2}^2 + \beta_1^2 a_{t-3}^2 + \cdots), \tag{33}$$

**exponential smoothing** of $a_t^2$ with discount factor $\beta_1$; the weights $(1 - \beta_1)\beta_1^{j-1}$ sum to 1. By (32) with $\alpha_0 = 0$, every horizon's forecast equals $\sigma_h^2(1)$.

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## 6 · C — ARCH in a regression (W §12.6, pp. 391–396)

Wooldridge's angle is the regression error. **Heteroskedasticity** in a time-series regression leaves OLS unbiased/consistent but invalidates the usual standard errors; heteroskedasticity-robust ones work as in cross-sections, provided there is **no serial correlation** (test for that first). The Breusch–Pagan regression (W eq. 12.46) $u_t^2 = \delta_0 + \delta_1 x_{t1} + \cdots + \delta_k x_{tk} + v_t$ needs homoskedastic, serially uncorrelated $v_t$.

**Worked — W Ex. 12.8, NYSE weekly returns, $n = 689$.** The EMH regression $return_t = \beta_0 + \beta_1 return_{t-1} + u_t$ gives $t_{\beta_1} = 1.55$: no predictability in the mean. Breusch–Pagan on $return_{t-1}$:

$$\widehat{u}_t^2 = 4.66 - 1.104\, return_{t-1}, \qquad (0.43)\ (0.201), \qquad R^2 = 0.042,$$

so $t = -1.104/0.201 = -5.5$: **the variance depends on the past even though the mean does not**, and it is higher after low returns (the leverage pattern of §1).

**ARCH(1) for errors** (W eq. 12.49): $E(u_t^2 \mid u_{t-1}, \dots) = a_0 + a_1 u_{t-1}^2$, which is the conditional variance only if $E(u_t \mid u_{t-1}, \dots) = 0$. Written as

$$u_t^2 = a_0 + a_1 u_{t-1}^2 + v_t, \qquad E(v_t \mid u_{t-1}, \dots) = 0 \tag{34}$$

(W eq. 12.50), it is an AR(1) in $u_t^2$, stable for $a_1 < 1$ — equation (13) with $m = 1$. Under ARCH errors with otherwise Gauss–Markov assumptions, **OLS is still BLUE and the usual statistics are asymptotically valid**. ARCH still matters for two reasons: weighted least squares or maximum likelihood based on (34) gives asymptotically **more efficient** estimators, and volatility itself is the object of interest in finance.

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### Links

- Neighbours: L08-05-seasonal-airline-model · L08-07-garch-family-asymmetry-tails (asymmetry, EGARCH, GARCH-M, non-Gaussian errors, Tiao's kurtosis formula).
- Unit: L08-00-map · L08-02-forecasting-ma-arma (the ARMA mean and forecast recursion used in (21)–(28)) · L08-03-unit-roots-dickey-fuller (the unit root behind IGARCH) · L08-08-stochastic-volatility (the alternative with its own shock) · L08-09-bayesian-garch (the same GARCH(1,1)-$t$ by MCMC) · L08-10-formulas · L08-11-pitfalls · L08-12-python (GARCH MLE with `scipy.optimize`).
- Class 1: L01-07 §2 (squared-return Ljung–Box), §6 (half-life).
- Rules: rules/05 (ARCH/GARCH).
- **Practice (number and printed page only):** T 3.3 and 3.4 (p. 168); T 3.6 (p. 169); T 3.11 and 3.12 (p. 170); W 12.4 (p. 397). Solutions: `Listas/lista-L08-book-exercises.pdf`.

## 3.7–3.9, 3.16 · The GARCH family: leverage, log-variance, risk premia and fat-tailed innovations

> **Claim.** GARCH treats $+a$ and $-a$ alike. **Asymmetric models give bad news a bigger coefficient.** GJR/TGARCH do it with an indicator, APARCH with $(|a| - \gamma a)^\delta$, EGARCH with a signed term in $\ln \sigma_t^2$. **What the site's output reports is not always what its formula says**: fGarch's `leverage=TRUE` fits the APARCH form, so the printed $\gamma_1 = 0.208$ for Petrobras means bad-news and good-news coefficients 0.135 and 0.058 (ratio 2.33), not "$\alpha+\gamma$ against $\alpha$". GARCH-M puts $\sigma_t^2$ in the mean (a risk premium). **Fat-tailed innovations matter more than any asymmetry**: the Student-$t$ fit adds 141 log-likelihood points to Gaussian GARCH, against 22 for GJR. Tiao's formula splits the total kurtosis into the part the innovations bring and the part the volatility dynamics bring. Prev: L08-06-arch-garch · Next: L08-08-stochastic-volatility · Map: L08-00-map

**Notation.** As in L08-06-arch-garch: $a_t = \sigma_t \varepsilon_t$, $\varepsilon_t$ iid, mean 0, variance 1; $\sigma_\infty^2$ the unconditional variance. In addition:

| Symbol | Meaning | Range |
|---|---|---|
| $\omega$ | the site's constant (Tsay's $\alpha_0$) | |
| $\gamma$ | asymmetry parameter (meaning depends on the model, §2–§4) | |
| $\delta$ | APARCH power | $\delta > 0$ |
| $N_{t-1}$ | indicator $1\{a_{t-1} < 0\}$ (T eq. 3.34) | $\{0, 1\}$ |
| $g(\varepsilon)$ | EGARCH's weighted innovation | |
| $\theta$ | EGARCH sign coefficient (rugarch's $\alpha_1$) | |
| $c$ | GARCH-M risk-premium parameter | |
| $K$ | excess kurtosis of $\varepsilon_t$ | $K = 6/(\nu - 4)$ for $t_\nu$ |
| $K_a$, $K_a^{(g)}$ | excess kurtosis of $a_t$, and its value with Gaussian $\varepsilon_t$ | |
| $\nu$, $\xi$ | Student-$t$ degrees of freedom; skew parameter | $\nu > 2$; $\xi > 0$ |

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### 1 · U — the news-impact curve

Fix yesterday's volatility at its unconditional level, $\sigma_{t-1} = \sigma_\infty$, and plot today's variance against yesterday's standardized shock $\varepsilon_{t-1}$. That is the **news-impact curve** (NIC). For GARCH(1,1), L08-06 (23) gives

$$NIC(\varepsilon) = \alpha_0 + \alpha_1 \sigma_\infty^2 \varepsilon^2 + \beta_1 \sigma_\infty^2, \tag{1}$$

a parabola symmetric about 0. The leverage fact (L08-06 §1, fact 4) says the left branch should be steeper. Each model below bends one branch.

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## 2 · C — threshold models: GJR, TGARCH, TS-GARCH (garchmodeling.html; T §3.9, p. 149)

**GJR / TGARCH($m, s$)** (Glosten–Jagannathan–Runkle 1993, Zakoian 1994; T eq. 3.34):

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^{s}(\alpha_i + \gamma_i N_{t-i})a_{t-i}^2 + \sum_{j=1}^{m}\beta_j \sigma_{t-j}^2, \qquad N_{t-i} = 1\{a_{t-i} < 0\}. \tag{2}$$

A positive $a_{t-1}$ contributes $\alpha_1 a_{t-1}^2$, a negative one $(\alpha_1 + \gamma_1)a_{t-1}^2$, with $\gamma_1 > 0$ for leverage. Threshold 0 separates the regimes.

**Worked — IBM monthly log returns 1926–2003 (T eq. 3.35).** With GED innovations (shape 1.51):

$$\sigma_t^2 = 3.45 \times 10^{-4} + (0.0658 + 0.0843 N_{t-1})a_{t-1}^2 + 0.8182\,\sigma_{t-1}^2.$$

Good news: 0.0658. Bad news: $0.0658 + 0.0843 = 0.1501$, **2.3 times larger**. Checks $Q(12) = 18.34$ ($p = 0.106$) for $\tilde{a}_t$, $Q(12) = 5.36$ ($p = 0.95$) for $\tilde{a}_t^2$; the leverage term is significant at 5%.

The site's page writes (2) for GJR and its $\sigma_t$ analogue for T-GARCH,

$$\sigma_t = \omega + \alpha_1 |a_{t-1}| + \gamma_1 |a_{t-1}| N_{t-1} + \beta_1 \sigma_{t-1}, \tag{3}$$

and the **Taylor–Schwert** model is (3) with $\gamma_1 = 0$: $\sigma_t = \omega + \alpha_1 |a_{t-1}| + \beta_1 \sigma_{t-1}$. Standard deviations, not variances, follow the recursion.

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## 3 · C — APARCH, and what fGarch actually estimates (garchmodeling.html; Ding, Granger and Engle 1993)

The **asymmetric power** GARCH:

$$\sigma_t^\delta = \omega + \alpha_1 \big(|a_{t-1}| - \gamma_1 a_{t-1}\big)^\delta + \beta_1 \sigma_{t-1}^\delta, \qquad \delta > 0,\ \gamma_1 \in (-1, 1). \tag{4}$$

**The term $|a| - \gamma a$, split by sign.** For $a \geq 0$, $|a| - \gamma a = a(1 - \gamma)$. For $a < 0$, $|a| = -a$, so $|a| - \gamma a = -a - \gamma a = |a|(1 + \gamma)$. Therefore

$$\big(|a| - \gamma a\big)^\delta = \begin{cases} (1-\gamma)^\delta |a|^\delta, & a \geq 0 \text{ (good news)} \\ (1+\gamma)^\delta |a|^\delta, & a < 0 \text{ (bad news).} \end{cases} \tag{5}$$

**Special cases** (the page's list, with the sign of $\gamma$ made explicit):

| Model | $\delta$ | $\gamma_1$ | $\beta_1$ |
|---|---|---|---|
| ARCH | 2 | 0 | 0 |
| GARCH | 2 | 0 | free |
| TS-GARCH | 1 | 0 | free |
| T-GARCH (asymmetric, in $\sigma$) | 1 | $\in (0, 1)$ | free |

| Model | $\delta$ | $\gamma_1$ | $\beta_1$ |
|---|---|---|---|
| GJR (asymmetric, in $\sigma^2$) | 2 | $\in (0,1)$ | free |

**The mapping.** `garchFit(~garch(1,1), delta=2, leverage=TRUE)` and `delta=1, leverage=TRUE` fit (4) with $\delta$ fixed. So the printed `gamma1` is **APARCH's** $\gamma_1$, not the indicator coefficient of (2) and (3). With $\delta = 2$, (5) turns (4) into (2) with

$$\alpha_1^{GJR} = \alpha_1(1-\gamma_1)^2, \qquad \alpha_1^{GJR} + \gamma_1^{GJR} = \alpha_1(1+\gamma_1)^2,$$

$$\gamma_1^{GJR} = \alpha_1\big[(1+\gamma_1)^2 - (1-\gamma_1)^2\big] = \alpha_1\big[(1+2\gamma_1+\gamma_1^2) - (1-2\gamma_1+\gamma_1^2)\big] = 4\alpha_1\gamma_1. \tag{6}$$

**Persistence.** With symmetric $\varepsilon$, $E(|\varepsilon|\varepsilon) = 0$, so

$$E(|\varepsilon| - \gamma\varepsilon)^2 = E(\varepsilon^2) - 2\gamma E(|\varepsilon|\varepsilon) + \gamma^2 E(\varepsilon^2) = 1 + \gamma^2,$$

and taking expectations of (4) with $\delta = 2$ as in L08-06 (22):

$$\sigma_\infty^2 = \frac{\omega}{1 - \alpha_1(1+\gamma_1^2) - \beta_1}. \tag{7}$$

**Worked — Petrobras GJR, $\delta = 2$ (garchmodeling.html).** $\hat\omega = 2.327 \times 10^{-5}$, $\hat\alpha_1 = 0.09234$, $\hat\gamma_1 = 0.20835$, $\hat\beta_1 = 0.88178$, log-lik 11174.01.

- Good news: $0.09234 \times (1 - 0.20835)^2 = 0.09234 \times 0.62671 = \mathbf{0.05787}$.
- Bad news: $0.09234 \times (1+0.20835)^2 = 0.09234 \times 1.46011 = \mathbf{0.13483}$. Ratio $1.46011/0.62671 = 2.33$.
- Indicator coefficient (6): $4 \times 0.09234 \times 0.20835 = \mathbf{0.07696}$, and indeed $0.05787 + 0.07696 = 0.13483$.
- Persistence: $0.09234 \times (1 + 0.04341) + 0.88178 = 0.09635 + 0.88178 = \mathbf{0.97813}$.

Reading the page's formula literally ($\alpha_1 + \gamma_1 = 0.30$ against $\alpha_1 = 0.09$) would give persistence $0.09234 + 0.20835/2 + 0.88178 = 1.078$, **a non-stationary model that was never fitted**.

**Worked — Petrobras T-GARCH, $\delta = 1$.** $\hat\omega = 0.000725$, $\hat\alpha_1 = 0.09641$, $\hat\gamma_1 = 0.34588$, $\hat\beta_1 = 0.90265$. In the $\sigma_t$ recursion, a bad shock enters with $\alpha_1(1+\gamma_1) = 0.09641 \times 1.34588 = \mathbf{0.12976}$ and a good one with $\alpha_1(1-\gamma_1) = 0.09641 \times 0.65412 = \mathbf{0.06306}$: ratio 2.06 in the standard deviation.

**Worked — Petrobras APARCH, $\delta$ estimated.** $\hat\delta = 1.436$, $\hat\gamma_1 = 0.2713$, $\hat\alpha_1 = 0.0990$, $\hat\beta_1 = 0.8927$. By (5), the bad/good ratio in $\sigma^\delta$ is $(1.2713/0.7287)^{1.436} = 1.7446^{1.436} = \exp(1.436 \times 0.5566) = \exp(0.7993) = 2.22$. The power sits between the $\sigma$ ($\delta = 1$) and $\sigma^2$ ($\delta = 2$) versions, and the log-likelihood (11170.95) is lower than GJR's (11174.01) despite one more parameter, which says the two are close optima and $\delta = 2$ is not rejected.

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## 4 · C — EGARCH (T §3.8, pp. 143–148; garchmodeling.html via rugarch)

Nelson (1991) builds asymmetry into a **weighted innovation** (T eq. 3.24):

$$g(\varepsilon_t) = \theta\varepsilon_t + \gamma\big[|\varepsilon_t| - E|\varepsilon_t|\big], \qquad E[g(\varepsilon_t)] = 0. \tag{8}$$

Split by sign, with $|\varepsilon| = \varepsilon$ for $\varepsilon \geq 0$ and $-\varepsilon$ for $\varepsilon < 0$:

$$g(\varepsilon_t) = \begin{cases} (\theta+\gamma)\varepsilon_t - \gamma E|\varepsilon_t|, & \varepsilon_t \geq 0 \\ (\theta-\gamma)\varepsilon_t - \gamma E|\varepsilon_t|, & \varepsilon_t < 0, \end{cases} \tag{9}$$

so the slope is $\theta+\gamma$ on good news and $\theta-\gamma$ on bad news; $\theta < 0$ makes the bad-news branch steeper. For Gaussian $\varepsilon$, $E|\varepsilon| = \sqrt{2/\pi} = 0.7979$. For the standardized $t_\nu$ of L08-06 (16) (T p. 143):

$$E|\varepsilon_t| = \frac{2\sqrt{\nu-2}\,\Gamma(\frac{\nu+1}{2})}{(\nu-1)\Gamma(\frac{\nu}{2})\sqrt{\pi}}. \tag{10}$$

At $\nu = 7.04$ (the Petrobras $t$ fit) this is 0.7595: the heavier tails put more mass near 0 and in the tails, so the mean absolute value falls below the Gaussian 0.7979.

The EGARCH$(m, s)$ model (T eq. 3.25) is an ARMA in $\ln\sigma_t^2$ driven by $g$:

$$a_t = \sigma_t\varepsilon_t, \qquad \ln\sigma_t^2 = \alpha_0 + \frac{1+\beta_1 B + \cdots + \beta_{s-1}B^{s-1}}{1-\alpha_1 B - \cdots - \alpha_m B^m}\, g(\varepsilon_{t-1}). \tag{11}$$

**Why the log.** $\sigma_t^2 = \exp(\cdot) > 0$ for any coefficients, so **no positivity constraints** are needed. Since $Eg = 0$, the unconditional mean of $\ln\sigma_t^2$ is $\alpha_0$.

**rugarch's EGARCH(1,1)**, the one the page fits:

$$\ln\sigma_t^2 = \omega + \alpha_1\varepsilon_{t-1} + \gamma_1(|\varepsilon_{t-1}| - E|\varepsilon_{t-1}|) + \beta_1 \ln\sigma_{t-1}^2. \tag{12}$$

Its $\alpha_1$ is Nelson's $\theta$. Taking expectations with $|\beta_1| < 1$:

$$E\ln\sigma_t^2 = \frac{\omega}{1-\beta_1}. \tag{13}$$

**The bad/good ratio.** Compare a shock $\varepsilon_{t-1} = -c$ with $+c$, same $\sigma_{t-1}$. In (12) the $\gamma_1$ terms are equal and cancel:

$$\ln\sigma_t^2(-c) - \ln\sigma_t^2(+c) = \alpha_1(-c) - \alpha_1(c) = -2\alpha_1 c \quad\Longrightarrow\quad \frac{\sigma_t^2(-c)}{\sigma_t^2(+c)} = e^{-2\alpha_1 c}. \tag{14}$$

**Worked — Petrobras EGARCH(1,1) (garchmodeling.html, with a mean $\mu = 0.00028$, insignificant).** $\hat\omega = -0.15392$, $\hat\alpha_1 = -0.05549$, $\hat\beta_1 = 0.97741$, $\hat\gamma_1 = 0.18138$, log-lik 11173.14.

- By (13): $E\ln\sigma_t^2 = -0.15392/(1-0.97741) = -0.15392/0.02259 = -6.814$, so $e^{-6.814/2} = 0.0331$, a typical daily volatility of 3.3%, matching GARCH's 0.0336.
- Relative to a zero shock, by (12): $\varepsilon = -1$ raises $\ln\sigma_t^2$ by $-\alpha_1 + \gamma_1 = 0.05549 + 0.18138 = 0.23687$; $\varepsilon = +1$ by $\alpha_1 + \gamma_1 = -0.05549 + 0.18138 = 0.12589$.
- By (14): at $c = 1$, $e^{0.11098} = 1.117$; at $c = 2$, $e^{0.22196} = 1.249$. **A two-sigma drop raises next-day variance 25% more than a two-sigma rise.**
- $\hat\beta_1 = 0.977$: log-volatility persistence close to GARCH's 0.980.

**Diagnostics printed by rugarch** (read briefly). Weighted Ljung–Box on $\tilde{a}_t$: lag-1 statistic 12.77, $p = 0.0004$, so some serial correlation is left in the **mean** (the model has none). On $\tilde{a}_t^2$: $p$ from 0.11 to 0.48, so the volatility dynamics are captured. ARCH LM: $p \geq 0.49$. Sign bias: negative-size bias $t = 2.04$ ($p = 0.042$), joint $p = 0.049$, so a little asymmetry is left. Nyblom joint statistic $1.68 > 1.47$ (5% critical value): parameter instability over 20 years. Adjusted Pearson goodness of fit: $p \approx 10^{-9}$, **the Gaussian innovation is rejected**, which leads to §6.

**Forecasting (G, T p. 148).** Exponentiating (12) one step ahead gives $\sigma_{h+1}^2$ exactly. Two steps ahead requires $E_h \exp[g(\varepsilon_{h+1})]$, which for Gaussian $\varepsilon$ is $\exp(-\gamma\sqrt{2/\pi})[e^{(\theta+\gamma)^2/2}\Phi(\theta+\gamma) + e^{(\theta-\gamma)^2/2}\Phi(\gamma-\theta)]$; the $j$-step forecast follows by recursion. Not examinable.

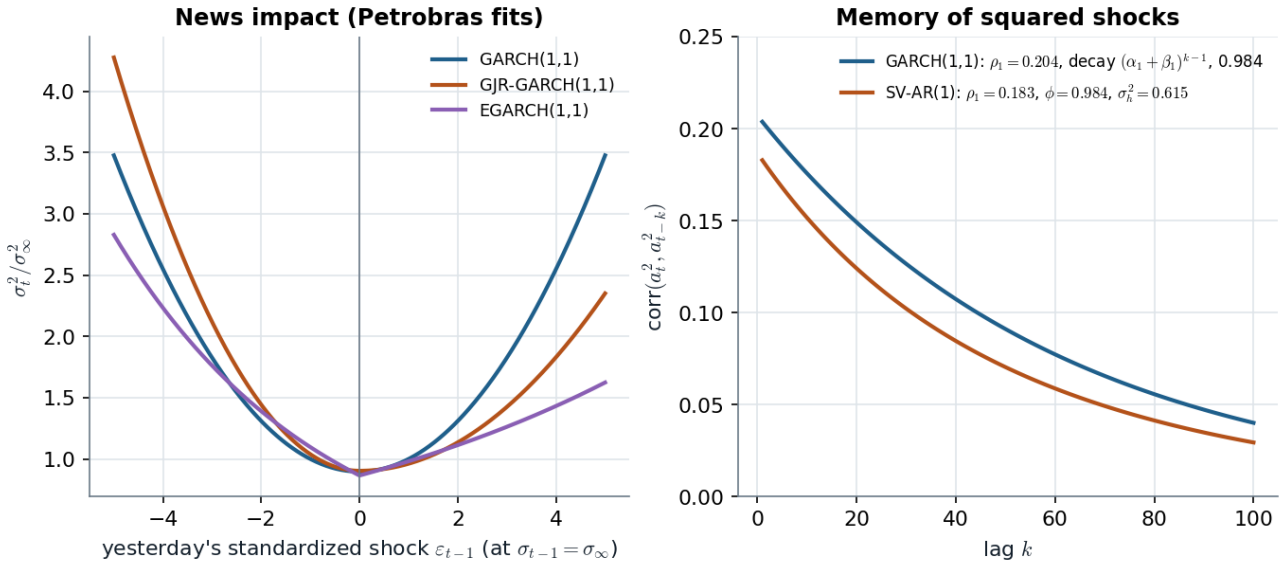



*Figure — left: news-impact curves $\sigma_t^2/\sigma_\infty^2$ at $\sigma_{t-1} = \sigma_\infty$ for the Petrobras fits of garchmodeling.html: GARCH (1), GJR in its fitted APARCH form (4) with $\delta = 2$ and $\sigma_\infty^2$ from (7), EGARCH (12) at $\ln\sigma_{t-1}^2$ equal to (13). The right panel belongs to L08-08-stochastic-volatility.*

GARCH's parabola is symmetric. GJR keeps the parabola shape but makes the left branch 2.33 times steeper, so a $-4\sigma$ day lifts variance far more than a $+4\sigma$ day. EGARCH is an exponential in $|\varepsilon|$, flatter near 0 and asymmetric by the factor (14). Near $\varepsilon = 0$ all three agree: they differ only in how they treat large news.

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## 5 · C — GARCH-M: volatility in the mean (T §3.7, p. 142)

$$r_t = \mu + c\,\sigma_t^2 + a_t, \qquad a_t = \sigma_t\varepsilon_t, \qquad \sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \beta_1\sigma_{t-1}^2. \tag{15}$$

$c$ is the **risk-premium parameter**: $c > 0$ means expected return rises with volatility. Variants use $c\,\sigma_t$ or $c\ln\sigma_t^2$. Because $\sigma_t^2$ is serially correlated, (15) **makes $r_t$ serially correlated**, which is another source of the small autocorrelations in historical returns.

**Worked — S&P 500 monthly excess returns 1926–1991 (Tsay, S-Plus):** $r_t = 0.0055 + 1.09\sigma_t^2 + a_t$, $\sigma_t^2 = 8.76 \times 10^{-5} + 0.123a_{t-1}^2 + 0.849\sigma_{t-1}^2$. The premium is positive but $t = 1.09/0.818 = 1.33$: not significant at 5%. At the unconditional variance 0.00314 of L08-06 §4.4, it adds $1.09 \times 0.00314 = 0.0034$ a month to the expected excess return.

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## 6 · C — non-Gaussian innovations and the kurtosis decomposition (T eqs. 3.7–3.10, pp. 121–122; §3.16, pp. 165–166)

**Distributions on offer** (fGarch's `cond.dist`): `norm`, `std` (standardized $t$, L08-06 (16)), `sstd` (skew-$t$), `ged`, `snorm`, `sged`.

- **Skew-$t$** (Fernández–Steel, T eq. 3.9): the standardized $t$ is stretched by $\xi$ above the mode and by $1/\xi$ below it, then re-centred and re-scaled to mean 0, variance 1. $\xi^2$ is the ratio of probability mass above to below the mode: $\xi < 1$ means a longer left tail.
- **GED** (T eq. 3.10): $f(x) = \dfrac{\nu \exp\left(-\frac{1}{2}|x/\lambda|^\nu\right)}{\lambda\, 2^{1+1/\nu}\Gamma(1/\nu)}$, $\lambda = \left[2^{-2/\nu}\Gamma(1/\nu)/\Gamma(3/\nu)\right]^{1/2}$. $\nu = 2$ is the normal, $\nu < 2$ has heavier tails, and $\nu = 1$ is the **Laplace** (double exponential).

**Why a heavy-tailed $\varepsilon$ lowers $\hat{\alpha}_1$ (T p. 126).** Under Gaussian $\varepsilon$, every large $|a_t|$ must be explained by a large $\sigma_t$, so the fit pushes $\alpha_1$ up. A $t$ innovation can produce the outlier itself. Intel: $\hat{\alpha}_1 = 0.356 \to 0.2845$ (L08-06-arch-garch §3.4); Petrobras: $0.1032 \to 0.0743$ below.

**Tiao's formula (T p. 166).** With $E(\varepsilon_t^4) = K + 3$, the excess kurtosis of $a_t$ from L08-06 (24) can be written as

$$K_a = \frac{K + K_a^{(g)} + \frac{5}{6}KK_a^{(g)}}{1 - \frac{1}{6}KK_a^{(g)}}, \qquad K_a^{(g)} = \frac{6\alpha_1^2}{1 - 2\alpha_1^2 - (\alpha_1 + \beta_1)^2}. \tag{16}$$

**Total tails = innovation tails $K$ + dynamics tails $K_a^{(g)}$ + an interaction.** With $\alpha_1 = 0$, $K_a^{(g)} = 0$ and $K_a = K$. For the standardized $t_\nu$, $E(\varepsilon^4) = 3 + 6/(\nu - 4)$, so $K = 6/(\nu - 4)$ for $\nu > 4$, and substituting into (16):

$$K_a = \frac{6 + (\nu + 1)K_a^{(g)}}{\nu - 4 - K_a^{(g)}}, \qquad \text{finite iff} \quad 1 - 2\alpha_1^2\frac{\nu - 1}{\nu - 4} - (\alpha_1 + \beta_1)^2 > 0. \tag{17}$$

**Worked — Petrobras GARCH(1,1)-$t$ (garchmodeling.html):** $\hat{\alpha}_1 = 0.07429$, $\hat{\beta}_1 = 0.90947$, $\hat{\nu} = 7.04$ (precisely 7.0426), log-lik 11292.37.

- Persistence $0.07429 + 0.90947 = 0.98376$; $\pi^2 = 0.96778$; $2\alpha_1^2 = 2(0.0055190) = 0.011038$.
- Dynamics part, (16): $1 - 0.011038 - 0.96778 = 0.021184$, so $K_a^{(g)} = 6(0.0055190)/0.021184 = 0.033114/0.021184 =$ **1.563**.
- Innovation part: $K = 6/(7.0426 - 4) = 6/3.0426 =$ **1.972**.
- Existence, (17): $2\alpha_1^2(\nu - 1)/(\nu - 4) = 0.011038 \times 6.0426/3.0426 = 0.011038 \times 1.98600 = 0.021922$, and $1 - 0.021922 - 0.96778 = 0.0103 > 0$: **the fourth moment exists, barely.**
- Total, (17): $\dfrac{6 + 8.0426 \times 1.5631}{3.0426 - 1.5631} = \dfrac{6 + 12.571}{1.4795} = \dfrac{18.571}{1.4795} =$ **12.55**. Neither part alone (1.56 or 1.97) comes close: the interaction dominates when the moment condition is nearly binding.

**The other non-Gaussian fits (same page).** Skew-$t$: $\hat{\xi} = 0.9407$ (se 0.017, so $\xi^2 = 0.885$: slightly more mass below the mode), $\hat{\nu} = 7.01$, $\hat{\alpha}_1 = 0.07299$, $\hat{\beta}_1 = 0.91130$, log-lik 11297.85. $t$ with $\nu$ fixed at 7: virtually identical to the free fit (log-lik 11292.37), since $\hat{\nu} = 7.04$ anyway. Laplace (GED shape fixed at 1): $\hat{\alpha}_1 = 0.09340$, $\hat{\beta}_1 = 0.90081$, log-lik 11192.49. *The page's legend labels the $\nu = 7$ fit "GARCH t(3)".*

### 6.1 · Reading the log-likelihood ladder

AIC $= -2\ell + 2q$ ($q$ = number of estimated parameters; no mean term except EGARCH's $\mu$):

| Model | $q$ | log-lik | AIC | gain over GARCH |
|---|---|---|---|---|
| ARCH(1) | 2 | 10672.32 | −21340.6 | −479.4 |
| TS-GARCH | 3 | 11107.02 | −22208.0 | −44.7 |
| GARCH(1,1) | 3 | 11151.72 | −22297.4 | 0 |
| T-GARCH | 4 | 11121.07 | −22234.1 | −30.7 |
| APARCH | 5 | 11170.95 | −22331.9 | +19.2 |
| EGARCH (with $\mu$) | 5 | 11173.14 | −22336.3 | +21.4 |
| GJR | 4 | 11174.01 | −22340.0 | +22.3 |
| GARCH, Laplace | 3 | 11192.49 | −22379.0 | +40.8 |
| GARCH-$t$, $\nu = 7$ fixed | 3 | 11292.37 | −22578.7 | +140.7 |
| GARCH-$t$ | 4 | 11292.37 | −22576.7 | +140.7 |
| GARCH skew-$t$ | 5 | 11297.85 | **−22585.7** | +146.1 |

Three readings. (i) **Modelling $\sigma$ instead of $\sigma^2$ hurts here** (TS-GARCH and T-GARCH below GARCH): $\delta = 2$ suits Petrobras, as APARCH's $\widehat{\delta} = 1.44$ with a lower likelihood than GJR confirms. (ii) **Asymmetry is worth about 22 points for one parameter** (GJR), a clear AIC gain. (iii) **The innovation distribution is worth 141 points** for one parameter, six times the asymmetry; skew adds 5.5 more. The rugarch EGARCH reports Akaike −4.3229 per observation, and $-4.3229 \times 5167 = -22336.4$ reproduces its row. The page's $\sigma_t$ comparison shows all these models tracing nearly the same volatility path: they disagree about tails and asymmetry, not about when volatility is high.

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## Links

- Neighbours: L08-06-arch-garch (GARCH(1,1), kurtosis (24)–(25), forecasts) · L08-08-stochastic-volatility (the right panel of the figure; a second shock in the variance).
- Unit: L08-00-map · L08-09-bayesian-garch (GARCH(1,1)-$t$ by MCMC) · L08-10-formulas · L08-11-pitfalls · L08-12-python.
- Class 1: L01-07 §2 (uncorrelated ≠ independent).
- Rules: rules/05 (ARCH/GARCH variants).
- **Practice (number and printed page only):** T 3.14 (p. 171). Related, from the ARCH/GARCH block: T 3.4 (p. 168, Student-$t$ likelihood). Solutions: `Listas/lista-L08-book-exercises.pdf`.

## Stochastic volatility · a second shock for the variance, its moments, and why it needs MCMC

**Claim. A stochastic-volatility (SV) model gives the log variance its own AR(1) shock**, $\ln \sigma_t^2 = \alpha_0 + \alpha_1 \ln \sigma_{t-1}^2 + v_t$, instead of GARCH's exact function of past squared returns. **With Gaussian shocks every moment is a lognormal moment**: $E(a_t^2) = e^{\mu_h + \sigma_h^2/2}$, kurtosis $3e^{\sigma_h^2} > 3$, and $\text{corr}(a_t^2, a_{t-i}^2) = (e^{\sigma_h^2 \alpha_1^i} - 1)/(3e^{\sigma_h^2} - 1)$, which decays at rate $\alpha_1$, like GARCH's at rate $\alpha_1 + \beta_1$. **The price is the likelihood**: $\sigma_t$ is not known given the past, so $f(R \mid \theta)$ is an $n$-dimensional integral over the volatility path. **Bayesian MCMC treats the path as unknowns** and samples it one $h_t$ at a time from (12.23), a density built from three factors. On Petrobras, SV and GARCH share the same persistence (0.984); SV needs thinner-tailed errors ($\nu \approx 12$ against 7) because its own volatility shock already supplies kurtosis. Prev: L08-07-garch-family-asymmetry-tails · Next: L08-09-bayesian-garch · Map: L08-00-map

**Notation.** Two sources, two meanings of $h_t$. **Tsay §12.7 writes $h_t$ for the variance $\sigma_t^2$; the site pages and the R package `stochvol` write $h_t$ for the log variance $\ln \sigma_t^2$.** This note keeps both, and says which at every use.

| Symbol | Meaning | Range |
|---|---|---|
| $r_t$, $a_t$ | return and its shock (mean removed); Tsay's notation | |
| $y_t$ | the site's (demeaned) return | |
| $\varepsilon_t$ | standardized return shock, iid $N(0,1)$ (or $t_\nu$ in SVt) | |
| $\sigma_t^2$ | conditional variance of $a_t$ given the volatility | $> 0$ |
| $\alpha_0, \alpha_1, \sigma_v^2$ | Tsay's log-variance AR(1): intercept, persistence, shock variance | $\lvert\alpha_1\rvert < 1$ |
| $v_t$ | log-variance shock, iid $N(0, \sigma_v^2)$, independent of $\varepsilon_t$ | |
| $\mu_h, \sigma_h^2$ | stationary mean and variance of $\ln \sigma_t^2$ | |
| $h_t$ (stochvol) | $\ln \sigma_t^2$; $\mu, \phi, \sigma$ its level, persistence and shock sd | $\lvert\phi\rvert < 1$, $\sigma > 0$ |
| $h_t$ (Tsay §12.7) | $\sigma_t^2$ itself | |
| $\rho$ | $\text{corr}(\varepsilon_t, u_t)$, the leverage parameter | $(-1, 1)$ |
| $\nu$ | Student-$t$ degrees of freedom | $> 2$ |
| $H = (h_1, \ldots, h_n)'$ | the volatility path, treated as unknowns | |
| $\theta$ | the static parameters | |

## 1 · U — what SV adds to GARCH

Tsay sorts volatility models into two families (T §3.2, p. 113). In the first, "an exact function" governs $\sigma_t^2$. That is GARCH: given $F_{t-1}$, $\sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2$ is a known number. In the second, "a stochastic equation" describes it. That is SV, where $\sigma_t^2$ has a shock of its own, unknown even after seeing every past return.

| | GARCH(1,1) | SV-AR(1) |
|---|---|---|
| volatility driven by | past observations: **observation-driven** | its own shock $v_t$: **parameter-driven** |
| shocks per observation | one ($\varepsilon_t$) | two ($\varepsilon_t$, $v_t$) |
| $\sigma_t^2$ given $F_{t-1}$ | known | random |
| positivity | needs $\alpha_0 > 0$, $\alpha_1, \beta_1 \geq 0$ | automatic (logs) |
| likelihood | product of one-step densities, exact | $n$-dimensional integral (§5) |
| estimation | MLE (L08-06) | quasi-likelihood via the Kalman filter, or MCMC (§6) |

"Adding the innovation $v_t$ substantially increases the flexibility of the model in describing the evolution of $\sigma_t^2$, but it also increases the difficulty in parameter estimation" (T, p. 153).

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## 2 · C — the model, in Tsay's form and in `stochvol`'s form (T §3.12, p. 153; site pages)

**Tsay (3.40)** with $m = 1$:

$$a_t = \sigma_t \varepsilon_t, \qquad (1 - \alpha_1 B) \ln \sigma_t^2 = \alpha_0 + v_t, \tag{1}$$

that is, $\ln \sigma_t^2 = \alpha_0 + \alpha_1 \ln \sigma_{t-1}^2 + v_t$, with $\varepsilon_t$ iid $N(0,1)$, $v_t$ iid $N(0, \sigma_v^2)$, the two independent, and $|\alpha_1| < 1$. The log keeps $\sigma_t^2 > 0$ with no constraint on the coefficients, as in EGARCH (L08-07).

**The site and `stochvol`** (sp500-gaussian-student-SV-models.html §5):

$$y_t = e^{h_t/2} \varepsilon_t, \qquad h_{t+1} = \mu + \phi(h_t - \mu) + \sigma u_t, \qquad \operatorname{corr}(\varepsilon_t, u_t) = \rho. \tag{2}$$

Here $h_t = \ln \sigma_t^2$, since $e^{h_t/2} = \sigma_t$. Matching (1) to (2): $\phi = \alpha_1$, $\sigma^2 = \sigma_v^2$, and $\alpha_0 = (1 - \phi)\mu$, so $\mu$ is the *mean* of the log variance (the "intercept versus mean" distinction of L01-07 (6)–(7)).

**The four models of `stochvol`**, as the page lists them:

| Model | $\varepsilon_t$ | $u_t$ | $\rho$ | $\nu$ |
|---|---|---|---|---|
| SV | $N(0,1)$ | $N(0,1)$ | 0 | — |
| SVt | $t_\nu(0,1)$ | $N(0,1)$ | 0 | unknown |
| SVl (leverage) | $N(0,1)$ | $N(0,1)$ | unknown | — |
| SVtl | $t_\nu(0,1)$ | $N(0,1)$ | unknown | unknown |

*Erratum.* The page's SVtl line reads "$\varepsilon_t \sim N(0,1)$", the SVl specification copied. The $t$ in SVtl, and the unknown $\nu$ in the same line, mean $\varepsilon_t \sim t_\nu$.

**Leverage.** With $\rho < 0$, a negative return shock $\varepsilon_t$ comes with a positive $u_t$ on average, so $h_{t+1}$ rises after bad news. This is SV's version of the asymmetry that GJR and EGARCH build into the variance equation (L08-07). The timing matters: $u_t$ moves $h_{t+1}$, the *next* variance, so the correlation is not a contemporaneous link between the return and its own variance.

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## 3 · C — the stationary law of the log variance (T p. 154)

$\ln \sigma_t^2$ in (1) is a Gaussian AR(1). By L01-07 (16), for $|\alpha_1| < 1$ it settles into

$$\ln \sigma_t^2 \sim N\Big(\frac{\alpha_0}{1-\alpha_1},\ \frac{\sigma_v^2}{1-\alpha_1^2}\Big) \equiv N(\mu_h, \sigma_h^2). \tag{3}$$

In `stochvol`'s notation (2), $\mu_h = \mu$ and $\sigma_h^2 = \sigma^2/(1-\phi^2)$. Its autocovariance at lag $i$ is, by L01-07 (11),

$$\operatorname{Cov}(\ln \sigma_t^2, \ln \sigma_{t-i}^2) = \alpha_1^i \sigma_h^2. \tag{4}$$

This is also the law `stochvol` uses to start the chain: $h_1 \sim N(\mu, \sigma^2/(1-\phi^2))$ (§8).

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## 4 · C — moments of $a_t$: kurtosis and the ACF of $a_t^2$ (T p. 154, derived)

The one tool is the **lognormal moment**. If $X \sim N(m, s^2)$ then $E(e^{kX}) = e^{km+k^2s^2/2}$ (the normal moment-generating function at $k$).

**Second moment.** $\varepsilon_t$ is independent of $\sigma_t$ and $E(\varepsilon_t^2) = 1$:

$$E(a_t^2) = E(\sigma_t^2)E(\varepsilon_t^2) = E(e^{\ln \sigma_t^2})$$

Apply the lognormal moment with $k = 1$, $m = \mu_h$, $s^2 = \sigma_h^2$ from (3):

$$E(a_t^2) = \exp\left(\mu_h + \sigma_h^2/2\right). \tag{5}$$

**Fourth moment.** $E(\varepsilon_t^4) = 3$ for a standard normal:

$$E(a_t^4) = E(\sigma_t^4)E(\varepsilon_t^4) = 3E(e^{2\ln \sigma_t^2})$$

Apply the lognormal moment with $k = 2$:

$$E(a_t^4) = 3\exp\left(2\mu_h + 2\sigma_h^2\right). \tag{6}$$

*Erratum, T p. 154.* The book prints "$E(a_t^4) = 3\exp(2\mu_h^2 + 2\sigma_h^2)$". The square on $\mu_h$ is a typo: by (6) the exponent is linear in $\mu_h$, and only the linear version makes the kurtosis below free of $\mu_h$, as a ratio of scale-free moments must be.

**Kurtosis.** Divide (6) by the square of (5):

$$\frac{E(a_t^4)}{[E(a_t^2)]^2} = \frac{3\exp(2\mu_h + 2\sigma_h^2)}{\exp(2\mu_h + \sigma_h^2)} = 3e^{\sigma_h^2} > 3. \tag{7}$$

**SV has fat tails even with Gaussian $\varepsilon_t$.** A normal with a random variance is a scale mixture of normals, and every such mixture has kurtosis above 3. The excess is $3(e^{\sigma_h^2} - 1)$: it grows with the variability of the log variance, not with its level.

With Student-$t$ shocks of kurtosis $\kappa_\varepsilon = E(\varepsilon_t^4)$, the same steps give kurtosis $\kappa_\varepsilon e^{\sigma_h^2}$. For a standardized $t_\nu$, $\kappa_\varepsilon = 3(\nu-2)/(\nu-4)$, $\nu > 4$ (from T §3.16, p. 166: excess $6/(\nu-4)$).

**Autocorrelation of $a_t^2$.** For $i \geq 1$, $\varepsilon_t$ and $\varepsilon_{t-i}$ are independent of each other and of the volatility path:

$$E(a_t^2 a_{t-i}^2) = E(\sigma_t^2 \sigma_{t-i}^2) E(\varepsilon_t^2) E(\varepsilon_{t-i}^2) = E(e^{\ln \sigma_t^2 + \ln \sigma_{t-i}^2}).$$

The sum of the two logs is normal with mean $2\mu_h$ and, by (3)–(4), variance $\sigma_h^2 + \sigma_h^2 + 2\alpha_1^i \sigma_h^2$:

$$E(a_t^2 a_{t-i}^2) = \exp\left(2\mu_h + \sigma_h^2(1 + \alpha_1^i)\right). \tag{8}$$

Subtract the product of means, $[E(a_t^2)]^2 = \exp(2\mu_h + \sigma_h^2)$ from (5):

$$\mathrm{Cov}(a_t^2, a_{t-i}^2) = \exp(2\mu_h + \sigma_h^2)(e^{\sigma_h^2 \alpha_1^i} - 1).$$

The variance is (6) minus the same product:

$$\mathrm{Var}(a_t^2) = 3\exp(2\mu_h + 2\sigma_h^2) - \exp(2\mu_h + \sigma_h^2) = \exp(2\mu_h + \sigma_h^2)(3e^{\sigma_h^2} - 1).$$

The ratio cancels $\exp(2\mu_h + \sigma_h^2)$:

$$\boxed{\mathrm{corr}(a_t^2, a_{t-i}^2) = \frac{e^{\sigma_h^2 \alpha_1^i} - 1}{3e^{\sigma_h^2} - 1}} \tag{9}$$

which is T's formula on p. 154. For small $\sigma_h^2 \alpha_1^i$, $e^x - 1 \approx x$, so (9) $\approx \sigma_h^2 \alpha_1^i / (3e^{\sigma_h^2} - 1)$: **geometric decay at rate** $\alpha_1$, the same shape as GARCH(1,1)'s $\rho_k = \rho_1(\alpha_1 + \beta_1)^{k-1}$ (L08-06). With $t$ shocks, replace the 3 in the denominator by $\kappa_\varepsilon$.

### 4.1 · Worked — Tsay Example 12.3 (monthly S&P 500 log returns in %, 1962–2009; T p. 642)

Posterior means: $\alpha_0 = 0.454$, $\alpha_1 = 0.837$, $\sigma_v^2 = 0.086$ (Tsay's $\ln h_t = \ln \sigma_t^2$). By (3):

$$\mu_h = \frac{0.454}{1 - 0.837} = \frac{0.454}{0.163} = 2.785, \qquad \sigma_h^2 = \frac{0.086}{1 - 0.837^2} = \frac{0.086}{1 - 0.700569} = \frac{0.086}{0.299431} = 0.2872.$$

By (5): $E(a_t^2) = \exp(2.785 + 0.2872/2) = \exp(2.785 + 0.1436) = \exp(2.929) = 18.71$ (%²), so the unconditional sd is $\sqrt{18.71} = 4.33\%$ a month. By (7): kurtosis $3e^{0.2872} = 3 \times 1.3327 = 3.998$. By (9) at $i = 1$: $(e^{0.2872 \times 0.837} - 1)/(3 \times 1.3327 - 1) = (e^{0.2404} - 1)/2.998 = 0.2717/2.998 = 0.0906$.

The Gaussian GARCH(1,1) of the same data (T (12.26), p. 640) has $h_t = 0.878 + 0.125a_{t-1}^2 + 0.837h_{t-1}$ (Tsay's $h_t = \sigma_t^2$): persistence $0.125 + 0.837 = 0.962$, unconditional variance $0.878/0.038 = 23.1$ (%²), sd 4.81%. The two models agree on the order of magnitude of volatility.

### 4.2 · Worked — Petrobras SV-t (GARCH11-SVar1.html §5)

Posterior means on standardized returns: $\mu = -0.419$, $\phi = 0.9842$, $\sigma = 0.1388$, $\nu = 12.13$. By (3) in `stochvol` form:

$$\sigma_h^2 = \frac{0.1388^2}{1 - 0.9842^2} = \frac{0.019253}{0.031326} = 0.6146.$$

With Gaussian $\varepsilon_t$, (7) gives kurtosis $3e^{0.6146} = 3 \times 1.849 = 5.547$, and (9) gives the lag-1 correlation

$$\frac{e^{0.6146 \times 0.9842} - 1}{3e^{0.6146} - 1} = \frac{e^{0.6049} - 1}{3 \times 1.8490 - 1} = \frac{0.8311}{4.5469} = 0.1828.$$

The GARCH(1,1)-$t$ fitted by ML to the same series ($\alpha_1 = 0.073478$, $\beta_1 = 0.910218$) has $\rho_1 = \alpha_1(1 - \alpha_1\beta_1 - \beta_1^2)/(1 - 2\alpha_1\beta_1 - \beta_1^2) = 0.2037$ (L08-06), and decays at $\alpha_1 + \beta_1 = 0.9837$, against SV's $\phi = 0.9842$.

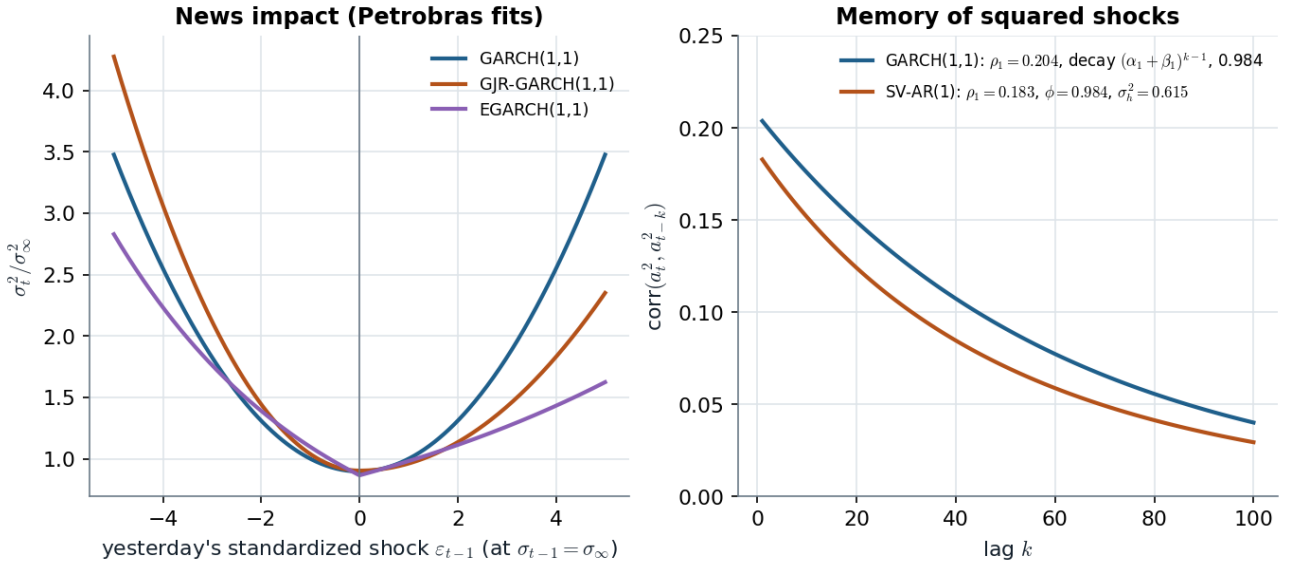



*Figure — right panel:* corr$(a_t^2, a_{t-k}^2)$ *for GARCH(1,1),* $\rho_1(\alpha_1 + \beta_1)^{k-1}$*, and for SV-AR(1), equation (9), at the Petrobras calibrations of GARCH11-SVar1.html. The left panel is read in L08-07-garch-family-asymmetry-tails.*

The two curves are almost parallel on the right: the persistence parameters (0.9837 and 0.9842) set the decay, and only the level of the first autocorrelation differs. **Squared-return autocorrelations alone cannot tell the two models apart.** What separates them is how the kurtosis is produced (§9) and whether the variance is known one step ahead (§1).

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## 5 · C — why the likelihood is hard (T §12.7, p. 637)

Write $R = (r_1, \ldots, r_n)'$ and the latent path $H = (h_1, \ldots, h_n)'$ (Tsay's $h_t = \sigma_t^2$). Given $H$, the returns are independent normals. Given $\theta$, $H$ is a Gaussian AR(1) in logs. The likelihood of $\theta$ alone integrates

the path out:

$$f(R \mid \theta) = \int f(R \mid H, \theta)\, f(H \mid \theta)\, dH, \tag{10}$$

"a mixture over the $n$-dimensional $H$ distribution" (T p. 637). For daily data $n$ is in the thousands, and there is no closed form, so (10) cannot be maximised as the GARCH likelihood is.

**Two ways out.**

1. **Quasi-likelihood by linearisation** (named in T p. 153; Harvey, Ruiz and Shephard 1994). Square and take logs of $a_t = \sigma_t \varepsilon_t$:

$$\ln a_t^2 = \ln \sigma_t^2 + \ln \varepsilon_t^2. \tag{11}$$

$\ln \varepsilon_t^2$ is a $\ln \chi_1^2$ variable with mean $\psi(1/2) + \ln 2 = -1.2704$ and variance $\pi^2/2 = 4.935$. So (11) plus (1) is a linear state-space model: an observed series equal to an AR(1) state plus non-Gaussian noise. The Kalman filter treats the noise as Gaussian and gives a quasi-likelihood. It is consistent but inefficient, because $\ln \chi_1^2$ is far from normal (long left tail), and it fails when some $a_t = 0$. T (3.41)'s long-memory SV uses the same decomposition (p. 154). [G] 2. **Bayesian MCMC** (Jacquier, Polson and Rossi 1994; T §12.7): put $H$ among the unknowns and sample $(\theta, H)$ jointly from the posterior (§6). The integral (10) is never computed: the sampler draws $H$ and averages over the draws.

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## 6 · C — the Gibbs sampler of Tsay §12.7.1 (pp. 637–640)

The model of T (12.20)–(12.21), in Tsay's notation ($h_t = \sigma_t^2$):

$$r_t = \beta_0 + \beta_1 x_{1t} + \cdots + \beta_p x_{pt} + a_t, \qquad a_t = \sqrt{h_t}\, \varepsilon_t, \tag{12}$$

$$\ln h_t = \alpha_0 + \alpha_1 \ln h_{t-1} + v_t, \qquad v_t \sim N(0, \sigma_v^2). \tag{13}$$

Priors are independent across the mean and volatility blocks, $p(\beta, \omega) = p(\beta)p(\omega)$ with $\omega = (\alpha_0, \alpha_1, \sigma_v^2)$. The sampler cycles through three conditionals: $f(\beta \mid R, X, H, \omega)$, $f(H \mid R, X, \beta, \omega)$, $f(\omega \mid R, X, \beta, H)$.

### 6.1 · $\beta \mid H$: weighted regression

Given $H$, (12) is a regression with known heteroskedasticity. Divide by $\sqrt{h_t}$:

$$r_{o,t} = x_{o,t}'\beta + \varepsilon_t, \qquad r_{o,t} = r_t/\sqrt{h_t}, \quad x_{o,t} = x_t/\sqrt{h_t}, \tag{14}$$

a homoskedastic regression with unit error variance. With prior $\beta \sim N(\beta_o, A_o)$ the posterior is $N(\beta_*, A_*)$ (T p. 638):

$$A_*^{-1} = \sum_t x_{o,t} x_{o,t}' + A_o^{-1}, \qquad \beta_* = A_*\Big( \sum_t x_{o,t} r_{o,t} + A_o^{-1}\beta_o \Big). \tag{15}$$

In Example 12.3 the mean equation is $r_t = \mu + a_t$ ($x_t = 1$), so (15) reads: precision $\sum_t 1/h_t + 1/4$ (prior $\mu \sim N(0,4)$), mean $\big(\sum_t r_t/h_t\big)\big/\big(\sum_t 1/h_t + 1/4\big)$, a precision-weighted average in which quiet months count more.

### 6.2 · $h_t \mid h_{t-1}, h_{t+1}$: three factors (equation (12.23))

Only three densities in the joint posterior contain $h_t$ ($1 < t < n$): $a_t$'s, which has variance $h_t$; the transition into $\ln h_t$ from $\ln h_{t-1}$; and the transition out of $\ln h_t$ to $\ln h_{t+1}$. So

$$f(h_t \mid \cdot) \propto f(a_t \mid h_t)\, f(h_t \mid h_{t-1}, \omega)\, f(h_{t+1} \mid h_t, \omega). \tag{16}$$

Take the factors one at a time, writing $x = \ln h_t$.

- $a_t \mid h_t \sim N(0, h_t)$: $\quad f(a_t \mid h_t) \propto h_t^{-1/2} \exp\{-a_t^2/(2h_t)\}$.
- $\ln h_t \mid \ln h_{t-1} \sim N(\alpha_0 + \alpha_1 \ln h_{t-1}, \sigma_v^2)$ by (13). That is a density in $x$. As a density in $h_t$ it picks up the Jacobian $dx/dh_t = h_t^{-1}$: $\quad f(h_t \mid h_{t-1}) \propto h_t^{-1} \exp\{-(x-a)^2/(2\sigma_v^2)\}$ with $a = \alpha_0 + \alpha_1 \ln h_{t-1}$.
- $\ln h_{t+1} \mid \ln h_t \sim N(\alpha_0 + \alpha_1 x, \sigma_v^2)$: $\quad f(h_{t+1} \mid h_t) \propto \exp\{-(\ln h_{t+1} - \alpha_0 - \alpha_1 x)^2/(2\sigma_v^2)\}$, with no Jacobian in $h_t$ (that one belongs to $h_{t+1}$).

Rewrite the third exponent as a square in $x$. Factor out $\alpha_1$:

$$(\ln h_{t+1} - \alpha_0 - \alpha_1 x)^2 = \alpha_1^2 (x - b)^2, \qquad b = \frac{\ln h_{t+1} - \alpha_0}{\alpha_1}.$$

The two exponents then sum to $-[(x-a)^2 A + (x-b)^2 C]/(2\sigma_v^2)$ with $A = 1$ and $C = \alpha_1^2$. **Combine the two squares** (T p. 638, a scalar version of Box and Tiao's Lemma 1). Expand:

$$(x-a)^2 A + (x-b)^2 C = (A+C)x^2 - 2(Aa + Cb)x + Aa^2 + Cb^2.$$

Complete the square with $c = (Aa + Cb)/(A + C)$:

$$= (A+C)(x-c)^2 + Aa^2 + Cb^2 - (A+C)c^2,$$

and the constant simplifies:

$$Aa^2 + Cb^2 - \frac{(Aa+Cb)^2}{A+C} = \frac{(A+C)(Aa^2+Cb^2) - (Aa+Cb)^2}{A+C} = \frac{AC(a-b)^2}{A+C}.$$

So

$$(x-a)^2 A + (x-b)^2 C = (x-c)^2 (A+C) + (a-b)^2 \frac{AC}{A+C}. \tag{17}$$

The last term does not contain $h_t$ and drops into the proportionality constant. Substituting $A = 1$, $a = \alpha_0 + \alpha_1 \ln h_{t-1}$, $C = \alpha_1^2$, $b = (\ln h_{t+1} - \alpha_0)/\alpha_1$ into $c$:

$$c = \frac{(\alpha_0 + \alpha_1 \ln h_{t-1}) + \alpha_1^2 \cdot \dfrac{\ln h_{t+1} - \alpha_0}{\alpha_1}}{1 + \alpha_1^2} = \frac{\alpha_0 + \alpha_1 \ln h_{t-1} + \alpha_1 \ln h_{t+1} - \alpha_1 \alpha_0}{1 + \alpha_1^2},$$

$$\mu_t \equiv c = \frac{\alpha_0(1-\alpha_1) + \alpha_1(\ln h_{t+1} + \ln h_{t-1})}{1+\alpha_1^2}, \qquad \sigma^2 \equiv \frac{\sigma_v^2}{A+C} = \frac{\sigma_v^2}{1+\alpha_1^2}. \tag{18}$$

Multiplying the three factors, $h_t^{-1/2} \cdot h_t^{-1} = h_t^{-1.5}$:

$$f(h_t \mid \cdot) \propto h_t^{-1.5} \exp\Big\{ -\frac{a_t^2}{2h_t} - \frac{(\ln h_t - \mu_t)^2}{2\sigma^2} \Big\}, \tag{19}$$

which is T (12.23). **Read (18):** the transition terms alone say $\ln h_t \sim N(\mu_t, \sigma^2)$, a smoother between its two neighbours (T p. 640 recovers it as the least-squares estimate of a "missing" $\ln h_t$ in an AR(1)). The factor $h_t^{-1/2} e^{-a_t^2/(2h_t)}$ then pulls $h_t$ toward $a_t^2$: a large return today raises today's volatility draw. (19) is not a standard density. Tsay draws from it by **Griddy Gibbs** (evaluate on a grid, sample from the discretised CDF); Jacquier, Polson and Rossi use a **Metropolis** step.

**End points** (T p. 639). For $t = 1$, fix $h_1$ (or predict $h_0$ backwards using time reversibility, $\ln h_0$ from a 2-step backcast); for $t = n$ use only $\ln h_n \sim N(\alpha_0 + \alpha_1 \ln h_{n-1}, \sigma_v^2)$, or forecast $h_{n+1}$.

### 6.3 · $\omega \mid H$: an AR(1) regression and an inverted $\chi^2$

Given $H$, (13) is a Gaussian AR(1) in $\ln h_t$, so L01-07 §5's regression applies. With $z_t = (1, \ln h_{t-1})'$, $\alpha = (\alpha_0, \alpha_1)'$ and prior $\alpha \sim N(\alpha_o, C_o)$ (T pp. 638–639):

$$C_*^{-1} = \sum_{t=2}^{n} \frac{z_t z_t'}{\sigma_v^2} + C_o^{-1}, \qquad \alpha_* = C_*\Big( \sum_{t=2}^{n} \frac{z_t \ln h_t}{\sigma_v^2} + C_o^{-1}\alpha_o \Big). \tag{20}$$

With $v_t = \ln h_t - \alpha_0 - \alpha_1 \ln h_{t-1}$ and prior $m\lambda/\sigma_v^2 \sim \chi^2_m$:

$$\frac{m\lambda + \sum_{t=2}^{n} v_t^2}{\sigma_v^2} \;\Big|\; H, \alpha \;\sim\; \chi^2_{m+n-1}. \tag{21}$$

> **Every block is a regression once $H$ is given.** The latent path turns one intractable likelihood into three conjugate updates plus a one-dimensional draw per $h_t$.

### 6.4 · Example 12.3, the run (T pp. 640–642)

Data: 575 monthly S&P 500 log returns (%), January 1962 to December 2009. Model (12.27): $r_t = \mu + a_t$, $\ln h_t = \alpha_0 + \alpha_1 \ln h_{t-1} + v_t$. Priors: $\mu \sim N(0, 4)$; $\alpha \sim N[(0, 0.6)', \mathrm{diag}(0.25, 0.04)]$; $10 \times 0.1/\sigma_v^2 \sim \chi^2_{10}$ ($m = 10$, $\lambda = 0.1$). Start: the GARCH(1,1) volatilities of (12.26). $h_t$ by Griddy Gibbs with 400 grid points. 2500 iterations, the first 500 discarded.

| | $\mu$ | $\alpha_0$ | $\alpha_1$ | $\sigma_v^2$ |
|---|---|---|---|---|
| posterior mean | 0.409 | 0.454 | 0.837 | 0.086 |
| posterior sd | 0.157 | 0.068 | 0.025 | 0.007 |

$\alpha_1 = 0.837$ confirms "strong serial dependence in the volatility series", and is smaller than Jacquier, Polson and Rossi's daily estimate (T p. 642). Its implied moments are §4.1's. *Minor inconsistency in Tsay:* the text says the posterior mean of $h_t$ is taken "over the 5000 iterations", while the run keeps 2000 and the captions of Figures 12.5–12.6 say 2000.

## 7 · C — the Student-$t$ question on the S&P 500 (sp500-gaussian-student-SV-models.html §1–4)

Daily S&P 500 log returns, May 2017 to May 2022, demeaned $(y_t)$.

**Model 1, iid** $N(0,\sigma^2)$**.** The MLE of $\sigma$ is $\hat\sigma_1 = \sqrt{\sum y_t^2/n} = 0.012729$.

**Model 2, iid** $\sigma t_\nu$ (not standardized: $y_t/\sigma \sim t_\nu$). The page profiles the log-likelihood $\sum_t \ln f_{t_\nu}(y_t/\sigma) - n\ln\sigma$ over a grid of 1000 values of $\sigma \in [0.001, 0.15]$, for $\nu \in \{1,\dots,30,35,\dots,100,200,\dots,1000\}$, and keeps the best pair: $\hat\nu = 2$, $\hat\sigma_2 = 0.005922$.

Reading it:

- $\hat\sigma_2$ is a **scale, not a standard deviation**. A $t_2$ has infinite variance, so $\hat\sigma_2 < \hat\sigma_1$ says nothing about which model sees less risk.
- $\hat\nu = 2$ means the iid model needs tails heavy enough to have **no finite variance**. That is the iid model's way of absorbing the 2020 COVID cluster: an iid law can only fit a cluster of large returns by making large returns likely every day.
- The page's quantile plot puts the empirical quantiles of the returns beside those of $N(0,\hat\sigma_1^2)$ and of $\hat\sigma_2 t_2$ at tail probabilities 0.01, 0.03, …, 0.99: the comparison to read is in the outer quantiles, where a Gaussian with the sample variance has the least mass.

**The SV reading.** Volatility clustering is serial dependence in the scale, not a property of the marginal. Once $h_t$ moves, each day's conditional law can be close to normal, and the marginal still has kurtosis $3e^{\sigma_h^2}$ by (7). That is why the page then fits SV, SVt, SVl and SVtl and compares the four volatility paths with $0.1|r_t|$.

## 8 · C — `stochvol`'s default priors (sp500 page §5.2; Kastner 2016)

$$\mu \sim N(0,100^2), \quad \frac{\phi+1}{2} \sim \text{Beta}(5,1.5), \quad \sigma^2 \sim \text{Gamma}(\tfrac12,\tfrac12), \quad h_1 \sim N\Big(\mu, \frac{\sigma^2}{1-\phi^2}\Big), \quad \frac{\rho+1}{2} \sim \text{Beta}(4,4), \quad \nu \sim \text{Ex} \tag{22}$$

What each says:

- $\mu$: flat on the scale of log variances.
- $\phi$: the Beta is on $(\phi+1)/2 \in (0,1)$, so $\phi \in (-1,1)$: **stationarity is imposed by the prior**. $E[(\phi+1)/2] = 5/6.5 = 0.769$, so $E(\phi) = 2(0.769) - 1 = 0.538$. The mode of the Beta is $(5-1)/(5+1.5-2) = 4/4.5 = 0.889$, so the prior mode of $\phi$ is 0.778. The prior leans toward persistence, and the data move it to 0.98.
- $\sigma^2$: `stochvol`'s `priorsigma` $= B_\sigma$ means $\sigma^2 \sim B_\sigma\chi_1^2 = \text{Gamma}(\frac12, \text{ rate } \frac{1}{2B_\sigma})$, equivalently $\sigma \sim |N(0,B_\sigma)|$. The default $B_\sigma = 1$ gives the page's $\text{Gamma}(1/2,1/2) = \chi_1^2$. A half-normal on $\sigma$ keeps mass near 0, which lets the data shrink toward constant volatility (Kastner's argument against an inverse-gamma prior here). [G]
- $h_1$: the stationary law (3).
- $\rho$: Beta(4,4) on $(\rho+1)/2$ is symmetric, so $E(\rho) = 0$. Its variance is $4\cdot 4/(8^2\cdot 9) = 0.0278$, so $\text{sd}(\rho) = 2\sqrt{0.0278} = 0.333$.
- $\nu$: exponential with rate 0.1, $E(\nu) = 10$.

**How `stochvol` samples** [G]: it uses the linearisation (11) with $\ln \varepsilon_t^2$ approximated by a finite mixture of normals (the auxiliary-mixture sampler), draws the whole path $H$ in one block, and alternates centred and non-centred parametrisations (ASIS, Kastner and Frühwirth-Schnatter 2014). This explains the warning in garch-sv-petrobras.html: the simple returns contain values "very close to zero", $\ln y_t^2 \to -\infty$, so the package adds an offset of $3.1 \times 10^{-6}$, and suggests demeaning first.

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## 9 · C — GARCH vs SV on Petrobras (GARCH11-SVar1.html)

Petrobras ADR, August 2000 to March 2021, log returns demeaned and divided by their sd (so $V = 1$ in sample). Three fits:

| | $\alpha_0$ / $\mu$ | $\alpha_1$ / $\phi$ | $\beta_1$ / $\sigma$ | $\nu$ | persistence |
|---|---|---|---|---|---|
| GARCH(1,1)-$t$, ML (fGarch) | 0.014364 | 0.073478 | 0.910218 | 7.026 | 0.9837 |
| GARCH(1,1)-$t$, Bayes (bayesGARCH), posterior means | 0.019209 | 0.085027 | 0.894024 | 6.984 | 0.9791 |
| SV-AR(1)-$t$ (`svtsample`), posterior means | $\mu = -0.419$ | $\phi = 0.9842$ | $\sigma = 0.1388$ | 12.13 | 0.9842 |

**The volatility paths** (page §5.1) are overlaid over the full sample and over the last 1000 days (the data are downloaded by the page and not on disk, so this note does not reproduce the plot). The structural difference to look for: GARCH's $\sigma_t^2$ jumps by $\alpha_1 a_{t-1}^2$ the day *after* a large return and then decays at $\alpha_1 + \beta_1$, while SV's posterior $h_t$ is a smoother of its two neighbours, (18), pulled toward $a_t^2$ on the *same* day, (19).

**Why $\nu$ differs, 7 against 12.** Both models must reproduce the same sample kurtosis of the returns. In SV the volatility shock alone supplies the factor $e^{\sigma_h^2} = 1.849$ of (7). With $\nu = 12.13$ the $t$ shocks supply $\kappa_\varepsilon = 3(12.13 - 2)/(12.13 - 4) = 3 \times 10.13/8.13 = 3.738$, and the total is $3.738 \times 1.849 = 6.91$. GARCH's $\alpha_1 a_{t-1}^2$ feedback also fattens tails, but only through $\alpha_1$ (L08-06: Gaussian excess $6\alpha_1^2/(1 - 2\alpha_1^2 - (\alpha_1 + \beta_1)^2) = 1.50$ at the ML values), so GARCH loads more of the kurtosis on $\varepsilon_t$: $\nu = 7$, and total kurtosis 11.87 by Tiao's formula (L08-07).

> **Fat tails can come from the shock or from the volatility.** SV splits them between two sources, GARCH mostly loads them on the shock. A fitted $\nu$ is not comparable across the two models.

**The page's residual check** (§5.2): returns divided by each model's median $\sigma_t$, with histograms and normal QQ plots. A well-specified model leaves standardized returns that look iid $t_\nu$ (or normal for SV, whose extra kurtosis lives in $h_t$).

*Errata in GARCH11-SVar1.html.* Section 3 is titled "standard GARCH(1,1) with Gaussian errors", but the call is `garchFit(..., cond.dist="std")`, a Student-$t$ fit (hence the "shape" 7.026). Its SV table prints a "p-value" of 1.9987 for $\mu$. The code computes $2(1 - \Phi(m/se))$ with $m/se = -3.22 < 0$;

the two-sided value is $2\Phi(-3.22) = 0.0013$. (A p-value for a posterior mean is a frequentist reading of a Bayesian summary anyway.)

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## 10 · G — beyond the class

- **Long-memory SV** (T (3.41), p. 154): $\sigma_t = \sigma \exp(u_t/2)$ with $(1-B)^d u_t = \eta_t$; by (11), $\ln a_t^2$ is a Gaussian long-memory signal plus non-Gaussian white noise. Median $\hat{d} \approx 0.38$ in the studies Tsay cites. Long memory is in L08-04-long-memory-arfima.
- **Particle filters** evaluate (10) by sequential Monte Carlo (named only: the syllabus lists SMC, no book on disk covers it).
- **Multivariate and factor SV**: Class 9 (`Aula/class-09/materials/sv-fsv-US-BR.html`).

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## Links

- Neighbours: L08-07-garch-family-asymmetry-tails (EGARCH's log variance, leverage, Tiao's kurtosis) · L08-09-bayesian-garch (Bayesian GARCH, the observation-driven counterpart).
- Unit: L08-00-map · L08-06-arch-garch (GARCH moments and the ACF of $a_t^2$) · L08-10-formulas · L08-11-pitfalls · L08-12-python (SV simulation: moments (5)–(9) checked by Monte Carlo).
- Class 1: L01-07 (the AR(1) equilibrium law (16) behind (3), the regression of §5 behind (20)) · L01-04 (the first conjugate update). Classes 3+4 (Gibbs sampling, Metropolis–Hastings) supply the MCMC machinery used in §6.
- Rules: rules/05 (state space, volatility).
- **Practice (number and printed page only).** No Class 8 practice item is specifically about SV. The volatility items are T 3.14 (p. 171) and W 12.4 (p. 397); the rest of the volatility list is in L08-06-arch-garch. Solutions: `Listas/lista-L08-book-exercises.pdf`.

## Bayesian GARCH · conjugate warm-ups, a GARCH-t posterior, and what $\Pr(\alpha_1 + \beta > 1 \mid \text{data})$ says

**Claim. The deck builds up to Bayesian GARCH through three models that are conjugate**: iid $N(0, \sigma^2)$ (one inverse-gamma update), iid $N(\mu, \sigma^2)$ and an AR(1) on log prices (Gibbs with normal and inverse-gamma conditionals). Their priors are **elicited from a training sample** (2001–2006) and the posterior uses 2007–2013. **GARCH(1,1) with Student-$t$ errors is not conjugate**: $h_t$ depends on $(\alpha_0, \alpha_1, \beta)$ through a recursion, so the posterior is sampled by Metropolis–Hastings steps (`bayesGARCH`, Ardia and Hoogerheide 2010). The payoff is **probability statements about functions of the parameters**: $\Pr(\alpha_1 + \beta > 1 \mid \text{data}) = 0.0034$ on Petrobras, and a posterior band for every $\sigma_t$. On the 2000–2021 Petrobras series the posterior means put persistence at 0.9791 against the MLE's 0.9837, because a posterior mean averages along a curved likelihood ridge that the mode does not see. Prev: L08-08-stochastic-volatility · Next: L08-10-formulas · Map: L08-00-map

**Notation.** The deck writes the GARCH variance as $h_t$ (Tsay's $\sigma_t^2$; *not* `stochvol`'s log variance of L08-08-stochastic-volatility) and the GARCH coefficient as $\beta$ (Tsay's $\beta_1$).

| Symbol | Meaning | Range |
|---|---|---|
| $p_t$, $y_t = \ln p_t$ | Petrobras price and log price | |
| $r_t = y_t - y_{t-1}$ | log return | |
| $n_0$, $n$ | training sample (2001–2006) 1506 days; posterior sample (2007–2013) 1760 days | |
| $m_{\tilde{r}}$, $V_{\tilde{r}}$, $m_{\tilde{r}^2}$ | training mean and variance of $r_t$, training mean of $r_t^2$ | |
| $(a, b, s)$, $(s_a, s_b)$ | training OLS estimates of the AR(1) on log prices and their standard errors | |
| $\text{IG}(a, b)$ | inverse gamma, density $\propto (\sigma^2)^{-(a+1)} e^{-b/\sigma^2}$, mean $b/(a-1)$ | $a, b > 0$ |
| $\mu, \sigma^2$ | mean and variance of the iid models | |
| $\alpha, \beta$ in M2 | intercept and slope of $y_t \mid y_{t-1}$ | |
| $\alpha_0, \alpha_1, \beta$ in M3 | GARCH(1,1) constant, ARCH and GARCH coefficients | $\alpha_0 > 0$, $\alpha_1, \beta \geq 0$ |
| $h_t$ | conditional variance of $r_t$ in M3 | $> 0$ |
| $\nu$, $\varrho = (\nu - 2)/\nu$ | Student-$t$ degrees of freedom and the scaling that makes $V(r_t \mid h_t) = h_t$ | $\nu > 2$ |
| $\lambda, \delta$ | rate and shift of the translated-exponential prior on $\nu$ | $\lambda > 0$, $\delta \geq 2$ |
| $\varpi_t$ | auxiliary mixing variable of the $t$ as a scale mixture of normals | $> 0$ |

## 1 · C — the data and the training-sample design (slides 1–8)

Petrobras log prices, 12/29/2000 to 12/31/2013 ($n = 3268$ days), and log returns, 01/02/2001 to 12/31/2013 (3267 days). The scatter of $y_{t-1}$ against $y_t$ (slide 3) lies on the 45-degree line: log prices behave like a random walk (L08-03-unit-roots-dickey-fuller).

**Split.** The first $n_0 = 1506$ days (2001–2006) are used only to build the priors. The last $n = 1760$ days (2007–2013) give the posterior. Training-sample summaries (slide 7):

$$(m_{\tilde{r}}, V_{\tilde{r}}) = (0.00112605,\ 0.0006609406), \qquad m_{\tilde{r}^2} = 0.0006617694,$$

and OLS of $y_t$ on $(1, y_{t-1})$ in the training sample:

$$(a, b, s) = (0.0005451564,\ 1.0003342482,\ 0.02569926), \qquad (s_a, s_b) = (0.0018491322,\ 0.0009933758).$$

> **A training sample turns a subjective prior into a reproducible one.** The prior is the posterior of an earlier period, inflated so that the later data dominate. Every prior below multiplies a training variance by 100 or uses an inverse gamma with only 5 "prior observations".

**Three conjugate models** (slide 6):

- **M0**: $r_1, \ldots, r_n$ iid $N(0, \sigma^2)$.
- **M1**: $r_1, \ldots, r_n$ iid $N(\mu, \sigma^2)$.
- **M2**: $y_t \mid y_{t-1} \sim N(\alpha + \beta y_{t-1}, \sigma^2)$, $t = 2, \ldots, n$, on log prices.

---

## 2 · C — M0: one inverse-gamma update, every step

**Prior** (slide 8): $\sigma^2 \sim \text{IG}(5,\ 4m_{\tilde{r}^2})$, that is

$$p(\sigma^2) \propto (\sigma^2)^{-(5+1)} \exp\Big\{ -\frac{4m_{\tilde{r}^2}}{\sigma^2} \Big\}. \tag{1}$$

Its mean is $b/(a-1)$:

$$E(\sigma^2) = \frac{4m_{\tilde{r}^2}}{5-1} = m_{\tilde{r}^2} = 0.0006617694,$$

so **the prior is centred at the training-sample mean square**, the training MLE of $\sigma^2$ under M0. Its variance is $b^2/[(a-1)^2(a-2)]$, so

$$\text{sd}(\sigma^2) = \frac{E(\sigma^2)}{\sqrt{a-2}} = \frac{0.0006617694}{\sqrt{3}} = 0.000382.$$

**Likelihood.** For iid $N(0, \sigma^2)$:

$$L(\sigma^2) = \prod_{t=1}^{n} (2\pi\sigma^2)^{-1/2} \exp\Big\{ -\frac{r_t^2}{2\sigma^2} \Big\} \propto (\sigma^2)^{-n/2} \exp\Big\{ -\frac{\sum_t r_t^2}{2\sigma^2} \Big\}. \tag{2}$$

Setting $d\ln L/d\sigma^2 = -n/(2\sigma^2) + \sum r_t^2/(2\sigma^4) = 0$ gives the MLE

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{t=1}^{n} r_t^2. \tag{3}$$

**Posterior.** Multiply (1) by (2) and collect powers of $\sigma^2$ and terms in $1/\sigma^2$:

$$p(\sigma^2 \mid r) \propto (\sigma^2)^{-(5+1)} e^{-4m_{\tilde{r}^2}/\sigma^2} \cdot (\sigma^2)^{-n/2} e^{-\sum r_t^2/(2\sigma^2)} = (\sigma^2)^{-(5+n/2+1)} \exp\Big\{ -\frac{4m_{\tilde{r}^2} + \sum r_t^2/2}{\sigma^2} \Big\},$$

which is the kernel of

$$\sigma^2 \mid r \sim \mathrm{IG}\Big(5 + \frac{n}{2},\ 4m_{\tilde{r}^2} + \frac{1}{2}\sum_{t=1}^{n} r_t^2\Big). \tag{4}$$

With $n = 1760$: shape $5 + 880 = 885$.

**The posterior mean as a weighted average.** By (4) and $\sum r_t^2 = n\hat{\sigma}^2$ from (3):

$$E(\sigma^2 \mid r) = \frac{4m_{\tilde{r}^2} + \frac{n}{2}\hat{\sigma}^2}{5 + \frac{n}{2} - 1} = \frac{4}{4 + \frac{n}{2}}\, m_{\tilde{r}^2} + \frac{\frac{n}{2}}{4 + \frac{n}{2}}\, \hat{\sigma}^2. \tag{5}$$

With $n = 1760$ the prior weight is $4/(4 + 880) = 4/884 = 0.0045$: **0.45% prior, 99.55% data**. The prior counts as $2(a - 1) = 8$ observations, against 1760. This is the inverse-gamma version of the Gamma–Poisson shrinkage $a/(n + a)$ of L01-04. Slide 9 plots prior, posterior and MLE: the posterior is a narrow spike near 0.0011, on the MLE, while the prior, centred at the 2001–2006 value 0.00066, is far wider. The 2007–2013 returns were more volatile than the training years, and with 0.45% weight the prior cannot hold the posterior back.

---

## 3 · C — M1: Gibbs with two conjugate conditionals (slides 8, 10–11)

**Priors** (independent):

$$\mu \sim N\Big(m_{\tilde{r}},\ \frac{100V_{\tilde{r}}}{n_0 - 1}\Big), \qquad \sigma^2 \sim \mathrm{IG}(5,\ 4V_{\tilde{r}}). \tag{6}$$

$V_{\tilde{r}}/(n_0 - 1)$ is about the squared standard error of the training mean, so the prior sd of $\mu$ is 10 such standard errors:

$$\tau_0^2 = \frac{100 \times 0.0006609406}{1505} = 4.3916 \times 10^{-5}, \qquad \tau_0 = 0.006627,$$

about six times the training mean 0.001126. $E(\sigma^2) = 4V_{\tilde{r}}/4 = V_{\tilde{r}}$, as in M0.

$\mu \mid \sigma^2, r$**.** As a function of $\mu$, the likelihood is $\exp\{-\sum(r_t - \mu)^2/(2\sigma^2)\} \propto \exp\{-n(\mu - \bar{r})^2/(2\sigma^2)\}$ (expand the square around $\bar{r}$; the remainder does not contain $\mu$). Times the normal prior, the exponent

is $-\frac{1}{2}[(\mu-\bar r)^2 n/\sigma^2 + (\mu - m_{\tilde r})^2/\tau_0^2]$. Completing the square as in L08-08 (17), with $A = n/\sigma^2$ and $C = 1/\tau_0^2$:

$$\mu \mid \sigma^2, r \sim N\Big(\frac{n\bar r/\sigma^2 + m_{\tilde r}/\tau_0^2}{n/\sigma^2 + 1/\tau_0^2},\ \frac{1}{n/\sigma^2 + 1/\tau_0^2}\Big). \tag{7}$$

The mean is **precision-weighted**: data precision $n/\sigma^2$ against prior precision $1/\tau_0^2$.

$\sigma^2 \mid \mu, r$**.** Repeat §2 with $r_t - \mu$ in place of $r_t$:

$$\sigma^2 \mid \mu, r \sim \mathrm{IG}\Big(5 + \frac{n}{2},\ 4V_{\tilde r} + \frac{1}{2}\sum_t (r_t - \mu)^2\Big). \tag{8}$$

**The sampler.** Start at $\sigma^{2(0)}$; for $j = 1, \ldots, M$ draw $\mu^{(j)}$ from (7) given $\sigma^{2(j-1)}$, then $\sigma^{2(j)}$ from (8) given $\mu^{(j)}$. Slide 10 shows 100 000 iterations: traces that look like noise and ACFs at zero from lag 1, i.e. practically independent draws (the two blocks barely interact: with $\bar r \approx 0$ small next to the spread of $r_t$, $\sum_t (r_t - \mu)^2$ is almost flat in $\mu$). Slide 11: the prior of $\mu$ spans about $\pm 0.02$, the posterior about $\pm 0.002$.

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## 4 · C — M2: the AR(1) on log prices, a Bayesian unit-root look (slides 8, 12–13)

**Priors:**

$$\alpha \sim N(a, 100 s_a^2), \qquad \beta \sim N(b, 100 s_b^2), \qquad \sigma^2 \sim \mathrm{IG}(5, 4s^2). \tag{9}$$

Numbers: $\mathrm{sd}(\alpha) = 10 \times 0.0018491 = 0.01849$; $\mathrm{sd}(\beta) = 10 \times 0.00099338 = 0.00993$; $E(\sigma^2) = s^2 = 0.02569926^2 = 0.00066045$.

**Conditionals.** Given $\sigma^2$, M2 is a normal linear regression of $y_t$ on $x_t = (1, y_{t-1})'$, with a normal prior $N(m_0, D_0)$, $m_0 = (a, b)'$, $D_0 = \mathrm{diag}(100 s_a^2, 100 s_b^2)$. The posterior has the same form as L08-08 (15) with $x_{o,t} = x_t/\sigma$:

$$(\alpha, \beta)' \mid \sigma^2, y \sim N\Big(D_*(\textstyle\sum_t x_t y_t/\sigma^2 + D_0^{-1} m_0),\ D_*\Big), \qquad D_*^{-1} = \textstyle\sum_t x_t x_t'/\sigma^2 + D_0^{-1}, \tag{10}$$

and, as in (8),

$$\sigma^2 \mid \alpha, \beta, y \sim \mathrm{IG}\Big(5 + \frac{n-1}{2},\ 4s^2 + \frac{1}{2}\sum_{t=2}^{n} (y_t - \alpha - \beta y_{t-1})^2\Big). \tag{11}$$

**The unit-root reading.** The training OLS slope is $b = 1.00033$, above 1, with $(b-1)/s_b = 0.000334/0.000993 = 0.34$: nowhere near a rejection of $\beta = 1$, and the Dickey–Fuller caveat of L08-03 applies to that $t$ anyway. The prior $N(1.00033, 0.00993^2)$ gives $\Pr(\beta > 1) = 0.513$. A Bayesian does not need the Dickey–Fuller distribution: the posterior of $\beta$ is a statement about $\beta$ given the data, whatever the sampling distribution of $\hat\beta$ would be (Sims's argument, named). Slide 13 shows the posterior of $\beta$ as a spike at about 1 inside the prior's 0.97–1.03 spread. Slide 12 shows the cost: the draws of $\alpha$ and of $\beta$ have ACFs still near 0.5 at lag 50, while those of $\sigma^2$ are uncorrelated. With a near-unit-root regressor $y_{t-1}$ whose mean is far from 0, $\alpha$ and $\beta$ are almost collinear ($\hat\alpha \approx \bar y(1 - \hat\beta)$), and a sampler that updates them one at a time crawls along that ridge.

## 5 · C — M3: GARCH(1,1) with Student-$t$ errors (slides 14–16)

**Model** (slide 14):

$$r_t \mid h_t \sim t_\nu(0, \varrho h_t), \qquad h_t = \alpha_0 + \alpha_1 r_{t-1}^2 + \beta h_{t-1}, \qquad \varrho = \frac{\nu-2}{\nu}, \tag{12}$$

with $\alpha_0 > 0$, $\alpha_1 \geq 0$, $\beta > 0$, and $h_0 = 0$ "for convenience" (so $r_0 = 0$, and $h_1 = \alpha_0$ starts the recursion).

**Why $\varrho$.** A $t_\nu(0, s^2)$ variable is $s$ times a standard $t_\nu$, whose variance is $\nu/(\nu-2)$. With $s^2 = \varrho h_t$:

$$V(r_t \mid h_t) = \varrho h_t \cdot \frac{\nu}{\nu-2} = \frac{\nu-2}{\nu}\, h_t \, \frac{\nu}{\nu-2} = h_t. \tag{13}$$

So $h_t$ is the conditional variance, exactly as in Tsay's standardized $t$ of T (3.7) (p. 121) with $\varepsilon_t = r_t/\sqrt{h_t}$.

**Likelihood.** The $t_\nu(0, s^2)$ density with $s^2 = \varrho h_t$ and $\nu\varrho = \nu - 2$:

$$f(r_t \mid h_t, \nu) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)\sqrt{(\nu-2)\pi h_t}} \left(1 + \frac{r_t^2}{(\nu-2)h_t}\right)^{-(\nu+1)/2}, \tag{14}$$

and the conditional likelihood is $L(\alpha_0, \alpha_1, \beta, \nu) = \prod_{t=1}^n f(r_t \mid h_t, \nu)$ with $h_t$ from the recursion in (12): the same object fGarch maximises (L08-07-garch-family-asymmetry-tails, L08-06-arch-garch).

**Priors** (slide 15; the `bayesGARCH` call on slide 16). With $\alpha = (\alpha_0, \alpha_1)'$:

$$\alpha \sim N_2(\mu_\alpha, \Sigma_\alpha)\, \mathbb{1}\{\alpha > 0\}, \qquad \beta \sim N(\mu_\beta, \Sigma_\beta)\, \mathbb{1}\{\beta > 0\}, \qquad p(\nu) = \lambda e^{-\lambda(\nu-\delta)} \mathbb{1}\{\nu > \delta\}. \tag{15}$$

The translated exponential has $E(\nu) = \delta + 1/\lambda$. The call used, `bayesGARCH(r, mu.alpha=c(0,0), Sigma.alpha=1000*diag(1,2), mu.beta=0, Sigma.beta=1000, lambda=0.01, delta=2)`, gives:

- normal priors with variance 1000 truncated to positive values: **flat on the region of interest** ($\alpha_1, \beta < 1$ while the prior sd is 31.6);
- $\nu - 2 \sim \text{Exp}(0.01)$: $E(\nu) = 2 + 1/0.01 = 102$, sd 100. Vague, and $\nu > 2$ keeps the variance finite. The deck notes that $\lambda = 100$, $\delta = 500$ forces $\nu \approx 500$: the **normal case**.

  **The prior does not impose $\alpha_1 + \beta < 1$.** That is what makes $\Pr(\alpha_1 + \beta > 1 \mid \text{data})$ a question the data answer (§7).

## 6 · C/G — why no conjugacy, and how `bayesGARCH` samples

In M0–M2 every conditional was normal or inverse gamma because the parameters entered the likelihood linearly (in the mean) or as a single scale. In (12), $h_t = \alpha_0 \sum_{j<t} \beta^j + \alpha_1 \sum_{j<t} \beta^j r_{t-1-j}^2$: $\beta$ **enters through powers, and every $h_t$ depends on all three coefficients**. No prior makes $p(\alpha_0, \alpha_1, \beta \mid r)$ a known family. Three routes:

1. **Random-walk Metropolis–Hastings** (C): propose $\theta^* = \theta^{(j-1)} + \text{noise}$, accept with probability $\min\{1,\ p(\theta^* \mid r)/p(\theta^{(j-1)} \mid r)\}$; the posterior is needed only up to its constant, i.e. (14) times (15), one pass of the recursion per evaluation. L08-12-python runs this sampler on simulated GARCH(1,1)-$t$ data.

```
for j in 1..M:
    θ* = θ(j-1) + N(0, c·Σ)                          # proposal
    if θ* violates α0>0, α1≥0, β>0, ν>δ: reject
    log A = log p(r|θ*) + log p(θ*) - log p(r|θ(j-1)) - log p(θ(j-1))
    accept θ* with probability min(1, exp(log A))
```

2. **Griddy Gibbs** (Bauwens and Lubrano 1998): draw each coefficient from its conditional evaluated on a grid, as Tsay does for $h_t$ in L08-08 §6.2.
3. **Ardia and Hoogerheide (2010)**, the method in `bayesGARCH` [G]. Write the $t$ as a **scale mixture of normals**:

$$r_t \mid \varpi_t, h_t \sim N(0, \varpi_t \varrho h_t), \qquad \varpi_t \sim \mathrm{IG}(\tfrac{\nu}{2}, \tfrac{\nu}{2}). \tag{16}$$

Integrating $\varpi_t$ out returns $t_\nu(0, \varrho h_t)$. Given $\varpi = (\varpi_1, \dots, \varpi_n)$, each $\varpi_t$ has a conjugate IG conditional, and $(\alpha, \beta)$ are drawn by MH with proposals from an approximating GLS regression on the ARMA form of GARCH, $r_t^2 = \alpha_0 + (\alpha_1 + \beta) r_{t-1}^2 + w_t - \beta w_{t-1}$ (L08-06-arch-garch; Nakatsuma 2000). $\nu$ is drawn by an optimised rejection step. This is T §12.7's device again: augment latent variables until each block is easy.

**The run** (slide 17): 10 000 burn-in draws discarded, 10 000 kept, one chain, on the returns $r_0$ of the first part of the data.

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## 7 · C — reading the output (slides 18–21)

**Traces and marginals** (slides 18–19). Posterior draws span roughly $\alpha_0 \in (1, 5) \times 10^{-5}$, $\alpha_1 \in (0.04, 0.16)$, $\beta \in (0.82, 0.94)$, $\nu \in (4, 12)$. The ACFs of the draws decay slowly and reach zero only after roughly 100–150 lags: MH chains on GARCH are sticky, so 10 000 draws carry far fewer effective draws.

**The persistence** (slide 20). Each draw gives $\alpha_1^{(j)} + \beta^{(j)}$, and the histogram of these sums is the posterior of the persistence. The deck reports

$$\Pr(\alpha_1 + \beta > 1 \mid \text{data}) = \frac{1}{M} \sum_{j=1}^{M} \mathbb{1}\{\alpha_1^{(j)} + \beta^{(j)} > 1\} = 0.0034, \tag{17}$$

34 of the 10 000 kept draws. Its Monte Carlo standard error is at least $\sqrt{0.0034 \times 0.9966 / 10\,000} = 0.00058$ (iid draws; more with autocorrelation).

> **This is a direct probability that the model is covariance stationary: 99.66%.** A frequentist test of $H_0 : \alpha_1 + \beta = 1$ gives a p-value, which is a probability about the data under $H_0$, not about the hypothesis (L01-08). And because the prior (15) was flat on both sides of the line, the 0.34% is the data's verdict, not the prior's.

**The volatility with uncertainty** (slide 21). For each draw $j$, run the recursion (12) to get $h_t^{(j)}$, $t = 1, \dots, n$; the 2.5%, 50% and 97.5% quantiles of $\{\sqrt{h_t^{(j)}}\}_j$ at each $t$ are a posterior band for $\sigma_t$.

GARCH11-SVar1.html §4 does exactly this (its loop `sig2s[,t] = draws[,1] + draws[,2]*ret[t-1]^2 + draws[,3]*sig2s[,t-1]`). Any function of the parameters (unconditional variance, half-life, a VaR) gets its posterior the same way: compute it draw by draw.

---

## 8 · C — Bayes vs ML on the same data (GARCH11-SVar1.html §3–4.1)

Petrobras 2000–2021, returns demeaned and standardized. `bayesGARCH` with the priors of §5, 40 000 iterations, the first 20 000 discarded; fGarch's ML with `cond.dist="std"`.

| | $\alpha_0$ | $\alpha_1$ | $\beta$ | $\nu$ |
|---|---|---|---|---|
| ML estimate (s.e.) | 0.014364 (0.003164) | 0.073478 (0.009216) | 0.910218 (0.011001) | 7.026 (0.622) |
| posterior mean (sd) | 0.019209 (0.003783) | 0.085027 (0.010200) | 0.894024 (0.012120) | 6.984 (0.601) |
| (mean − ML)/posterior sd | 1.28 | 1.13 | −1.34 | −0.07 |

Derived quantities, persistence first:

$$\text{Bayes: } 0.085027 + 0.894024 = 0.979051, \qquad \text{ML: } 0.073478 + 0.910218 = 0.983696.$$

Unconditional variance $\alpha_0/(1-\alpha_1-\beta)$:

$$\text{Bayes: } \frac{0.019209}{1-0.979051} = \frac{0.019209}{0.020949} = 0.9169, \qquad \text{ML: } \frac{0.014364}{0.016304} = 0.8810,$$

both near the sample variance 1 of standardized returns. Half-life $\ln 0.5/\ln(\alpha_1+\beta)$: Bayes $-0.69315/-0.021171 = 32.7$ days, ML $-0.69315/-0.016438 = 42.2$ days. (Plugging posterior means into a nonlinear function is not the posterior mean of the function; the draw-by-draw rule of §7 is the right one. The posterior mean of the persistence itself *is* 0.979051, by linearity.)

**Why the means move away from the MLE.** With a flat prior the MLE is the posterior **mode**. The GARCH likelihood has a **curved ridge**: the data pin down the unconditional variance $\alpha_0/(1-\alpha_1-\beta)$ and the persistence much better than each coefficient, so $\alpha_0$ and $1-\alpha_1-\beta$ trade off along a ridge, and the posterior is skewed along it. Averaging over a skewed posterior moves the mean off the mode: here toward a larger $\alpha_0$, a larger $\alpha_1$ and a smaller $\beta$, together about one posterior sd each, and toward lower persistence. $\nu$, whose posterior is nearly symmetric, barely moves (−0.07 sd). Page §4.1 overlays the two $\sigma_t$ paths (full sample and last 500 days). Expect them close: the fitted $\sigma_t$ depends on the coefficients mostly through the well-determined directions (persistence, unconditional variance), not along the ridge.

*A reading note.* The page's Bayes table labels the columns "t value" and "Pr(>|t|)", computing $m/\text{sd}$ and $2(1-\Phi(m/\text{sd}))$. Those are frequentist labels on posterior summaries; the Bayesian statement is a credible interval or a posterior probability such as (17).

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### 9 · U — what the Bayesian route buys here

| Question | ML | Bayes |
|---|---|---|
| Is the model covariance stationary? | test $\alpha_1 + \beta = 1$ (boundary, nonstandard) | $\Pr(\alpha_1 + \beta < 1 \mid r) = 0.9966$ |
| Uncertainty in $\sigma_t$, half-life, VaR | delta method, case by case | quantiles of draws, one rule for all |
| Prior information | none | training sample (M0–M2) |
| Small samples, parameters near a boundary | asymptotic normality fails | exact finite-sample posterior |
| Cost | one optimisation | MCMC, sticky chains, diagnostics |

The deck's references (slide 22), named: Bauwens and Lubrano (1998, Griddy Gibbs), Nakatsuma (2000, MH for ARMA-GARCH), Vrontos, Dellaportas and Politis (2000, full Bayesian GARCH and EGARCH), Asai (2006, comparison of MCMC methods), Ardia and Hoogerheide (2010, `bayesGARCH`), Virbickaite, Ausín and Galeano (2015, survey).

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## Links

- Neighbours: L08-08-stochastic-volatility (SV by MCMC: the same latent-variable logic) · L08-10-formulas.
- Unit: L08-00-map · L08-06-arch-garch (GARCH moments, the ARMA form behind the GLS proposal) · L08-07-garch-family-asymmetry-tails (Student-$t$ likelihood, fGarch's output) · L08-03-unit-roots-dickey-fuller (the unit root behind M2) · L08-11-pitfalls · L08-12-python (conjugate M0 update and random-walk MH for GARCH(1,1)-$t$, with asserts).
- Class 1: L01-04 (the first conjugate update and the shrinkage weight) · L01-07 (the AR(1) regression behind M2) · L01-08 (a p-value is not a posterior probability). Classes 3+4 (conjugate analysis, Gibbs sampling, Metropolis–Hastings) supply the machinery used here.
- Rules: rules/05.
- **Practice (number and printed page only).** No Class 8 practice item is specifically Bayesian. The volatility items are T 3.14 (p. 171) and W 12.4 (p. 397); T 3.3 and 3.4 (p. 168) ask for the conditional likelihoods that §5's (14) writes down for the $t$ case. Solutions: `Listas/lista-L08-book-exercises.pdf`.

# Class 8 · Formulas — the exam specification, every result, every number

Prev: L08-09-bayesian-garch · Next: L08-11-pitfalls · Map: L08-00-map

Equations are cited by the **section note's own numbering**: "L08-06 (25)" is equation (25) of L08-06-arch-garch. Every number in §2 is asserted in L08-12-python (script `L08_python_checks.py`) or in `figures/make_figures.py`.

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## 0 · The exam specification, mapped

### 0.1 · The site's description of Class 8 (`Aula/COURSE.md`)

"Introduction to univariate time series econometrics", in five items:

| # | Site item (verbatim) | Where it is settled |
|---|---|---|
| i | "Autoregressive moving average models" | L08-01-ar2-arp-cycles-identification (AR(2), AR(p), cycles, PACF, AIC/BIC); L08-02-forecasting-ma-arma (forecasts, MA, ARMA, identification) |
| ii | "Unit root econometrics" | L08-03-unit-roots-dickey-fuller (random walk, drift, trend-stationarity, ARIMA, DF/ADF, spurious regression); L08-04-long-memory-arfima (fractional integration) |
| iii | "Seasonal models" | L08-05-seasonal-airline-model |
| iv | "ARCH/GARCH and related models" | L08-06-arch-garch (ARCH, GARCH, IGARCH); L08-07-garch-family-asymmetry-tails (GJR, TGARCH, APARCH, EGARCH, GARCH-M, non-Gaussian errors); L08-09-bayesian-garch (Bayesian GARCH) |
| v | "Stochastic volatility models" | L08-08-stochastic-volatility |

### 0.2 · The goals in `Map/class-plan.md`, item by item

| Goal | Note and section |
|---|---|
| AR, MA and ARMA: identification | L08-01 §8 (PACF), §9 (AIC/BIC); L08-02 §5 (MA ACF), §8 (identification table) |
| … estimation | L08-01 §10 (conditional LS); L08-02 §6 (conditional vs exact likelihood); L01-07 §5 (AR(1) OLS = MLE) |

| Goal | Note and section |
|---|---|
| … $h$-step forecasts | L08-02 §2–§3, §7, §9 |
| business cycles from complex AR roots | L08-01 §4–§6 |
| random walk, drift, unit-root tests | L08-03 §1–§7 |
| long memory | L08-04 |
| seasonal (airline) models | L08-05 §3–§6 |
| ARCH/GARCH and variants | L08-06; L08-07 |
| stochastic volatility (Gaussian and Student-$t$) | L08-08 §2–§8 |
| GARCH(1,1) vs SV-AR(1) | L08-08 §4.2, §9; figure `L08-news-impact-sv-memory` (right panel) |
| Bayesian GARCH | L08-09 |

**Final exam** (30%, no code; `Map/class-plan.md`): derivations of AR/MA/ARMA moments, forecast means and variances, ARCH/GARCH moments and forecasts, conjugate updates, and the reading of printed `arima`/`garchFit`/`adf.test`/`bayesGARCH` output.

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## 1 · Every definition and result

### 1.1 · AR(2), AR(p), PACF, criteria → L08-01-ar2-arp-cycles-identification

| Result | Statement | Where |
|---|---|---|
| AR(2) mean | $\mu = \phi_0/(1-\phi_1-\phi_2)$, if $\phi_1+\phi_2 \neq 1$ | (2) |
| Moment equations / Yule–Walker | $\gamma_\ell = \phi_1\gamma_{\ell-1} + \phi_2\gamma_{\ell-2}$, $\rho_\ell = \phi_1\rho_{\ell-1} + \phi_2\rho_{\ell-2}$ $(\ell > 0)$ (T 2.13) | (4), (6) |
| First autocorrelations | $\rho_1 = \phi_1/(1-\phi_2)$; $\rho_2 = \phi_1^2/(1-\phi_2) + \phi_2$ | (7), (8) |
| Variance | $\gamma_0 = \dfrac{(1-\phi_2)\sigma^2}{(1+\phi_2)[(1-\phi_2)^2 - \phi_1^2]}$ | (9) |
| Characteristic equation | $1-\phi_1 x - \phi_2 x^2 = 0$; stationary iff both solutions have $\lvert x\rvert > 1$ (T 2.14) | (10)–(11) |
| Triangle | $\phi_2 < 1-\phi_1$, $\phi_2 < 1+\phi_1$, $\phi_2 > -1$ | (15)–(18) |
| Complex roots, cycle | complex iff $\phi_1^2 + 4\phi_2 < 0$; $k = 2\pi/\cos^{-1}[\phi_1/(2\sqrt{-\phi_2})]$ | (13)–(14) |
| AR(p) | stationary iff all solutions of $1-\phi_1 x - \cdots - \phi_p x^p = 0$ exceed 1 in modulus | (21)–(22) |
| PACF | $\hat\phi_{\ell,\ell}$ from the AR($\ell$) regression; for AR(p): $\hat\phi_{p,p} \to \phi_p$, $\hat\phi_{\ell,\ell} \to 0$ and $V \approx 1/T$ for $\ell > p$ | (23), §8 |
| AIC, BIC | $\text{AIC} = -\frac{2}{T}\ln L + \frac{2}{T}k$; Gaussian AR($\ell$): $\ln\tilde\sigma_\ell^2 + 2\ell/T$ and $\ln\tilde\sigma_\ell^2 + \ell\ln T/T$ (T 2.16) | (24)–(25) |

### 1.2 · Forecasts, MA, ARMA → L08-02-forecasting-ma-arma

| Result | Statement | Where |
|---|---|---|
| Optimal forecast | $\hat{r}_h(\ell) = E(r_{h+\ell} \mid F_h)$ minimises the conditional MSE | (1)–(2) |
| AR recursion | $\hat{r}_h(\ell) = \phi_0 + \sum \phi_i \hat{r}_h(\ell - i)$, $\hat{r}_h(j) = r_{h+j}$ for $j \le 0$; $e_h(2) = a_{h+2} + \phi_1 a_{h+1}$ | (3)–(6) |
| AR(2) forecasts | $E(y_{n+2}) = (\phi_1^2 + \phi_2) y_n + \phi_1 \phi_2 y_{n-1}$; $V_2 = (1 + \phi_1^2)\sigma^2$; $V_3 = [1 + \phi_1^2 + (\phi_1^2 + \phi_2)^2]\sigma^2$ | (7)–(10) |
| MA(q) | $r_t = c_0 + a_t - \sum \theta_i a_{t-i}$ (Tsay's sign); $E = c_0$, $V = (1 + \sum \theta_i^2)\sigma^2$; always stationary | (15)–(16) |
| MA ACF | MA(1): $\rho_1 = -\theta_1/(1 + \theta_1^2)$, $\lvert\rho_1\rvert \le \frac{1}{2}$; MA(2): $\rho_1 = \frac{-\theta_1 + \theta_1\theta_2}{1 + \theta_1^2 + \theta_2^2}$, $\rho_2 = \frac{-\theta_2}{1 + \theta_1^2 + \theta_2^2}$; zero beyond $q$ | (17)–(18) |
| Invertibility | MA(1) invertible iff $\lvert\theta_1\rvert < 1$ ($\theta$ and $1/\theta$ give the same ACF) | (19) |
| MA forecasts | MA(1): $\hat{r}_h(1) = c_0 - \theta_1 a_h$, $\hat{r}_h(\ell) = c_0$ for $\ell \ge 2$ | (20) |
| ARMA(1,1) | $\gamma_0 = \frac{1 - 2\phi_1\theta_1 + \theta_1^2}{1 - \phi_1^2}\sigma^2$; $\rho_1 = \frac{(\phi_1 - \theta_1)(1 - \phi_1\theta_1)}{1 - 2\phi_1\theta_1 + \theta_1^2}$; $\rho_\ell = \phi_1 \rho_{\ell-1}$, $\ell \ge 2$ | (21)–(25) |
| Forecast-error variance | $V[e_h(\ell)] = (1 + \psi_1^2 + \cdots + \psi_{\ell-1}^2)\sigma^2 \to \gamma_0$ (T 2.34) | (26)–(28) |
| Identification | AR(p): PACF cuts off at $p$; MA(q): ACF cuts off at $q$; ARMA: neither (EACF) | §8 |

### 1.3 · Unit roots → L08-03-unit-roots-dickey-fuller

| Result | Statement | Where |
|---|---|---|
| Random walk | $\hat{p}_h(\ell) = p_h$; $e_h(\ell) = \sum_{j=1}^{\ell} a_{h+j}$; $V = \ell\sigma^2$; $\psi_i = 1$ | (1)–(6) |
| RW with drift | $p_t = p_0 + \mu t + \sum a_i$: mean $p_0 + \mu t$, sd $\sigma\sqrt{t}$; crossover $t^* = (\sigma/\mu)^2$ | (7)–(10) |
| Trend-stationary | $p_t = \beta_0 + \beta_1 t + r_t$: mean $\beta_0 + \beta_1 t$, variance $V(r_t)$ constant | (11) |
| Constant term | MA: the mean; stationary AR/ARMA: $\phi_0 = \mu(1 - \sum \phi_i)$; RW with drift: the slope | §3 |

| Result | Statement | Where |
|---|---|---|
| ARIMA(p,1,q) | $(1-B)y_t$ is a stationary invertible ARMA(p,q) | (12)–(13) |
| DF | $p_t = \phi_1 p_{t-1} + e_t$ (or with $\phi_0$); $H_0 : \phi_1 = 1$ vs $< 1$; $t = \sum p_{t-1}e_t / (\hat{\sigma}_e \sqrt{\sum p_{t-1}^2})$, non-normal law | (14)–(21) |
| ADF | $\Delta x_t = c_t + \beta x_{t-1} + \sum \tilde{\phi}_i \Delta x_{t-i} + e_t$; AR(3): $\beta = \phi_1 + \phi_2 + \phi_3 - 1$, $\tilde{\phi}_1 = -(\phi_2 + \phi_3)$, $\tilde{\phi}_2 = -\phi_3$; unit root iff $\sum \phi_i = 1$ | (22)–(29) |
| Critical values (W Tables 18.2–18.3) | no trend: $-3.43$, $-3.12$, $-2.86$, $-2.57$ (1, 2.5, 5, 10%); trend: $-3.96$, $-3.66$, $-3.41$, $-3.12$ | §5.5 |
| Spurious regression | regressing one independent RW on another: residual is a RW, $t$ diverges, $R^2$ random | (30) |

### 1.4 · Long memory → L08-04-long-memory-arfima

| Result | Statement | Where |
|---|---|---|
| Fractional difference | $(1-B)^d x_t = a_t$, $-\frac{1}{2} < d < \frac{1}{2}$; $(1-B)^d = 1 - dB - \frac{d(1-d)}{2}B^2 - \ldots$ | (1)–(3) |
| Representations | $\psi_k = \frac{(k+d-1)!}{k!(d-1)!}$ (stationary if $d < \frac{1}{2}$); $\pi_k = \frac{(k-d-1)!}{k!(-d-1)!}$ (invertible if $d > -\frac{1}{2}$) | (4)–(6) |
| ACF | $\rho_k = \prod_{j=1}^{k} \frac{j-1+d}{j-d}$; $\rho_1 = \frac{d}{1-d}$; $\rho_k \sim \frac{(-d)!}{(d-1)!} k^{2d-1}$ | (7)–(11) |
| PACF, spectrum | $\phi_{k,k} = d/(k-d)$; $f(\omega) \sim \omega^{-2d}$ as $\omega \to 0$ | §4 |

### 1.5 · Seasonal models → L08-05-seasonal-airline-model

| Result | Statement | Where |
|---|---|---|
| Seasonal difference | $\Delta_s x_t = (1-B^s)x_t$; $\Delta_s \Delta x_t = x_t - x_{t-1} - x_{t-s} + x_{t-s-1}$ | (1)–(3) |
| Airline model | $(1-B^s)(1-B)x_t = (1-\theta B)(1-\Theta B^s)a_t$ (T 2.41) | (4)–(5) |
| Its autocovariances | $\gamma_0 = (1+\theta^2)(1+\Theta^2)\sigma^2$, $\gamma_1 = -\theta(1+\Theta^2)\sigma^2$, $\gamma_s = -\Theta(1+\theta^2)\sigma^2$, $\gamma_{s\pm1} = \theta\Theta\sigma^2$ | (6)–(10) |

| Result | Statement | Where |
|---|---|---|
| ACF | $\rho_1 = -\theta/(1+\theta^2)$, $\rho_s = -\Theta/(1+\Theta^2)$, $\rho_{s\pm1} = \rho_1\rho_s$, all others 0 | (11)–(12) |
| Non-multiplicative | $w_t = (1-\theta B - \Theta B^s)a_t$: $\rho_{s+1} = 0$ | (13)–(14) |
| SARIMA | $\phi_P(B^s)\phi(B)\Delta_s^D\Delta^d x_t = \theta_0 + \theta_Q(B^s)\theta(B)a_t$; ARIMA(1,1,1)×(1,1,1)$_{12}$ = constrained ARMA(26,13) | (15)–(18) |
| Deterministic seasonality | $\Theta = 1$: seasonal dummies (W §10.5e) | (19) |

### 1.6 · ARCH and GARCH → L08-06-arch-garch

| Result | Statement | Where |
|---|---|---|
| Structure | $r_t = \mu_t + a_t$, $\sigma_t^2 = V(a_t \mid F_{t-1})$ | (1)–(3) |
| ARCH test | Engle LM: regress $a_t^2$ on $m$ lags, $F$; McLeod–Li: $Q(m)$ on $a_t^2$ | (4)–(5) |
| ARCH(m) | $a_t = \sigma_t\varepsilon_t$, $\sigma_t^2 = \alpha_0 + \sum\alpha_i a_{t-i}^2$ (T 3.5) | (6) |
| ARCH(1) moments | $Ea_t = 0$; $V = \alpha_0/(1-\alpha_1)$; $m_4 = \frac{3\alpha_0^2(1+\alpha_1)}{(1-\alpha_1)(1-3\alpha_1^2)}$; kurtosis $3\frac{1-\alpha_1^2}{1-3\alpha_1^2} > 3$, needs $\alpha_1^2 < \frac{1}{3}$ | (7)–(12) |
| Likelihoods | Gaussian $-\frac{1}{2}\sum[\ln\sigma_t^2 + a_t^2/\sigma_t^2]$; standardized $t$ density (T 3.7) and (3.8) | (14)–(17) |
| GARCH(m,s) | $\sigma_t^2 = \alpha_0 + \sum\alpha_i a_{t-i}^2 + \sum\beta_j\sigma_{t-j}^2$; ARMA form for $a_t^2$ with $\eta_t = a_t^2 - \sigma_t^2$ (T 3.15) | (20)–(21) |
| Unconditional variance | $\sigma_\infty^2 = \alpha_0/[1 - \sum(\alpha_i + \beta_i)]$ | (22) |
| GARCH(1,1) kurtosis | $K_a^{(g)} = \dfrac{6\alpha_1^2}{1 - 2\alpha_1^2 - (\alpha_1+\beta_1)^2}$ (Gaussian $\varepsilon$) | (24)–(25) |
| Forecasts | $\sigma_h^2(\ell) = \alpha_0 + (\alpha_1+\beta_1)\sigma_h^2(\ell-1)$; $= \sigma_\infty^2 + (\alpha_1+\beta_1)^{\ell-1}[\sigma_h^2(1) - \sigma_\infty^2]$ (T 3.17) | (26)–(28) |
| Half-life | $\ln 0.5/\ln(\alpha_1+\beta_1)$ | (29) |
| IGARCH | $\alpha_1+\beta_1 = 1$: $\sigma_h^2(\ell) = \sigma_h^2(1) + (\ell-1)\alpha_0$ (T 3.22); $\alpha_0 = 0$: RiskMetrics exponential smoothing | (31)–(33) |
| ARCH in a regression | $u_t^2 = \alpha_0 + \alpha_1 u_{t-1}^2 + v_t$ (W 12.50): OLS still consistent, WLS/ML more efficient | (34) |

### 1.7 · Asymmetry and tails → L08-07-garch-family-asymmetry-tails

| Result | Statement | Where |
|---|---|---|
| News impact | $\sigma_t^2$ as a function of $a_{t-1}$ at $\sigma_{t-1} = \sigma_\infty$ | (1) |
| GJR/TGARCH | $\sigma_t^2 = \alpha_0 + (\alpha_1 + \gamma_1 N_{t-1})a_{t-1}^2 + \beta_1 \sigma_{t-1}^2$, $N = 1\{a < 0\}$ (T 3.34) | (2) |
| APARCH | $\sigma_t^\delta = \omega + \alpha(\lvert a_{t-1}\rvert - \gamma a_{t-1})^\delta + \beta\sigma_{t-1}^\delta$; what `fGarch leverage=TRUE` fits | (4)–(5) |
| $\delta = 2$ mapping | good news $\alpha(1-\gamma)^2$, bad news $\alpha(1+\gamma)^2$, indicator coefficient $4\alpha\gamma$; $\sigma_\infty^2 = \omega/[1-\alpha(1+\gamma^2)-\beta]$ | (6)–(7) |
| EGARCH | $g(\varepsilon) = \theta\varepsilon + \gamma[\lvert\varepsilon\rvert - E\lvert\varepsilon\rvert]$, $E\lvert\varepsilon\rvert = \sqrt{2/\pi}$ (normal); $\ln\sigma_t^2$ ARMA in $g$ (T 3.24–3.25); bad/good ratio $e^{-2\alpha c}$ in rugarch's form | (8)–(14) |
| GARCH-M | $r_t = \mu + c\sigma_t^2 + a_t$ (T 3.23) | (15) |
| Tiao's kurtosis | $K_a = \dfrac{K + K_a^{(g)} + \frac{5}{6}KK_a^{(g)}}{1 - \frac{1}{6}KK_a^{(g)}}$; $t_\nu$: $K = \frac{6}{\nu-4}$, $K_a = \frac{6+(\nu+1)K_a^{(g)}}{\nu-4-K_a^{(g)}}$ | (16)–(17) |

### 1.8 · Stochastic volatility → L08-08-stochastic-volatility

| Result | Statement | Where |
|---|---|---|
| SV-AR(1) | $a_t = \sigma_t\varepsilon_t$, $\ln\sigma_t^2 = \alpha_0 + \alpha_1 \ln\sigma_{t-1}^2 + v_t$ (T 3.40); `stochvol`: $y_t = e^{h_t/2}\varepsilon_t$, $h_{t+1} = \mu + \phi(h_t - \mu) + \sigma u_t$ | (1)–(2) |
| Log-variance law | $N(\mu_h, \sigma_h^2)$, $\mu_h = \alpha_0/(1-\alpha_1)$, $\sigma_h^2 = \sigma_v^2/(1-\alpha_1^2)$ | (3)–(4) |
| Moments | $Ea^2 = e^{\mu_h + \sigma_h^2/2}$; $Ea^4 = 3e^{2\mu_h + 2\sigma_h^2}$; kurtosis $3e^{\sigma_h^2}$ | (5)–(7) |
| ACF of $a^2$ | $\dfrac{e^{\sigma_h^2\alpha_1^i} - 1}{3e^{\sigma_h^2} - 1}$ | (8)–(9) |
| Likelihood | $f(R \mid \theta) = \int f(R \mid H) f(H \mid \theta)\, dH$, $n$-dimensional | (10) |
| Linearisation | $\ln a_t^2 = \ln\sigma_t^2 + \ln\varepsilon_t^2$, $E\ln\varepsilon^2 = -1.2704$, $V = \pi^2/2$ | (11) |

| Result | Statement | Where |
|---|---|---|
| Gibbs | $\beta \mid H$ weighted regression; $h_t \mid h_{t\pm1}$ (T 12.23) with $\mu_t = \frac{\alpha_0(1-\alpha_1)+\alpha_1(\ln h_{t+1}+\ln h_{t-1})}{1+\alpha_1^2}$, variance $\frac{\sigma_v^2}{1+\alpha_1^2}$; $\alpha \mid H$ AR regression; $\sigma_v^2 \mid H$ inverted $\chi^2$ | (12)–(21) |

### 1.9 · Bayesian GARCH → L08-09-bayesian-garch

| Result | Statement | Where |
|---|---|---|
| M0 update | $\sigma^2 \sim IG(a,b)$, $r_i \sim N(0,\sigma^2) \Rightarrow IG(a+n/2,\ b+\sum r_i^2/2)$; posterior mean = weighted average with prior weight $(a-1)/(a-1+n/2)$ | (1)–(5) |
| M1 Gibbs | $\mu \mid \sigma^2$ normal, precision-weighted; $\sigma^2 \mid \mu$ inverse gamma | (6)–(8) |
| M2 | Bayesian regression of $y_t$ on $y_{t-1}$: normal and inverse-gamma conditionals | (9)–(11) |
| M3 | $r_t \sim t_\nu(0,\varrho h_t)$, $\varrho = (\nu-2)/\nu$ so $V(r_t \mid h_t) = h_t$; truncated-normal priors on $(\alpha,\beta)$, $p(\nu) = \lambda e^{-\lambda(\nu-\delta)}1\{\nu > \delta\}$ | (12)–(15) |
| Scale mixture | $r_t \mid \varpi_t \sim N(0, \varpi_t \varrho h_t)$, $\varpi_t \sim IG(\nu/2, \nu/2)$ | (16) |
| Stationarity, Bayesian | $\Pr(\alpha_1+\beta>1 \mid \text{data})$ read off the draws | (17) |

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## 2 · Every number and worked case

### 2.1 · AR(2) and AR(p) (L08-01)

| Case | Numbers |
|---|---|
| $\phi = (1.75, -0.80)$ | complex (disc $-0.1375$), solution modulus 1.118, $k = 30.09$; $\rho_1 = 0.9722$, $\rho_2 = 0.9014$, min $\rho_{15} = -0.1866$; $\gamma_0 = 50.704\sigma^2$, sd 7.1207 |
| $\phi = (0.90, 0.05)$ | real solutions 1.0499 and $-19.05$; $\rho_1 = 0.9474$; $\gamma_0 = 9.7812$, sd 3.1275 |
| GNP AR(3) (T 2.15) | $(1+0.521B)(1-0.869B+0.274B^2)$; disc $-0.341$; $k = 10.62$ quarters (R's roots: 10.656); $\phi_0 = 0.6150 \times 0.0077 = 0.0047355$ |
| Tsay Fig. 2.4(b) $(0.6, -0.4)$ | disc $-1.24$, $k = 5.84$; PACF 0.4286, $-0.4$, then 0 |

| Case | Numbers |
|---|---|
| PACF SE | $1/\sqrt{996} = 0.0317$; GNP band $\pm 2/\sqrt{176} = \pm 0.151$ |
| Table 2.1 | AIC $\to$ AR(9) ($-5.849$); BIC $\to$ AR(1) ($-5.833$), AR(3) $-5.831$ |

### 2.2 · Forecasts, MA, ARMA (L08-02)

| Case | Numbers |
|---|---|
| ar2-forecast ($y_n = -5.487150$, $y_{n-1} = -3.508837$) | $E$: $-6.7954$, $-7.5023$, $-7.6927$, $-7.4603$; $V$: 1, 4.0625, 9.1814, 15.7318; $\psi$: 1, 1.75, 2.2625, 2.5594; sd(50) 7.1206 $\to$ 7.12069; 90% limits $\pm 11.71$ |
| MA(1) | $\theta = 0.6$: $\rho_1 = -0.4412$; $\theta = 0.5$ and 2: $-0.4$; bound $\frac{1}{2}$ at $\lvert\theta\rvert = 1$ |
| ARMA(1,1) $(0.8, 0.4)$ | $\gamma_0 = 1.4444$, $\rho_1 = 0.5231$, $\rho_2 = 0.4185$; $\psi$: 1, 0.4, 0.32, 0.256 |

### 2.3 · Unit roots (L08-03)

| Case | Numbers |
|---|---|
| 3M drift | $\mu = 0.0103$ (se 0.002318), $t = 4.44$, $\sigma = 0.0637$, crossover 38.2 months |
| DF MC ($T = 250$, seed 2026) | quantiles $-3.45$, $-3.16$, $-2.87$, $-2.58$; $-1.645$ rejects a true unit root 46.5% of the time |
| W examples | T-bill $t = -2.46$ ($\hat{\rho} = 0.909$, not rejected); inflation $-3.01$ (rejected at 5%); GDP with trend $-2.41$ (not rejected) |
| US GDP ADF | Tsay $-1.701$ (p 0.4297), $\hat{\rho} = 0.9992$; R/slides $-1.61$ (0.45); site page `tseries` $-0.370$ (0.987) |
| S&P 500 ADF | $-1.998$ (p 0.602); site page $k = 2$: $-2.018$ (0.571); differences $-37.5$ (p $\le$ 0.01) |
| Spurious regression | 66.2% ($n = 50$), 84.7% ($n = 250$) false rejections |

### 2.4 · Long memory (L08-04)

| $d$ | Numbers |
|---|---|
| 0.25 | $\rho_1, \rho_2, \rho_3 = 0.3333, 0.2381, 0.1948$; $\phi_{22} = 0.1429$ |
| 0.4 | $\rho_1 = 2/3$; $\rho_{100} = 0.2673$ vs AR(1) $(2/3)^{100} \approx 2.5 \times 10^{-18}$; log–log slope $2d - 1 = -0.2$ |

### 2.5 · Seasonal (L08-05)

| Case | Numbers |
|---|---|
| J&J airline $(0.678, 0.314)$, $s = 4$ | $\hat{\sigma} = 0.089$, $Q(12) = 10.0$ (p 0.44); $\rho_1 = -0.4645$, $\rho_4 = -0.2858$, $\rho_3 = \rho_5 = 0.1328$ |
| Decile 1 | exact: seasonal AR 0.951, MA 0.997 (cancel); January dummy 0.1253 (se 0.0115), intercept 0.0029, $R^2 = 0.2016$ |
| AirPassengers airline | $n = 96$: $(-0.3941, -0.6129)$, BIC $-284.27$; $n = 108$: $(-0.3864, -0.5885)$, BIC $-336.93$ (best of 16); $n = 144$: $(-0.4018, -0.5569)$, AIC $-483.4$; ACF $\rho_1 = -0.3362$, $\rho_{12} = -0.4371$, $\rho_{11} = \rho_{13} = 0.1470$ |
| Out of sample 1958–60 | RMSE airline 36.6 vs trend + 11 dummies 63.3 (our computation) |

### 2.6 · ARCH/GARCH (L08-06)

| Case | Numbers |
|---|---|
| Intel ARCH tests | $Q(12) = 18.26$ (p 0.11) on $r$; 89.85 on $a^2$; LM 53.62 |
| Intel ARCH(1) (3.12) | $\alpha_0 = 0.0111$, $\alpha_1 = 0.356$; uncond. sd 0.1313 (Tsay prints 0.1315); kurtosis 4.227 |
| Tsay Ex. 3.3 (3.19) | (0.000086, 0.1216, 0.8511); persistence **0.9727** (Tsay prints 0.9772); $\sigma^2_\infty = 0.003150$, sd 0.0561; half-life 25.04 months; kurtosis 6.654; Table 3.1 0.0536 … 0.0560 |
| IGARCH (Tsay) | (0.000119, 0.1941, 0.8059): forecasts rise by 0.000119 per step |
| Two-pass | $\hat{\alpha}_1 = 0.9583 - 0.8456 = 0.1127$ |
| Petrobras 2000–2021 | ARCH(1) sd 0.0333; GARCH (2.255e−5, 0.10317, 0.87681): persistence 0.97997, sd 0.03356, half-life 34.3 days, excess kurtosis 3.48; LR 958.8 |
| Petrobras 2000–2025 | GARCH normal persistence 0.97982; GARCH-$t$ 0.98694, $\nu = 6.23$ |

### 2.7 · Asymmetry and tails (L08-07)

| Case | Numbers |
|---|---|
| PBR GJR (APARCH $\delta = 2$) | good 0.05787, bad 0.13483, indicator 0.07696; persistence 0.97813 (literal reading 1.078) |
| PBR T-GARCH ($\delta = 1$) | bad 0.12976, good 0.06306 (ratio 2.06) |
| PBR EGARCH | $\alpha = -0.05549$, $\gamma = 0.18138$, $\beta = 0.97741$; bad/good ratio 1.117 at $c = 1$, 1.249 at $c = 2$ |
| $E\lvert\varepsilon\rvert$ | $\sqrt{2/\pi} = 0.7979$; standardized $t_{7.04}$: 0.7595 |
| PBR GARCH-$t$ kurtosis | $K_a^{(g)} = 1.563$, $K = 1.972$, total excess 12.55 |
| GARCH-M (Tsay) | $c = 1.09$ (se 0.818), $t = 1.33$ |

| Case | Numbers |
|---|---|
| AIC ladder | GARCH $-22297.4$ … skew-$t$ best ($-22585.7$) |

### 2.8 · Stochastic volatility (L08-08)

| Case | Numbers |
|---|---|
| Tsay Ex. 12.3 | posterior (0.409, 0.454, 0.837, 0.086); $\mu_h = 2.785$, $\sigma_h^2 = 0.2872$, $Ea^2 = 18.71$ %², sd 4.33%, kurtosis 3.998, lag-1 corr 0.0906 |
| PBR SV-$t$ | $\mu = -0.419$, $\phi = 0.9842$, $\sigma = 0.1388$, $\nu = 12.13$; $\sigma_h^2 = 0.6146$; kurtosis (Gaussian $\varepsilon$) 5.547; $\text{corr}(a_t^2, a_{t-1}^2)$ 0.1828 vs GARCH 0.2037 |
| S&P 500 iid | Gaussian $\hat{\sigma} = 0.012729$; $t$: scale 0.005922, $\hat{\nu} = 2$ |

### 2.9 · Bayesian GARCH (L08-09)

| Case | Numbers |
|---|---|
| M0 prior | IG(5, 4 × 0.0006617694): mean 0.0006617694, sd 0.000382; prior weight with $n = 1760$: $4/884 = 0.45\%$ |
| M1, M2 priors | sd($\mu$) 0.006627; sd($\alpha$) 0.01849, sd($\beta$) 0.00993 |
| M3 | $\Pr(\alpha_1 + \beta > 1 \mid \text{data}) = 0.0034$; prior $E\nu = \delta + 1/\lambda = 102$ |
| Bayes vs ML (PBR standardized) | persistence 0.97905 vs 0.98370; half-life 32.7 vs 42.2 days |

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## Links

- Unit: L08-00-map · L08-11-pitfalls · L08-12-python
- Class 1 review: L01-07 (AR(1), random walk, VAR(1))
- Rules: rules/04 · rules/05

# Class 8 · Pitfalls — the confusions, each with its minimal counterexample, and an original True/False self-test

Prev: L08-10-formulas · Next: L08-12-python · Map: L08-00-map

Equations are cited by the section note's own numbering; every number is asserted in L08-12-python or in `figures/make_figures.py`.

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## Part I — the confusions this material produces

### 1 · A coefficient above 1 vs a nonstationary process

**The confusion.** "$\phi_1 = 1.75 > 1$, so the AR(2) explodes."

**Counterexample** (slides' $\phi = (1.75, -0.80)$; L08-01 (15)–(18)). $\phi_2 = -0.80 < 1 - 1.75 = -0.75$, $-0.80 < 1 + 1.75$, $-0.80 > -1$: inside the triangle. Both solutions of $1 - 1.75x + 0.8x^2 = 0$ have modulus $1.118 > 1$, and $\gamma_0 = 50.7\sigma^2$ is finite.

> **Stationarity is a property of the roots, not of any single coefficient** (as with $\Phi_{22} = 1.1$ in L01-07 §9.1). → L08-01-ar2-arp-cycles-identification §5.

### 2 · Characteristic roots vs solutions of the characteristic equation

**The confusion.** Mixing "the roots must lie outside the unit circle" with "inside".

**Resolution** (L08-01 (10)–(12)). Tsay calls the **inverses** of the solutions of $1 - \phi_1 x - \phi_2 x^2 = 0$ the characteristic roots: stationarity needs the **solutions** outside ($|x| > 1$) and the **characteristic roots** inside ($|\omega| < 1$). For $(1.75, -0.80)$: $|x| = 1.118$, $|\omega| = 0.894$. Same statement, two languages; say which one you use.

### 3 · Complex roots vs oscillating data

**The confusion.** "A smooth, highly persistent ACF means real roots."

**Counterexample.** $(1.75, -0.80)$ has $\rho_1 = 0.972$ and a smooth ACF, yet the roots are complex ($1.75^2 + 4(-0.80) = -0.1375 < 0$): the cycle has length 30.1 and shows only as a dip to $\rho_{15} = -0.187$. The near-unit real root of $(0.90, 0.05)$ gives an ACF that never turns negative. → figure `L08-ar2-triangle-acf`.

### 4 · ACF cut-off vs PACF cut-off

**The confusion.** "The ACF dies at lag 2, so fit an AR(2)."

**Resolution** (L08-02 §8). A cut-off in the **ACF** identifies an **MA**; a cut-off in the **PACF** identifies an **AR**. AR(2) $(0.6, -0.4)$: ACF damped waves, PACF 0.4286, $-0.4$, then exactly 0. MA(1) $\theta = 0.6$: ACF $-0.441$ then 0, PACF decays. ARMA cuts off in neither.

### 5 · An MA(1) with $|\rho_1| = 0.6$

**The confusion.** Matching any lag-1 autocorrelation with an MA(1).

**Counterexample** (L08-02 (17)). $|\rho_1| = |\theta_1|/(1+\theta_1^2) \le \frac{1}{2}$, with equality at $|\theta_1| = 1$. A sample $\hat{\rho}_1 = 0.6$ with nothing after it is not an MA(1). And $\theta = 0.5$ and $\theta = 2$ give the same $\rho_1 = -0.4$: only the **invertible** one ($|\theta| < 1$) is reported.

### 6 · Sign conventions in printed MA coefficients

**The confusion.** Plugging R's `ma1 = -0.3864` into Tsay's $(1 - \theta B)$ as $\theta = -0.3864$.

**Resolution** (L08-02, L08-05 §3.1). Tsay writes $a_t - \theta a_{t-1}$; R, `forecast` and statsmodels print the coefficient of $+a_{t-1}$. R's $-0.3864$ is Tsay's $\theta = +0.3864$. The AirPassengers ACF $\rho_1 = -0.336$ comes out negative either way only if the conversion is done right.

### 7 · A long-horizon forecast interval vs the forecast of a stationary series

**The confusion.** "Forecast uncertainty always grows without bound."

**Counterexample** (L08-02 (28) vs L08-03 (6)). Stationary AR(2): $V[e_h(\ell)] = \sigma^2 \sum_{j<\ell} \psi_j^2 \to \gamma_0$ (sd 7.12 for the ar2-forecast model). Random walk: $V = \ell\sigma^2 \to \infty$. Bounded vs unbounded is exactly the stationary/unit-root distinction. → figure `L08-unit-root-bands-df`, left.

### 8 · Failing to reject a unit root vs accepting it

**The confusion.** "ADF p = 0.45, so GDP has a unit root."

**Counterexample** (W Ex. 18.2; L08-03 §6.3). For T-bills, $t = -2.46$ does not reject $\rho = 1$, but a standard $t$ test of $\rho = 0.9$ gives $t = (0.909 - 0.9)/0.037 = 0.24$ and does not reject either. Both cannot be "accepted". With $T = 35$ (W Ex. 18.4), $\hat{\rho} = 0.79$ still fails to reject: the test has low power near 1.

### 9 · Normal critical values for a DF statistic

**The confusion.** Rejecting $H_0 : \phi = 1$ when $t < -1.645$.

**Counterexample** (seeded MC, $T = 250$; L08-03 §5.5). Under a true unit root the DF $t$ has 5% quantile $-2.87$ (W: $-2.86$). Using $-1.645$ rejects a true unit root **46.5%** of the time.

### 10 · Same test name, different regressions

**The confusion.** "Tsay reports ADF = $-1.701$ for GDP but the site page prints $-0.370$: one is wrong."

**Resolution** (L08-03 §7). `tseries::adf.test` always includes a constant **and a trend**; Tsay's and the deck's runs use a constant only ($-1.701$ / $-1.61$). Different deterministic terms, different null distributions, same conclusion (no rejection).

### 11 · Trend-stationary vs random walk with drift

**The confusion.** "Both trend upward, so detrending and differencing are interchangeable."

**Counterexample** (L08-03 (9), (11)). RW with drift: variance $t\sigma^2$, shocks permanent; trend-stationary: variance $V(r_t)$ constant, shocks transitory. At $\ell = 120$ months (3M calibration) the $\pm 2$ sd band is $\pm 1.40$ vs $\pm 0.29$. Detrending a RW leaves a nonstationary residual; differencing a trend-stationary series creates a non-invertible MA unit root.

### 12 · A significant regression between two trending series

**The confusion.** $t = 5$ and $R^2 = 0.9$ between two price levels ⇒ a relation.

**Counterexample** (Granger–Newbold; W §18.3). Two **independent** random walks, $n = 50$: the nominal 5% $t$ test rejects 66% of the time (our MC: 66%), 84.7% at $n = 250$. Under $H_0$ the error is itself a random walk.

### 13 · Long memory vs unit root

**The confusion.** "A slowly decaying ACF means a unit root."

**Counterexample** (L08-04 (7)–(11)). ARFIMA with $d = 0.4$ is **stationary**, yet $\rho_{100} = 0.267$, while an AR(1) with the same $\rho_1 = 2/3$ has $\rho_{100} \approx 10^{-18}$. A unit root gives sample ACFs near 1 at all fixed lags; long memory gives small ACFs that decay hyperbolically ($k^{2d-1}$).

### 14 · Multiplicative vs additive seasonal MA

**The confusion.** "The airline model is just an MA with lags 1 and 12."

**Counterexample** (L08-05 (12)–(14)). $(1 - \theta B)(1 - \Theta B^{12})$ adds the term $\theta\Theta B^{13}$: nonzero $\rho_{11} = \rho_{13} = \rho_1\rho_{12} = 0.147$ (AirPassengers). The non-multiplicative $1 - \theta B - \Theta B^{12}$ has $\rho_{13} = 0$. Same two parameters, different ACF.

### 15 · Uncorrelated vs independent returns

**The confusion.** "Ljung–Box on returns does not reject, so returns are iid."

**Counterexample** (L01-07 §2; L08-06 §2.1). IBM: $Q(30) = 38.2$ (p 0.14) on returns, 182.1 on squares; Intel: $Q(12) = 18.26$ (p 0.11) on returns, 89.85 on $a_t^2$. Serially uncorrelated, conditionally heteroskedastic: that is exactly an ARCH process.

### 16 · Conditional vs unconditional variance (and kurtosis)

**The confusion.** "Gaussian innovations ⇒ Gaussian returns."

**Counterexample** (L08-06 (12), (25)). ARCH(1) with $\alpha_1 = 0.356$ and $N(0, 1)$ innovations has unconditional kurtosis 4.23; Tsay's GARCH(1,1) has 6.65. Conditionally normal, unconditionally fat-tailed.

### 17 · Persistence vs the size of the ARCH coefficient

**The confusion.** "$\alpha_1 = 0.12$ is small, so volatility shocks die fast."

**Counterexample** (L08-06 (28)–(29)). The decay rate is $\alpha_1 + \beta_1 = 0.9727$: half-life 25 months (Petrobras: 0.980, 34 days). $\alpha_1$ sets the immediate reaction; $\beta_1$ carries it.

### 18 · IGARCH vs "very persistent" GARCH

**The confusion.** "$\alpha_1 + \beta_1 = 0.98$ is basically IGARCH."

**Resolution** (L08-06 (31)–(32)). At 0.98 forecasts still revert to $\sigma^2_\infty$ (half-life 34); at exactly 1 there is no $\sigma^2_\infty$ and forecasts rise linearly by $\alpha_0$ per step. A Bayesian reports $\Pr(\alpha_1 + \beta > 1 \mid \text{data})$ (0.0034 for Petrobras 2007–2013) instead of guessing.

### 19 · The printed `gamma1` of `garchFit(leverage=TRUE)`

**The confusion.** Reading the glossary page's formula $\sigma_t^2 = \omega + \alpha\varepsilon^2 + \gamma\varepsilon^2 I(\varepsilon < 0) + \beta\sigma^2$ with its printed numbers.

**Counterexample** (L08-07 (4)–(7)). With $\alpha = 0.0923$, $\gamma = 0.2084$, $\beta = 0.8818$, the literal reading has persistence $\alpha + \gamma/2 + \beta = 1.078 > 1$. `fGarch` fits the APARCH form $\alpha(|a| - \gamma a)^2$: bad news 0.1348, good news 0.0579, persistence 0.978.

### 20 · Heavier-tailed innovations vs a smaller ARCH effect

**The confusion.** "The $t$ fit has a smaller $\alpha_1$, so less volatility clustering."

**Resolution** (T p. 126; L08-07 §6). Under normal errors the model inflates $\alpha_1$ to manufacture the tails; with $t$ errors part of the kurtosis comes from $\varepsilon$ (Intel: 0.356 → 0.285; Petrobras 0.103 → 0.074). Total kurtosis rises (PBR GARCH-$t$: 12.55).

### 21 · Observation-driven vs parameter-driven volatility

**The confusion.** "SV is GARCH with a log."

**Resolution** (L08-08 §1, §5). In GARCH, $\sigma_t^2$ is a known function of past data; in SV it has its own shock, so the likelihood integrates over the whole volatility path (MCMC). That extra shock is why the SV fit needs fewer tail degrees of freedom (Petrobras: $\nu = 12$ vs GARCH-$t$ 7).

### 22 · A posterior "p-value"

**The confusion.** GARCH11-SVar1.html prints "Pr(>|t|) = 1.9987" for $\mu$ in the SV table.

**Resolution** (L08-08 §9; L08-09 §8). It is $2(1 - \Phi(t))$ with $t < 0$: not a probability. Posterior summaries answer $\Pr(\mu < 0 \mid \text{data})$ directly; frequentist labels on MCMC output mislead.

---

## Part II — True/False self-test (original)

**Scoring** (rules/00): V/F score = $\max\{0,\ \text{correct} - \text{wrong}/2\}\times$ item value; blank items do not count as wrong. Difficulty: **[E]** easy · **[M]** medium · **[P]** hard. Numbers are illustrative unless a note is cited. Answers follow each block.

### Block A — AR, MA, ARMA

**Setting (items 1–6).** A central-bank analyst fits $y_t = 1.2y_{t-1} - 0.5y_{t-2} + a_t$, $\sigma = 1$, to quarterly output-gap data, and a colleague proposes $y_t = a_t + 0.9a_{t-1}$ instead.

1. **[E]** The AR(2) is stationary.
2. **[M]** The AR(2) implies stochastic cycles of average length about 11 quarters.
3. **[M]** The AR(2)'s lag-1 autocorrelation is 0.8.
4. **[M]** The colleague's MA(1) has lag-1 autocorrelation 0.9.
5. **[P]** The MA(1) $y_t = a_t + (1/0.9)a_{t-1}$ has the same ACF as the colleague's model, so the data cannot tell them apart; the convention is to report the invertible one.
6. **[M]** The 10-quarter-ahead forecast-error variance of the AR(2) exceeds its unconditional variance.

**Answers.**

1. **True.** $\phi_2 = -0.5 < 1 - 1.2 = -0.2$, $-0.5 < 2.2$, $-0.5 > -1$: inside the triangle (L08-01 (18)).
2. **True.** $\phi_1^2 + 4\phi_2 = 1.44 - 2 = -0.56 < 0$: complex. $\cos^{-1}[1.2/(2\sqrt{0.5})] = \cos^{-1}(0.8485) = 0.5576$, so $k = 2\pi/0.5576 = 11.27$ quarters (L08-01 (14)).
3. **True.** $\rho_1 = \phi_1/(1-\phi_2) = 1.2/1.5 = 0.8$ (L08-01 (7)).
4. **False.** $\rho_1 = 0.9/(1+0.81) = 0.497$, below the $\frac{1}{2}$ bound; no MA(1) reaches 0.9 (L08-02 (17)).
5. **True.** $\theta$ and $1/\theta$ give the same $\theta/(1+\theta^2)$: $1.111/(1+1.235) = 0.497$. Only $|\theta| < 1$ is invertible (L08-02 (19)).
6. **False.** $V[e_h(\ell)] = \sigma^2 \sum_{j<\ell} \psi_j^2$ increases **to** $\gamma_0$ and never exceeds it (L08-02 (28)).

**Block B — identification and forecasting**

7. [**E**] For an AR(3), the theoretical PACF is zero from lag 4 on.
8. [**M**] BIC penalises each parameter by $\ln T/T$ per observation, so with $T = 996$ it tends to choose smaller models than AIC.
9. [**M**] For an MA(2), all forecasts beyond two steps equal the unconditional mean.
10. [**P**] The ACF of an ARMA(1,1) with $\phi_1 = 0.8$ decays geometrically at rate 0.8 from lag 1, exactly as an AR(1) with $\phi = 0.8$.

**Answers.**

7. **True.** $\phi_{\ell,\ell} = 0$ for $\ell > p$ (T p. 47; L08-01 §8).
8. **True.** Per parameter: 2 for AIC vs $\ln 996 = 6.90$ for BIC; Table 2.1 has AIC $\to 9$, BIC $\to 1$ (L08-01 (25)).
9. **True.** $\hat{r}_h(\ell) = c_0$ for $\ell > q$ (T p. 63; L08-02 §7).
10. **False.** $\rho_\ell = \phi_1 \rho_{\ell-1}$ only from $\ell \geq 2$; $\rho_1$ differs (0.5231 for $(0.8, 0.4)$ vs 0.8) (L08-02 (25)).

**Block C — unit roots and long memory**

**Setting (items 11–16).** A pension fund's analyst runs `adf.test` on 60 years of log house prices and gets $t = -2.1$, p = 0.53.

11. [**E**] The analyst may conclude that house prices have a unit root.
12. [**M**] Had the analyst compared $t = -2.1$ with $-1.645$, he would have rejected a unit root, and that rejection would have been too frequent under the null.
13. [**M**] If log prices follow a random walk with drift, the conditional variance of the 20-year-ahead forecast is 20 times that of the 1-year-ahead forecast (annual data).
14. [**P**] In the ADF regression of an AR(3), the coefficient on $x_{t-1}$ in the $\Delta$ form equals $\phi_1 + \phi_2 + \phi_3 - 1$, so a unit root means $\phi_1 + \phi_2 + \phi_3 = 1$.
15. [**M**] A stationary ARFIMA(0, 0.3, 0) has an ACF that decays faster than any AR(1)'s at long lags.
16. [**M**] Regressing the log house-price index on an unrelated random walk would yield a significant slope more often than 5% of the time.

**Answers.**

11. **False.** Non-rejection is not acceptance; DF tests have little power near 1 (W p. 576; L08-03 §6.3).
12. **True.** $-2.1 < -1.645$ rejects; under a true unit root that rule rejects about 46% of the time (L08-03 §5.5).
13. **True.** $V[e_h(\ell)] = \ell\sigma^2$: $20\sigma^2$ vs $\sigma^2$ (L08-03 (6)); the drift shifts the mean only.
14. **True.** L08-03 (25)–(27): $\beta = -\phi(1)$, zero iff $\sum \phi_i = 1$.
15. **False.** It decays hyperbolically, $\rho_k \sim ck^{-0.4}$, **slower** than any geometric $\phi^k$ (L08-04 (11)).
16. **True.** Spurious regression: 66% rejections at $n = 50$ (W §18.3; L08-03 §8).

**Block D — seasonal models**

17. **[E]** In the airline model with $s = 12$, the ACF of $w_t = (1 - B^{12})(1 - B)x_t$ is nonzero only at lags 1, 11, 12, 13.
18. **[M]** In the airline model, $\rho_{13} = \rho_1\rho_{12}$.
19. **[P]** When the exact-likelihood estimates of a seasonal AR and a seasonal MA factor are both close to 1 and nearly cancel, the seasonality is better modelled with dummy variables.
20. **[M]** Taking logs of AirPassengers before fitting the airline model is meant to turn a multiplicative seasonal pattern into an additive one.

**Answers.**

17. **True.** L08-05 (6)–(12).
18. **True.** $\gamma_{13} = \theta\Theta\sigma^2$ and the product form of the variance give $\rho_{13} = \rho_1\rho_{12}$ (L08-05 (12)).
19. **True.** Decile 1: seasonal AR 0.951 and MA 0.997 cancel, and a January dummy (0.1253) removes lags 12, 24, 36 (T Ex. 2.4; L08-05 §5).
20. **True.** Amplitude grows with the level; the log stabilises it (airpassengers page; T p. 83).

**Block E — ARCH and GARCH**

**Setting (items 21–26).** A bank's risk desk fits a Gaussian GARCH(1,1) to daily returns: $\alpha_0 = 0.02$, $\alpha_1 = 0.10$, $\beta_1 = 0.85$ (returns in %).

21. **[E]** The unconditional daily variance is 0.4 (%²).
22. **[M]** A volatility shock has a half-life of about 13.5 days.
23. **[M]** The unconditional kurtosis of returns is finite and above 3.
24. **[P]** If today's 1-day-ahead variance forecast is 1.4, the 3-day-ahead forecast is 1.1.
25. **[M]** Because the innovations are Gaussian, the unconditional distribution of returns is Gaussian.
26. **[M]** The model implies that a $-3\%$ shock and a $+3\%$ shock raise tomorrow's variance by the same amount.

**Answers.**

21. **True.** $0.02/(1 - 0.95) = 0.4$ (L08-06 (22)).
22. **True.** $\ln 0.5/\ln 0.95 = 13.5$ (L08-06 (29)).
23. **True.** $1 - 2(0.01) - 0.95^2 = 0.0775 > 0$; $K_a^{(g)} = 0.06/0.0775 = 0.774$, kurtosis 3.77 (L08-06 (25)).
24. **False.** $\sigma^2(\ell) = 0.4 + 0.95^{\ell-1}(1.4 - 0.4)$, and $0.95^2 = 0.9025$, so $\sigma^2(3) = 0.4 + 0.9025 \times 1.0 = 1.3025$, not 1.1. The gap to $\sigma^2_\infty$ shrinks by only 5% per day (L08-06 (28)).
25. **False.** Conditionally Gaussian, unconditionally fat-tailed (item 23).
26. **True.** GARCH depends on $a^2_{t-1}$ only; asymmetry needs GJR/EGARCH (L08-07 §1).

**Block F — asymmetry, SV, Bayes**

27. **[M]** In an EGARCH with $\log\sigma_t^2 = \omega + \alpha z_{t-1} + \gamma(|z_{t-1}| - E|z|) + \beta\log\sigma^2_{t-1}$, a negative $\alpha$ means bad news raises volatility more than good news.
28. **[M]** In an SV-AR(1) with Gaussian $\varepsilon$, the unconditional kurtosis is $3e^{\sigma_h^2}$, where $\sigma_h^2$ is the stationary variance of the log variance.
29. **[P]** The SV likelihood can be evaluated by the same one-pass recursion as the GARCH likelihood, once the parameters are given.
30. **[M]** With an IG(5, b) prior and $n = 1760$ observations, the prior's weight in the posterior mean of $\sigma^2$ is below 1%.

**Answers.**

27. **True.** The bad/good ratio at $|z| = c$ is $e^{-2\alpha c} > 1$ when $\alpha < 0$ (Petrobras: 1.117 at $c = 1$) (L08-07 (14)).
28. **True.** $Ea^4/(Ea^2)^2 = 3e^{2\mu_h+2\sigma_h^2}/e^{2\mu_h+\sigma_h^2}$ (L08-08 (7)).
29. **False.** The volatility path is latent with its own shock; the likelihood is an $n$-dimensional integral (L08-08 (10)), handled by MCMC or filtering.
30. **True.** Weight $(a-1)/(a-1+n/2) = 4/884 = 0.45\%$ (L08-09 (5)).

---

## Part III — the distinctions that cost most points

| Pair | One-line test |
|---|---|
| solutions vs characteristic roots | outside vs inside the unit circle — say which |
| ACF vs PACF cut-off | MA vs AR |
| bounded vs growing forecast variance | stationary vs unit root |
| fail to reject vs accept | DF power is low near 1 |
| DF vs normal critical values | $-2.86$ vs $-1.645$ at 5% |
| hyperbolic vs geometric decay | long memory vs ARMA |
| multiplicative vs additive seasonal MA | $\rho_{s+1} = \rho_1\rho_s$ vs 0 |
| conditional vs unconditional kurtosis | GARCH turns normal shocks into fat tails |
| $\alpha_1$ vs $\alpha_1 + \beta_1$ | reaction vs persistence |
| printed `gamma1` vs GJR's $\gamma$ | APARCH form in `fGarch` |
| GARCH vs SV | observation- vs parameter-driven |

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## Links

- Unit: L08-00-map · L08-10-formulas · L08-12-python
- Notes: L08-01-ar2-arp-cycles-identification · L08-02-forecasting-ma-arma · L08-03-unit-roots-dickey-fuller · L08-04-long-memory-arfima · L08-05-seasonal-airline-model · L08-06-arch-garch · L08-07-garch-family-asymmetry-tails · L08-08-stochastic-volatility · L08-09-bayesian-garch
- Class 1 pitfalls on stationarity: L01-10 §10, §13

# Class 8 · Computation in Python — every routine, with the assert that must pass

> Every computation the unit uses, written in Python (numpy, scipy, statsmodels; no R, no `arch`, no PyMC), and each checked with an `assert` against the number printed in the deck, the site page or the book. The code below is the script `Notas/L08/L08_python_checks.py`, split into its sections. Run `python Notas/L08/L08_python_checks.py` (about one minute). It prints one `ok` line per section and ends with `ALL L08 CHECKS PASSED`, which is the output of the run on 2026-10-27. Prev: L08-11-pitfalls · Map: L08-00-map · Formulas: L08-10-formulas

**What is reused, not rewritten.**

| Already in the unit | What it computes | How this note uses it |
|---|---|---|
| `figures/make_figures.py` | `ar2_acf`, `ar2_gamma0`, `cycle_length`, `psi_weights`, `acf_linear`, `pacf_from_acf` (Durbin–Levinson), `arfima_acf`; eight figures with their own asserts | **imported** as `mf` (functions only; `build()` is not called) |
| `companions/check_companions.py` | `ar_python()` and `ga_python()`, the exact read-out strings of the AR(2) Triangle and the Volatility Dial | **imported** as `cc`; §1 and §6 check that the pages' defaults print the numbers derived here |
| statsmodels | `ArmaProcess.acf`, `arma_pacf`, `adfuller`, `SARIMAX`, `OLS` | independent path for every hand formula |

**What had to be written.** `arch` and PyMC are not installed, so the GARCH likelihoods (normal and standardized $t$), their maximisation, the GARCH simulator, and the random-walk Metropolis sampler for the Bayesian GARCH-$t$ are written here in NumPy/SciPy (the variance recursion is one call to `scipy.signal.lfilter`). The SV part uses simulation and closed forms only; a full SV Gibbs sampler (L08-08 §6) is described in the note but not coded.

**Conventions.** `near(x, printed, tol)` passes when `x` matches a printed value to `tol` (default 5e-4). Monte Carlo checks are seeded and allow about four standard errors. Equation numbers "(k)" are those of the section note named in each heading. Tsay's MA sign ($a_t - \theta a_{t-1}$) is used in formulas; statsmodels and R print $+\theta$.

---

## 0 · Setup

```
"""Class 8 (L08) reference implementation, with assert self-checks against the decks'
 ↪  and books' numbers.

Run:  python Notas/L08/L08_python_checks.py      (stops at the first failed assert;
 ↪  about 1-2 minutes)

Sections §1-§9 are the code blocks of L08-12-python.md, in the same order. The model
 ↪  functions
```

```
(AR(2) ACF and variance, psi weights, PACF by Durbin-Levinson, ARFIMA ACF) are
 ↪  imported from
figures/make_figures.py, and the two companions' read-out functions from
 ↪  companions/check_companions.py
(reuse, not copies). Equation numbers "(k)" refer to the section note named in each
 ↪  header.
Every Monte Carlo draw is seeded; MC checks use tolerances of about four standard
 ↪  errors.
Data: AirPassengers is printed on seasonal.pdf slide 17 (embedded below); CRSP Decile
 ↪  1 and the
daily S&P 500 returns are Tsay's files in Listas/data/ (read-only).
"""
import math
import pathlib
import sys
import warnings

import numpy as np
import statsmodels.api as sm
from scipy import integrate, optimize, signal, special, stats
from statsmodels.tsa.arima_process import ArmaProcess, arma_pacf
from statsmodels.tsa.statespace.sarimax import SARIMAX
from statsmodels.tsa.stattools import adfuller

HERE = pathlib.Path(__file__).resolve().parent
DATA = HERE.parents[1] / "Listas" / "data"
sys.path.insert(0, str(HERE / "figures"))
sys.path.insert(0, str(HERE / "companions"))
import make_figures as mf  # noqa: E402  model functions only; build() is not called
import check_companions as cc  # noqa: E402  the companions' read-out functions

warnings.filterwarnings("ignore")          # statsmodels' convergence chatter on the
 ↪  small SARIMA grid
sys.stdout.reconfigure(encoding="utf-8")


def near(x, printed, tol=5e-4):
    """True when x agrees with a printed value to within tol (default: half a unit in
     ↪  the 3rd decimal)."""
    return abs(float(x) - printed) <= tol
```

---

## 1 · AR(2), AR(p), PACF, information criteria → L08-01-ar2-arp-cycles-identification

The triangle conditions L08-01 (15)–(17) are checked against the root moduli at five points (stationary complex, stationary real, the GNP factor, a unit-root edge, an explosive complex pair). The recursion L08-01 (6) is checked against statsmodels' exact ACF, the GNP cycle against R's printed roots (10.656) and Tsay's rounded factor (10.62), the PACF against `arma_pacf` and against a long seeded sample, and the companion's default read-outs against the same numbers.

```
# === §1 AR(2), AR(p), PACF, information criteria (L08-01) ===
def ar2_roots(p1, p2):
    """Solutions x of 1 - p1 x - p2 x^2 = 0 (L08-01 (10)); stationary iff all |x| > 1."""
    return np.roots([-p2, -p1, 1.0])


def in_triangle(p1, p2):
    """Triangle conditions L08-01 (15)-(17)."""
    return p2 < 1 - p1 and p2 < 1 + p1 and p2 > -1


for (p1, p2), stat, cplx in [((1.75, -0.80), True, True), ((0.90, 0.05), True, False),
                             ((0.869, -0.274), True, True), ((0.5, 0.5), False, False), ((0.5, -1.2), False, True)]:
    x = ar2_roots(p1, p2)
    assert (np.abs(x).min() > 1) == stat == in_triangle(p1, p2), (p1, p2)      # roots <=> triangle
    assert (p1 ** 2 + 4 * p2 < 0) == cplx == bool(np.iscomplex(x).any())
r = mf.ar2_acf(1.75, -0.8, 40)
assert near(r[1], 0.9722, 5e-5) and near(r[2], 0.9014, 5e-5) and r.argmin() == 15 and near(r.min(), -0.1866, 5e-5)
assert near(math.sqrt(mf.ar2_gamma0(1.75, -0.8)), 7.12069, 5e-6) and near(abs(ar2_roots(1.75, -0.8)[0]), 1.1180, 5e-5)
assert near(math.sqrt(mf.ar2_gamma0(0.9, 0.05)), 3.1275, 5e-5) and near(ar2_roots(0.9, 0.05).real.min(), -19.05, 5e-3)
assert near(mf.cycle_length(1.75, -0.8), 30.09, 5e-3) and near(mf.cycle_length(0.6, -0.4), 5.836, 5e-4)
# The ACF recursion agrees with statsmodels' exact ACF.
assert np.allclose(ArmaProcess(ar=[1, -1.75, 0.8]).acf(41), r, atol=1e-12)
# GNP AR(3) (T eq. 2.15): R's unrounded coefficients give k = 10.656; Tsay's rounded factor gives 10.62.
roots = np.roots([0.1423, -0.1793, -0.3480, 1.0])                          # 1 - .348x - .1793x^2 + .1423x^3
z = roots[np.argmax(roots.imag)]                                           # 4-decimal coefficients: R used more digits
assert near(z.real, 1.590253, 5e-4) and near(abs(z), 1.913308, 5e-4) and near(abs(roots.real.min()), 1.920152, 5e-4)
assert near(2 * math.pi / math.acos(1.590253 / 1.913308), 10.65638, 5e-6)  # from R's printed roots (ar-ma.html)
assert near(2 * math.pi / math.acos(z.real / abs(z)), 10.656, 2e-3)
assert near(mf.cycle_length(0.869, -0.274), 10.62, 5e-3)
assert near((1 - .348 - .1793 + .1423) * 0.0077, 0.0047355, 5e-8)          # intercept from R's mean
assert np.allclose(np.polymul([1, 0.521], [1, -0.869, 0.274]), [1, -0.348, -0.178749, 0.142754])
# PACF: Durbin-Levinson on the exact ACF equals statsmodels' arma_pacf; AR(2) cuts off after lag 2.
pac = mf.pacf_from_acf(mf.ar2_acf(0.6, -0.4, 10), 10)
```

```
assert np.allclose(pac, arma_pacf([1, -0.6, 0.4], [1], lags=11)[1:], atol=1e-10)
assert near(pac[0], 0.4286, 5e-5) and near(pac[1], -0.4, 1e-10) and
 ↪  np.abs(pac[2:]).max() < 1e-10
# Sample PACF of a long seeded AR(2): within 4/sqrt(T) of the theory (SE 1/sqrt(T), T
 ↪  p. 47).
rng = np.random.default_rng(81)
T = 10000
e = rng.standard_normal(T + 500)
y = signal.lfilter([1.0], [1, -0.6, 0.4], e)[500:]
spac = sm.tsa.pacf(y, nlags=6)
assert np.all(np.abs(spac[1:3] - pac[:2]) < 4 / math.sqrt(T)) and
 ↪  np.all(np.abs(spac[3:]) < 4 / math.sqrt(T))
assert near(1 / math.sqrt(996), 0.0317, 5e-5) and near(2 / math.sqrt(176), 0.151,
 ↪  5e-4)
# Table 2.1 (T p. 47): AIC picks p = 9, BIC picks p = 1.
aic = [-5.838, -5.837, -5.846, -5.845, -5.847, -5.847, -5.846, -5.847, -5.849,
 ↪  -5.847, -5.845, -5.843]
bic = [-5.833, -5.827, -5.831, -5.825, -5.822, -5.818, -5.812, -5.807, -5.805,
 ↪  -5.798, -5.791, -5.784]
assert int(np.argmin(aic)) + 1 == 9 and int(np.argmin(bic)) + 1 == 1
# The AR(2) companion's default read-outs (imported) print the same numbers.
o, status = cc.ar_python(1.75, -0.8, 1.0, -5.48715, -3.508837, 50)
assert status == "stationary" and (o["rho1"], o["rho2"], o["sd"], o["k"]) ==
 ↪  ("0.972", "0.901", "7.121", "30.09")
print("§1 ok: AR(2) roots/triangle, gamma0, cycles 30.09/10.62/10.656, PACF, Table
 ↪  2.1, companion")
```

---

## 2 · Forecasts, MA, ARMA → L08-02-forecasting-ma-arma

The forecast recursion, the closed forms L08-02 (7)–(10) and the ψ-weight variance (28) must agree with the ar2-forecast page (E: −6.7954 … ; V: 1, 4.0625, …; limit sd 7.12069), and a seeded rerun of the page's 10 000-path Monte Carlo must land within MC error of the limit. Then the MA(1) bound $|\rho_1| \le \frac{1}{2}$ on a fine grid, and ARMA(1,1) $(0.8, 0.4)$: $\gamma_0 = 1.4444$, $\rho_1 = 0.5231$.

```
# === §2 Forecasts, MA, ARMA (L08-02) ===
p1, p2, yn, yn1 = 1.75, -0.8, -5.487150, -3.508837
psi = mf.psi_weights([p1, p2], [], 60)
E = [p1 * yn + p2 * yn1]
E.append(p1 * E[0] + p2 * yn)
for h in range(2, 50):
    E.append(p1 * E[-1] + p2 * E[-2])
V = np.cumsum(psi[:50] ** 2)                                    # L08-02 (28)
closed = [p1 * yn + p2 * yn1, (p1 ** 2 + p2) * yn + p1 * p2 * yn1,  # L08-02 (7)-(10)
          (p1 ** 3 + 2 * p1 * p2) * yn + (p1 ** 2 * p2 + p2 ** 2) * yn1,
          (p1 ** 4 + 3 * p1 ** 2 * p2 + p2 ** 2) * yn + (p1 ** 3 * p2 + 2 * p1 * p2
 ↪  ** 2) * yn1]
assert np.allclose(E[:4], closed) and np.allclose(E[:4], [-6.7954, -7.5023, -7.6927,
 ↪  -7.4603], atol=5e-5)
assert np.allclose(V[:4], [1, 4.0625, 9.1814, 15.7318], atol=5e-5)
```

```
assert near(V[1], 1 + p1 ** 2, 1e-12) and near(V[2], 1 + p1 ** 2 + (p1 ** 2 + p2) **
 ↪  2, 1e-12)
assert near(math.sqrt(V[-1]), 7.1206, 5e-5) and near(math.sqrt(mf.ar2_gamma0(p1,
 ↪  p2)), 7.12069, 5e-6)
# The page's Monte Carlo (M = 10 000 paths, 50 steps): our seeded run lands within MC
 ↪  error of 0 and 7.12.
rng = np.random.default_rng(1233)
M = 10000
yf = np.zeros((50, M))
yf[0] = p1 * yn + p2 * yn1 + rng.standard_normal(M)
yf[1] = p1 * yf[0] + p2 * yn + rng.standard_normal(M)
for h in range(2, 50):
    yf[h] = p1 * yf[h - 1] + p2 * yf[h - 2] + rng.standard_normal(M)
assert abs(yf[49].mean()) < 4 * 7.12 / math.sqrt(M) and abs(yf[49].std() - 7.1207) <
 ↪  0.2
# MA(1) in Tsay's sign r = a - theta a_{-1}: rho1 = -theta/(1+theta^2); |rho1| <= 1/2
 ↪  with the bound at theta = 1.
rho1 = lambda th: -th / (1 + th ** 2)
assert near(rho1(0.6), -0.4412, 5e-5) and rho1(0.5) == rho1(2.0) == -0.4
th = np.linspace(-5, 5, 100001)
assert near(np.abs(rho1(th)).max(), 0.5, 1e-12) and
 ↪  near(abs(th[np.argmax(np.abs(rho1(th)))]), 1.0, 1e-4)
assert np.allclose(ArmaProcess(ma=[1, -0.6]).acf(4), [1, rho1(0.6), 0, 0],
 ↪  atol=1e-12)
# ARMA(1,1), (phi, theta) = (0.8, 0.4): gamma0 = (1 - 2 phi theta + theta^2)/(1 -
 ↪  phi^2), rho1 (L08-02 (23)-(25)).
ph, th1 = 0.8, 0.4
g0 = (1 - 2 * ph * th1 + th1 ** 2) / (1 - ph ** 2)
r1 = (ph - th1) * (1 - ph * th1) / (1 - 2 * ph * th1 + th1 ** 2)
acf_arma = ArmaProcess(ar=[1, -ph], ma=[1, -th1]).acf(4)
assert near(g0, 1.4444, 5e-5) and near(r1, 0.5231, 5e-5) and
 ↪  np.allclose(acf_arma[1:3], [r1, ph * r1], atol=1e-12)
assert np.allclose(mf.psi_weights([ph], [th1], 4), [1, 0.4, 0.32, 0.256])
print("§2 ok: AR(2) forecast means/variances, 7.12 limit, MC, MA(1) bound, ARMA(1,1)
 ↪  0.5231")
```

---

### 3 · Unit roots, Dickey–Fuller, spurious regression → L08-03-unit-roots-dickey-fuller

`df_tstats` is the figure's simulation with the same seed, so the quantiles printed in the figure and here are identical; they must match Wooldridge's Table 18.2 to 0.06. `adfuller` with no lags equals the hand $t$ ratio of the regression $\Delta y_t = c + u\, y_{t-1} + e_t$. The spurious-regression rate (66.2% at $n = 50$, Davidson–MacKinnon via W p. 579) and the telescoping identity behind the DF numerator (L08-03 (21)) close the section.

```
# === §3 Unit roots, Dickey-Fuller, spurious regression (L08-03) ===
def df_tstats(T=250, R=20000, seed=2026):
    """t ratio of u in Delta y = c + u y_{-1} + e under H0 (driftless RW): the
     ↪  figure's simulation, same seed."""
    rng = np.random.default_rng(seed)
```

```
    y = rng.standard_normal((R, T)).cumsum(axis=1)
    dy, yl = np.diff(y, axis=1), y[:, :-1]
    ylc = yl - yl.mean(axis=1, keepdims=True)
    b = (ylc * dy).sum(1) / (ylc ** 2).sum(1)
    resid = dy - dy.mean(1, keepdims=True) - b[:, None] * ylc
    se = np.sqrt((resid ** 2).sum(1) / (T - 3) / (ylc ** 2).sum(1))
    return b / se


t = df_tstats()
q = np.quantile(t, [0.01, 0.025, 0.05, 0.10])
assert np.allclose(q, [-3.43, -3.12, -2.86, -2.57], atol=0.06)        # W Table 18.2
 ↪  (p. 575)
assert near((t < -1.645).mean(), 0.465, 0.01)
# One series: statsmodels' ADF agrees with the hand regression (same lag structure,
 ↪  constant only).
rng = np.random.default_rng(7)
rw = rng.standard_normal(500).cumsum()
stat, pval, *_ = adfuller(rw, maxlag=0, regression="c", autolag=None)
X = sm.add_constant(rw[:-1])
fit = sm.OLS(np.diff(rw), X).fit()
assert near(stat, fit.tvalues[1], 1e-8) and pval > 0.05
ar = signal.lfilter([1.0], [1, -0.5], rng.standard_normal(500))
assert adfuller(ar, maxlag=0, regression="c", autolag=None)[1] < 0.01
# Spurious regression (W p. 579): two independent RWs, n = 50, nominal 5% two-sided t
 ↪  test.
rng = np.random.default_rng(1974)
R, n = 10000, 50
xs = rng.standard_normal((R, n)).cumsum(1)
ys = rng.standard_normal((R, n)).cumsum(1)
xc, yc = xs - xs.mean(1, keepdims=True), ys - ys.mean(1, keepdims=True)
b1 = (xc * yc).sum(1) / (xc ** 2).sum(1)
u = yc - b1[:, None] * xc
tsp = b1 / np.sqrt((u ** 2).sum(1) / (n - 2) / (xc ** 2).sum(1))
assert near((np.abs(tsp) > stats.t.ppf(0.975, n - 2)).mean(), 0.662, 0.03)
# 3M drift (T eq. 2.37): t = 0.0103/0.002318 = 4.44; crossover (sigma/mu)^2 = 38.2
 ↪  months (L08-03 (10)).
assert near(0.0103 / (0.0637 / math.sqrt(755)), 4.44, 5e-3) and near((0.0637 /
 ↪  0.0103) ** 2, 38.2, 0.05)
# Telescoping (L08-03 (21)): sum p_{t-1} e_t = (p_T^2 - sum e_t^2)/2 exactly, p_0 =
 ↪  0.
e = rng.standard_normal(300)
p = e.cumsum()
assert near((np.r_[0, p[:-1]] * e).sum(), (p[-1] ** 2 - (e ** 2).sum()) / 2, 1e-9)
assert near(stats.chi2.cdf(1, 1), 0.683, 5e-4)                        # P(chi2_1 <
 ↪  1): DF numerator negative
print("§3 ok: DF quantiles", np.round(q, 2), "false rejection 46.5%, ADF = hand t,
 ↪  spurious 66%, telescoping")
```

## 4 · Long memory → L08-04-long-memory-arfima

The product recursion for $\rho_k$ is checked against the Gamma-function form for four values of $d$ (including a negative one, where $\Gamma(d) < 0$: hence `gammasgn`), the PACF $d/(k-d)$ is recovered from the ACF by Durbin–Levinson (T p. 102, property 4), and the asymptotic $k^{2d-1}$ law is within 1% at $k = 100$.

```
# === §4 Long memory (L08-04) ===
for d in (0.1, 0.25, 0.4, -0.25):
    rr = mf.arfima_acf(d, 200)
    k = np.arange(1, 201)
    gam = special.gammasgn(d) * np.exp(special.gammaln(k + d) + special.gammaln(1 -
 ↪  d)       # Gamma ratio, log scale
                                       - special.gammaln(k - d + 1) -
                                        ↪  special.gammaln(d))
    assert np.allclose(rr[1:], gam, rtol=1e-9) and near(rr[1], d / (1 - d), 1e-12)
     ↪  # L08-04 (7)-(10)
    assert np.allclose(mf.pacf_from_acf(rr, 8), d / (np.arange(1, 9) - d), atol=1e-9)
     ↪  # PACF d/(k-d), T p. 102
r25, r4 = mf.arfima_acf(0.25, 100), mf.arfima_acf(0.4, 100)
assert np.allclose(r25[1:4], [1 / 3, 0.2381, 0.1948], atol=5e-5)
asym = special.gamma(1 - 0.4) / special.gamma(0.4) * 100 ** (2 * 0.4 - 1)          #
 ↪  (-d)!/(d-1)! k^(2d-1)
assert near(r4[100], 0.2673, 5e-5) and abs(asym / r4[100] - 1) < 0.01 and (2 / 3) **
 ↪  100 < 3e-18
coef = np.array([1, -0.25, -0.25 * 0.75 / 2, -0.25 * 0.75 * 1.75 / 6])
 ↪  # (1-B)^d, L08-04 (3)
assert np.allclose(coef, [special.binom(0.25, j) * (-1) ** j for j in range(4)])
print("§4 ok: ARFIMA ACF = Gamma form, PACF d/(k-d), rho_100(0.4) = 0.267, binomial
 ↪  coefficients")
```

---

## 5 · Seasonal models → L08-05-seasonal-airline-model

statsmodels' `SARIMAX` reproduces the slides' and the page's airline fits on log AirPassengers for $n = 96$, 108 and 144 to the third decimal (log-likelihood 148.76, BIC −284.27 / −336.93 / −474.77), and the BIC search over the 16 models picks $(0,1,1)(0,1,1)_{12}$. The out-of-sample RMSE comparison (36.6 vs 63.3) is our computation, not the page's. The airline ACF L08-05 (11)–(12) is checked against the ACF of the expanded MA polynomial, the corrected ARMA(26,13) coefficients against `numpy.polymul`, and Tsay's January-effect regression against his own data file.

```
# === §5 Seasonal models (L08-05) ===
AIR = np.array([112, 118, 132, 129, 121, 135, 148, 148, 136, 119, 104, 118, 115, 126,
 ↪  141, 135, 125, 149, 170, 170, 158, 133, 114, 140,
                145, 150, 178, 163, 172, 178, 199, 199, 184, 162, 146, 166, 171, 180,
                 ↪  193, 181, 183, 218, 230, 242, 209, 191, 172, 194,
                196, 196, 236, 235, 229, 243, 264, 272, 237, 211, 180, 201, 204, 188,
                 ↪  235, 227, 234, 264, 302, 293, 259, 229, 203, 229,
                242, 233, 267, 269, 270, 315, 364, 347, 312, 274, 237, 278, 284, 277,
                 ↪  317, 313, 318, 374, 413, 405, 355, 306, 271, 306,
                315, 301, 356, 348, 355, 422, 465, 467, 404, 347, 305, 336, 340, 318,
                 ↪  362, 348, 363, 435, 491, 505, 404, 359, 310, 337,
```

```
                360, 342, 406, 396, 420, 472, 548, 559, 463, 407, 362, 405, 417, 391,
                 ↪  419, 461, 472, 535, 622, 606, 508, 461, 390, 432], float)
assert len(AIR) == 144 and AIR.sum() == 40363                          # seasonal.pdf
 ↪  slide 17 (Box-Jenkins Series G)
ly = np.log(AIR)


def airline(n):
    """ARIMA(0,1,1)x(0,1,1)_12 on log passengers 1949.1..(n); statsmodels prints
     ↪  +theta (Tsay's -theta)."""
    return SARIMAX(ly[:n], order=(0, 1, 1), seasonal_order=(0, 1, 1,
     ↪  12)).fit(disp=False)


printed = {96: (-0.3941, -0.6129, -284.27), 108: (-0.3864, -0.5885, -336.93), 144:
 ↪  (-0.4018, -0.5569, -474.77)}
for n, (ma1, sma1, bic) in printed.items():
    m = airline(n)
    assert near(m.params[0], ma1, 2e-3) and near(m.params[1], sma1, 2e-3) and
     ↪  near(m.bic, bic, 0.02), (n, m.params, m.bic)
assert near(airline(96).llf, 148.76, 5e-3) and near(airline(144).aic, -483.39, 5e-3)
grid = {}
for i in (0, 1):
    for j in (0, 1):
        for k in (0, 1):
            for l in (0, 1):
                grid[(i, j, k, l)] = SARIMAX(ly[:108], order=(i, 1, j),
 ↪  seasonal_order=(k, 1, l, 12)).fit(disp=False).bic
assert min(grid, key=grid.get) == (0, 1, 0, 1)                          # the page:
 ↪  BIC picks (0,1,1)(0,1,1)_12
pred = np.exp(airline(108).get_forecast(36).predicted_mean)
tt = np.arange(144)
D = np.column_stack([np.ones(144), tt] + [(tt % 12 == mth).astype(float) for mth in
 ↪  range(1, 12)])
bdet = np.linalg.lstsq(D[:108], ly[:108], rcond=None)[0]
rmse_air = math.sqrt(np.mean((AIR[108:] - pred) ** 2))
rmse_det = math.sqrt(np.mean((AIR[108:] - np.exp(D[108:] @ bdet)) ** 2))
assert near(rmse_air, 36.6, 0.05) and near(rmse_det, 63.3, 0.05)
# Airline ACF (T p. 84) vs statsmodels' ACF of the expanded MA(s+1) polynomial
 ↪  (statsmodels sign: 1 + ...).
for th, TH, s, vals in [(0.678, 0.314, 4, (-0.4645, -0.2858, 0.1328)), (0.3864,
 ↪  0.5885, 12, (-0.3362, -0.4371, 0.1470))]:
    poly = np.polymul([1, -th], np.r_[1, np.zeros(s - 1), -TH])
    acf = ArmaProcess(ma=poly).acf(s + 3)
    assert np.allclose([acf[1], acf[s], acf[s - 1], acf[s + 1]], [vals[0], vals[1],
     ↪  vals[2], vals[2]], atol=5e-5)
    assert near(acf[s + 1], acf[1] * acf[s], 1e-12) and np.abs(np.r_[acf[2:s - 1],
     ↪  acf[s + 2:]]).max() < 1e-12
# ARIMA(1,1,1)x(1,1,1)_12 = constrained ARMA(26,13) (L08-05 (16)-(18)): the corrected
 ↪  lag-12 and lag-13 coefficients.
```

```
phi, PHI = 0.3, 0.2
ar_full = np.polymul(np.polymul([1, -phi], np.r_[1, np.zeros(11), -PHI]),
 ↪  np.polymul([1, -1], np.r_[1, np.zeros(11), -1]))
assert len(ar_full) == 27                                                  # degree 26
assert near(-ar_full[12], 1 + PHI, 1e-12) and near(-ar_full[13], -(1 + phi) * (1 +
 ↪  PHI), 1e-12)
assert near(-ar_full[1], 1 + phi, 1e-12) and near(-ar_full[2], -phi, 1e-12) and
 ↪  near(-ar_full[26], -phi * PHI, 1e-12)
# CRSP Decile 1, January effect (T Ex. 2.4, p. 88): R_t = 0.002864 + 0.125251 Jan_t.
dec = np.loadtxt(DATA / "m-deciles08.txt", skiprows=1)
jan = ((dec[:, 0] // 100) % 100 == 1).astype(float)
ols = sm.OLS(dec[:, 1], sm.add_constant(jan)).fit()
assert len(dec) == 468 and np.allclose(ols.params, [0.002864, 0.125251], atol=5e-7)
assert np.allclose(ols.bse, [0.003333, 0.011546], atol=5e-7) and near(ols.rsquared,
 ↪  0.2016, 5e-5)
print(f"§5 ok: airline fits = slides, BIC winner, RMSE {rmse_air:.1f} vs
 ↪  {rmse_det:.1f}, ACF, ARMA(26,13), January 0.1253")
```

---

## 6 · ARCH and GARCH: moments, forecasts, maximum likelihood → L08-06-arch-garch

`garch_var` is the recursion (T 3.16) as one linear filter; `nll_garch` is the conditional log-likelihood, Gaussian or standardized $t$ (L08-06 (15), (16)–(17)); `fit_garch` maximises it by Nelder–Mead with $\sigma_1^2$ fixed at the sample variance (T p. 134). Checks: the Gaussian kurtosis formula (25) against a seeded million-draw simulation (with a small $\alpha_1$ so that the sample kurtosis converges), Tsay Ex. 3.3 and Table 3.1, the IGARCH line, the Petrobras numbers, recovery of the truth on simulated data, and a fit to Tsay's daily S&P 500 (1999–2008, $T = 2515$). On that file the Gaussian MLE gives $(\hat\alpha_0, \hat\alpha_1, \hat\beta_1) = (1.01 \times 10^{-6},\ 0.0710,\ 0.9238)$, persistence 0.9948, and the $t$ fit has the higher log-likelihood.

```
# === §6 ARCH and GARCH: moments, forecasts, maximum likelihood (L08-06) ===
def garch_var(a, w, al, be, s0):
    """sigma^2_t = w + al a^2_{t-1} + be sigma^2_{t-1} (T eq. 3.16) as one linear
     ↪  filter; sigma^2_1 = s0."""
    x = w + al * np.r_[0.0, a[:-1] ** 2]
    x[0] = s0
    return signal.lfilter([1.0], [1.0, -be], x)


def nll_garch(par, a, s0, dist="normal"):
    """Negative conditional log-likelihood, normal (L08-06 (15)) or standardized t
     ↪  (L08-06 (16)-(17))."""
    w, al, be = par[:3]
    if w <= 0 or al < 0 or be < 0 or al + be >= 1:
        return 1e12
    s2 = garch_var(a, w, al, be, s0)
    if dist == "normal":
        return 0.5 * np.sum(np.log(2 * np.pi * s2) + a ** 2 / s2)
    nu = par[3]
```

```
    if nu <= 2.05:
        return 1e12
    c = special.gammaln((nu + 1) / 2) - special.gammaln(nu / 2) - 0.5 * np.log((nu -
 ↪  2) * np.pi)
    return -np.sum(c - 0.5 * np.log(s2) - (nu + 1) / 2 * np.log1p(a ** 2 / ((nu - 2)
     ↪  * s2)))


def fit_garch(a, dist="normal"):
    """Conditional MLE by Nelder-Mead; sigma^2_1 fixed at the sample variance (T p.
     ↪  134)."""
    s0 = a.var()
    x0 = [0.05 * s0, 0.05, 0.90] + ([8.0] if dist == "t" else [])
    res = optimize.minimize(nll_garch, x0, args=(a, s0, dist), method="Nelder-Mead",
                            options=dict(xatol=1e-10, fatol=1e-10, maxiter=40000,
 ↪  maxfev=40000))
    return res.x, -res.fun


def sim_garch(n, w, al, be, rng, nu=None, burn=1000):
    """Simulate a GARCH(1,1) with N(0,1) or standardized t_nu innovations."""
    z = rng.standard_normal(n + burn) if nu is None else rng.standard_t(nu, n + burn)
 ↪  * math.sqrt((nu - 2) / nu)
    a, s2 = np.empty(n + burn), w / (1 - al - be)
    for t_ in range(n + burn):
        a[t_] = math.sqrt(s2) * z[t_]
        s2 = w + al * a[t_] ** 2 + be * s2
    return a[burn:]


assert np.allclose(garch_var(np.arange(5.0), 0.1, 0.2, 0.7, 1.0), [1.0, 0.8, 0.86,
 ↪  1.502, 2.9514])   # recursion by hand
# ARCH(1) (L08-06 (8), (12)): Intel (3.12) uncond. sd and kurtosis.
assert near(math.sqrt(0.0111 / (1 - 0.356)), 0.1313, 5e-5) and near(3 * (1 - 0.356 **
 ↪  2) / (1 - 3 * 0.356 ** 2), 4.227, 5e-4)
assert near(math.sqrt(0.00070027 / (1 - 0.36978713)), 0.0333, 5e-5)    # Petrobras
 ↪  ARCH(1)
# GARCH(1,1) Gaussian kurtosis (L08-06 (25)) against a long seeded simulation (alpha
 ↪  small: 8th moment finite).
w, al, be = 0.1, 0.05, 0.90
kg = 6 * al ** 2 / (1 - 2 * al ** 2 - (al + be) ** 2)
sim = sim_garch(1_000_000, w, al, be, np.random.default_rng(3))
assert near(np.mean(sim ** 2), w / (1 - al - be), 0.03) and abs(stats.kurtosis(sim)
 ↪  - kg) < 0.05
# Tsay Ex. 3.3 (3.19): persistence, sigma_inf, half-life, kurtosis; the 0.9772 on p.
 ↪  137 is 0.9727.
w, al, be = 0.000086, 0.1216, 0.8511
pers = al + be
assert near(pers, 0.9727, 1e-12) and near(w / (1 - pers), 0.003150, 5e-7) and
 ↪  near(math.sqrt(w / (1 - pers)), 0.0561, 5e-5)
```

```
assert near(math.log(0.5) / math.log(pers), 25.04, 5e-3) and near(3 + 6 * al ** 2 /
 ↪  (1 - 2 * al ** 2 - pers ** 2), 6.654, 5e-4)
s1 = 0.0536
fc = [s1 ** 2]
for _ in range(4):
    fc.append(w + pers * fc[-1])                                        # (3.17)
closed = [w / (1 - pers) + pers ** (l - 1) * (s1 ** 2 - w / (1 - pers)) for l in
 ↪  range(1, 6)]
assert np.allclose(fc, closed) and np.allclose(np.sqrt(fc), [0.0536, 0.0537, 0.0537,
 ↪  0.0538, 0.0538], atol=1.2e-4)
assert near(0.9583 - 0.8456, 0.1127, 1e-12)                             # two-pass
 ↪  estimate (T p. 140)
ig = [s1 ** 2]
for _ in range(9):
    ig.append(0.000119 + 1.0 * ig[-1])
assert np.allclose(np.diff(ig), 0.000119)                               # IGARCH
 ↪  forecasts are a line (3.22)
# Petrobras normal GARCH (garchmodeling.html): persistence, sigma_inf, half-life,
 ↪  excess kurtosis, LR.
w, al, be = 2.254954e-05, 0.1031668, 0.8768059
assert near(al + be, 0.97997, 5e-6) and near(math.sqrt(w / (1 - al - be)), 0.03356,
 ↪  5e-6)
assert near(math.log(0.5) / math.log(al + be), 34.26, 5e-3) and near(6 * al ** 2 / (1
 ↪  - 2 * al ** 2 - (al + be) ** 2), 3.477, 5e-4)
assert near(2 * (11151.72 - 10672.32), 958.8, 1e-9)
# MLE recovers the truth on a seeded sample (normal and t), and runs on Tsay's daily
 ↪  S&P 500 file.
rng = np.random.default_rng(11)
a = sim_garch(4000, 2e-5, 0.08, 0.90, rng)
th_n, ll_n = fit_garch(a)
assert abs(th_n[1] - 0.08) < 0.03 and abs(th_n[2] - 0.90) < 0.04
a_t = sim_garch(4000, 0.02, 0.08, 0.90, np.random.default_rng(12), nu=7)
th_t, ll_t = fit_garch(a_t, "t")
assert abs(th_t[3] - 7) < 2.5 and abs(th_t[1] + th_t[2] - 0.98) < 0.03 and ll_t >
 ↪  fit_garch(a_t)[1]
gs = np.loadtxt(DATA / "d-gmsp9908.txt", skiprows=1)
sp = np.log1p(gs[:, 2])
sp = sp - sp.mean()
th_sp, ll_sp = fit_garch(sp)
th_spt, ll_spt = fit_garch(sp, "t")
assert len(sp) == 2515 and 0.98 < th_sp[1] + th_sp[2] < 1 and 0.03 < th_sp[1] < 0.15
 ↪  and ll_spt > ll_sp
# The Volatility Dial's default read-outs (imported) agree.
o, regime, _ = cc.ga_python(0.86, 0.1216, 0.8511, 5.36, True, 7.04, 60)
assert regime == "stationary" and (o["pers"], o["hl"], o["sinf"], o["kg"]) ==
 ↪  ("0.9727", "25.04", "5.61", "3.654")
print("§6 ok: ARCH/GARCH moments, Table 3.1, IGARCH line, Petrobras numbers, MLE
 ↪  (normal, t), S&P 500:",
      np.round(th_sp, 6), "persistence", round(th_sp[1] + th_sp[2], 4))
```

## 7 · Asymmetry and tails → L08-07-garch-family-asymmetry-tails

The APARCH-to-GJR mapping L08-07 (6)–(7) is checked both numerically and as an identity between the two news-impact curves; the literal reading of the glossary's GJR formula is shown to be nonstationary. $E|\varepsilon|$ for the standardized $t$ (T p. 143) is checked against quadrature (0.7595 at $\nu = 7.04$), Tiao's two forms of the kurtosis (T p. 166) against each other (12.55 for the Petrobras GARCH-$t$), and the AIC ladder of the glossary page is recomputed.

```
# === §7 Asymmetry and tails (L08-07) ===
ag, gg, bg = 0.09234234, 0.2083503, 0.8817760                          # fGarch
 ↪  leverage, delta = 2 (APARCH form)
assert near(ag * (1 - gg) ** 2, 0.05787, 5e-6) and near(ag * (1 + gg) ** 2, 0.13483,
 ↪  5e-6)
assert near(4 * ag * gg, 0.07696, 5e-6) and near(ag * (1 + gg ** 2) + bg, 0.97813,
 ↪  5e-6)
assert ag + gg / 2 + bg > 1                                             # the literal
 ↪  indicator reading is not stationary
z = np.linspace(-6, 6, 2001)
nic_aparch = ag * (np.abs(z) - gg * z) ** 2
nic_gjr = (ag * (1 - gg) ** 2 + 4 * ag * gg * (z < 0)) * z ** 2       # the same curve
 ↪  written as GJR
assert np.allclose(nic_aparch, nic_gjr)
at_, gt_ = 0.0964078, 0.3458799                                         # T-GARCH
 ↪  (delta = 1)
assert near(0.09641 * 1.34588, 0.12976, 5e-6) and near(at_ * (1 + gt_), 0.12975,
 ↪  5e-6) and near(at_ * (1 - gt_), 0.06306, 5e-5)
ae, ge = -0.05549, 0.18138                                              # rugarch
 ↪  EGARCH: ratio exp(-2 alpha c)
assert near(math.exp(-2 * ae * 1), 1.117, 5e-4) and near(math.exp(-2 * ae * 2),
 ↪  1.249, 5e-4)
assert near(ae * -1 + ge * 1, 0.23687, 5e-6) and near(ae * 1 + ge * 1, 0.12589, 5e-6)
# E|eps|: sqrt(2/pi) for N(0,1); for the standardized t (T p. 143) by formula and by
 ↪  quadrature.
assert near(integrate.quad(lambda x: abs(x) * stats.norm.pdf(x), -np.inf, np.inf)[0],
 ↪  math.sqrt(2 / math.pi), 1e-9)
nu = 7.04
sc = math.sqrt((nu - 2) / nu)
eabs_formula = 2 * math.sqrt(nu - 2) * math.exp(special.gammaln((nu + 1) / 2) -
 ↪  special.gammaln(nu / 2)) / ((nu - 1) * math.sqrt(math.pi))
eabs_quad = integrate.quad(lambda x: abs(x) * stats.t.pdf(x / sc, nu) / sc, -np.inf,
 ↪  np.inf)[0]
assert near(eabs_formula, eabs_quad, 1e-8) and near(eabs_formula, 0.7595, 5e-5)
# Tiao's kurtosis formula (T p. 166): the general and the t forms agree; Petrobras
 ↪  GARCH-t total 12.55.
al, be, nu = 0.07428964, 0.9094674, 7.042551
kg = 6 * al ** 2 / (1 - 2 * al ** 2 - (al + be) ** 2)
K = 6 / (nu - 4)
ka1 = (K + kg + 5 / 6 * K * kg) / (1 - K * kg / 6)
```

```
ka2 = (6 + (nu + 1) * kg) / (nu - 4 - kg)
assert near(ka1, ka2, 1e-9) and near(kg, 1.563, 5e-4) and near(ka1, 12.55, 0.01)
assert 1 - 2 * al ** 2 * (nu - 1) / (nu - 4) - (al + be) ** 2 > 0
ll = {"GARCH": (11151.72, 3), "GJR": (11174.01, 4), "EGARCH": (11173.14, 5),
 ↪  "GARCH-t": (11292.37, 4), "skew-t": (11297.85, 5)}
aic_ = {k: -2 * v + 2 * p for k, (v, p) in ll.items()}
assert min(aic_, key=aic_.get) == "skew-t" and near(aic_["GARCH"], -22297.44, 1e-6)
assert near(-4.3229 * 5167, -22336.4, 0.1)                                  # rugarch's
 ↪  per-observation AIC, scaled back
print("§7 ok: APARCH -> GJR mapping, EGARCH ratios, E|eps| = 0.7595, Tiao 12.55, AIC
 ↪  ladder")
```

---

## 8 · Stochastic volatility → L08-08-stochastic-volatility

Closed-form moments L08-08 (5)–(9) for Tsay Ex. 12.3 and the Petrobras SV-$t$. The ACF formula is then checked on iid draws of the pair $(h_t, h_{t-k})$ from its bivariate normal: this isolates the formula from the very slow convergence of sample autocorrelations of $a_t^2$ when $\phi = 0.984$. A simulated SV path with Tsay's milder parameters checks kurtosis and the lag-1 correlation in the time domain. Last, the moments of $\ln \varepsilon^2$ used by the QML linearisation and the completing-the-square identity behind (12.23).

```
# === §8 Stochastic volatility (L08-08) ===
def sv_moments(mu_h, s2h, phi_):
    """E(a^2), kurtosis 3 e^{s2h}, corr(a_t^2, a_{t-1}^2): L08-08 (5)-(9), Gaussian
     ↪  eps."""
    return math.exp(mu_h + s2h / 2), 3 * math.exp(s2h), (math.exp(s2h * phi_) - 1) /
     ↪  (3 * math.exp(s2h) - 1)


mu_h, s2h = 0.454 / (1 - 0.837), 0.086 / (1 - 0.837 ** 2)                  # Tsay Ex.
 ↪  12.3
Ea2, kurt, c1 = sv_moments(mu_h, s2h, 0.837)
assert near(mu_h, 2.785, 5e-4) and near(s2h, 0.2872, 5e-5) and near(Ea2, 18.71, 5e-3)
 ↪  and near(kurt, 3.998, 5e-4)
assert near(math.sqrt(Ea2), 4.33, 5e-3) and near(c1, 0.0906, 1e-4)
phi_, sig_ = 0.9842135, 0.1387539                                          # PBR SV-t
 ↪  posterior means (stochvol h = log variance)
s2h_p = sig_ ** 2 / (1 - phi_ ** 2)
_, kurt_p, c1_p = sv_moments(-0.4190952, s2h_p, phi_)
assert near(s2h_p, 0.6146, 5e-5) and near(kurt_p, 5.547, 5e-4) and near(c1_p, 0.1828,
 ↪  5e-5)
# The ACF formula, checked on iid draws of (h_t, h_{t-k}) from their bivariate normal
 ↪  (no time-series convergence issue).
rng = np.random.default_rng(88)
nn = 4_000_000
for kk in (1, 20):
    cov = s2h_p * np.array([[1, phi_ ** kk], [phi_ ** kk, 1]])
    hh = rng.multivariate_normal([0, 0], cov, nn)
    a2 = np.exp(hh) * rng.standard_normal((nn, 2)) ** 2
    formula = (math.exp(s2h_p * phi_ ** kk) - 1) / (3 * math.exp(s2h_p) - 1)
```

```
    assert abs(np.corrcoef(a2[:, 0], a2[:, 1])[0, 1] - formula) < 0.02, kk
# A simulated SV path (Tsay 12.3 parameters): sample kurtosis and lag-1 corr of a^2
 ↪  near the formulas.
nT = 2_000_000
hT = signal.lfilter([1.0], [1, -0.837], 0.454 + math.sqrt(0.086) *
 ↪  rng.standard_normal(nT + 2000))[2000:]
aT = np.exp(hT / 2) * rng.standard_normal(nT)
assert abs(np.mean(aT ** 2) / Ea2 - 1) < 0.02 and abs(stats.kurtosis(aT,
 ↪  fisher=False) - kurt) < 0.15
assert abs(np.corrcoef(aT[1:] ** 2, aT[:-1] ** 2)[0, 1] - c1) < 0.01
# ln(eps^2) for eps ~ N(0,1): mean -1.2704, variance pi^2/2 (L08-08 (11)).
m1 = 2 * integrate.quad(lambda x: math.log(x * x) * stats.norm.pdf(x), 0, np.inf)[0]
 ↪  # symmetric: twice (0, inf)
m2 = 2 * integrate.quad(lambda x: math.log(x * x) ** 2 * stats.norm.pdf(x), 0,
 ↪  np.inf)[0]
assert near(m1, -1.2704, 5e-5) and near(m2 - m1 ** 2, math.pi ** 2 / 2, 1e-6)
# Completing the square (L08-08 (17)): (x-a)^2 A + (x-b)^2 C = (x-c)^2 (A+C) +
 ↪  (a-b)^2 AC/(A+C).
xg, A_, a_, C_, b_ = 0.7, 1.0, -0.3, 0.837 ** 2, 1.9
c_ = (A_ * a_ + C_ * b_) / (A_ + C_)
assert near((xg - a_) ** 2 * A_ + (xg - b_) ** 2 * C_, (xg - c_) ** 2 * (A_ + C_) +
 ↪  (a_ - b_) ** 2 * A_ * C_ / (A_ + C_), 1e-12)
print("§8 ok: SV moments (Tsay 12.3, PBR), ACF formula by MC, simulated path, E ln
 ↪  chi2_1, (12.23) identity")
```

---

## 9 · Bayesian GARCH → L08-09-bayesian-garch

The M0 prior and the prior weight 4/884; the conjugate update as a weighted average on a simulated sample. Then the GARCH(1,1)-$t$ posterior of `bayesGARCH` (L08-09 (12)–(15): $h_0 = r_0 = 0$, $V(r \mid h) = h$, truncated $N(0, 1000)$ priors, $\nu - 2 \sim \text{Exp}(0.01)$, no stationarity constraint) is sampled by a random-walk Metropolis started at the MLE, on 2000 simulated observations with truth $(0.02, 0.08, 0.90, \nu = 7)$. Run of 2026-10-27: acceptance 0.20; posterior means $(0.020, 0.079, 0.908, 5.84)$ after 5000 burn-in; $\Pr(\alpha_1 + \beta > 1 \mid \text{data}) = 0.041$. Every posterior mean is within four posterior sds of the truth. `bayesGARCH` itself uses a better proposal (Ardia–Hoogerheide's GLS approximation): this sampler is the simplest correct one, not a fast one.

```
# === §9 Bayesian GARCH (L08-09) ===
m_r2, n0, n = 0.0006617694, 1506, 1760                                  #
 ↪  bayes-garch.pdf slides 6-8
a0_, b0_ = 5.0, 4 * m_r2
assert near(b0_ / (a0_ - 1), m_r2, 1e-15) and near(math.sqrt(b0_ ** 2 / ((a0_ - 1) **
 ↪  2 * (a0_ - 2))), 0.000382, 5e-7)
w_prior = (a0_ - 1) / (a0_ - 1 + n / 2)
assert near(w_prior, 4 / 884, 1e-12) and near(w_prior, 0.0045, 5e-5)
# M0 conjugate update on a simulated sample: posterior IG(a + n/2, b + S/2) =
 ↪  weighted average of prior mean and S/n.
r0 = 0.033 * np.random.default_rng(5).standard_normal(n)
S = float((r0 ** 2).sum())
post_mean = (b0_ + S / 2) / (a0_ + n / 2 - 1)
```

```
assert near(post_mean, w_prior * m_r2 + (1 - w_prior) * S / n, 1e-15)
# M1 prior sd of mu: sqrt(100 V/(n0 - 1)) = 0.006627; M2 prior sds 10 s_a, 10 s_b.
assert near(math.sqrt(100 * 0.0006609406 / (n0 - 1)), 0.006627, 5e-7) and near(10 *
 ↪  0.0009933758, 0.00993, 5e-6)


def logpost_bgarch(par, r):
    """log posterior of bayesGARCH's GARCH(1,1)-t: h_0 = 0, r_0 = 0, V(r|h) = h
     ↪  (L08-09 (12)-(15))."""
    w, al, be, nu = par
    if w <= 0 or al <= 0 or be <= 0 or nu <= 2:
        return -np.inf
    x = w + al * np.r_[0.0, r[:-1] ** 2]
    h = signal.lfilter([1.0], [1.0, -be], x)                          # h_1 = w since
 ↪  h_0 = r_0 = 0
    c = special.gammaln((nu + 1) / 2) - special.gammaln(nu / 2) - 0.5 * np.log((nu -
 ↪  2) * np.pi)
    ll_ = np.sum(c - 0.5 * np.log(h) - (nu + 1) / 2 * np.log1p(r ** 2 / ((nu - 2) *
 ↪  h)))
    lp = -(w ** 2 + al ** 2 + be ** 2) / 2000 - 0.01 * (nu - 2)      # N(0, 1000)
 ↪  truncated; nu - 2 ~ Exp(0.01)
    return ll_ + lp


def rw_metropolis(r, start, steps, iters, seed):
    """Random-walk Metropolis on (w, alpha, beta, nu); returns draws and the
     ↪  acceptance rate."""
    rng_ = np.random.default_rng(seed)
    cur, lpc = np.array(start, float), logpost_bgarch(start, r)
    out, acc = np.empty((iters, 4)), 0
    for i in range(iters):
        prop = cur + steps * rng_.standard_normal(4)
        lpp = logpost_bgarch(prop, r)
        if math.log(rng_.uniform()) < lpp - lpc:
            cur, lpc, acc = prop, lpp, acc + 1
        out[i] = cur
    return out, acc / iters


true = (0.02, 0.08, 0.90, 7.0)                                        #
 ↪  standardized-return scale, like GARCH11-SVar1
rb = sim_garch(2000, *true[:3], np.random.default_rng(21), nu=true[3])
mle, _ = fit_garch(rb, "t")
draws, acc = rw_metropolis(rb, mle, np.array([0.004, 0.008, 0.010, 0.8]), 20000,
 ↪  seed=22)
post = draws[5000:]
pm, psd = post.mean(0), post.std(0)
assert 0.15 < acc < 0.6 and np.all(np.abs(pm - np.array(true)) < 4 * psd + 1e-3)
p_nonstat = float((post[:, 1] + post[:, 2] > 1).mean())
assert p_nonstat < 0.1 and abs(pm[1] + pm[2] - (mle[1] + mle[2])) < 0.02
```

```
# The page's Bayes vs ML numbers (GARCH11-SVar1.html): persistence and half-life.
pb, pm_ = 0.08502667 + 0.89402444, 0.073478 + 0.910218
assert near(pb, 0.979051, 5e-7) and near(pm_, 0.983696, 5e-7)
assert near(math.log(0.5) / math.log(pb), 32.74, 5e-3) and near(math.log(0.5) /
 ↪  math.log(pm_), 42.17, 5e-3)
print(f"§9 ok: IG update (prior weight 0.45%), MH acceptance {acc:.2f}, posterior
 ↪  means {np.round(pm, 3)},",
      f"Pr(alpha+beta>1) = {p_nonstat:.4f}, Bayes vs ML persistence")
```

---

## 10 · Run it

```
print("ALL L08 CHECKS PASSED")
```

Output of the run on 2026-10-27 (abridged to the section lines):

```
§1 ok: AR(2) roots/triangle, gamma0, cycles 30.09/10.62/10.656, PACF, Table 2.1,
 ↪  companion
§2 ok: AR(2) forecast means/variances, 7.12 limit, MC, MA(1) bound, ARMA(1,1) 0.5231
§3 ok: DF quantiles [-3.45 -3.16 -2.87 -2.58] false rejection 46.5%, ADF = hand t,
 ↪  spurious 66%, telescoping
§4 ok: ARFIMA ACF = Gamma form, PACF d/(k-d), rho_100(0.4) = 0.267, binomial
 ↪  coefficients
§5 ok: airline fits = slides, BIC winner, RMSE 36.6 vs 63.3, ACF, ARMA(26,13),
 ↪  January 0.1253
§6 ok: ARCH/GARCH moments, Table 3.1, IGARCH line, Petrobras numbers, MLE (normal,
 ↪  t), S&P 500: [1.00000e-06 7.10430e-02 9.23751e-01] persistence 0.9948
§7 ok: APARCH -> GJR mapping, EGARCH ratios, E|eps| = 0.7595, Tiao 12.55, AIC ladder
§8 ok: SV moments (Tsay 12.3, PBR), ACF formula by MC, simulated path, E ln chi2_1,
 ↪  (12.23) identity
§9 ok: IG update (prior weight 0.45%), MH acceptance 0.20, posterior means [0.02
 ↪  0.079 0.908 5.838], Pr(alpha+beta>1) = 0.0406, Bayes vs ML persistence
ALL L08 CHECKS PASSED
```

---

## Links

- Unit: L08-00-map · L08-10-formulas · L08-11-pitfalls
- The companions' own checks: `companions/check_companions.py`; the figures' asserts: `figures/make_figures.py`
- Class 1 computations (AR(1), VAR(1)): L01-11
- Rules: rules/04 · rules/05